ADDITIONAL TERMS AND CONDITIONS

1. **Pre-bid Conference:**

   a) a) It is hereby intimated that a Pre Bid Meeting in respect of "**Design, Development, Installation, Commissioning and Certification of Electrical Power and Generation System(EPGS) Test Rig"** will be held on **30 Jul 2025** by **10:30 Hrs** at **ERAS**, **Conference Hall, CABS, Bangalore -37.**

   b) All Prospective Vendors / bidders are to forward their queries, if any, by **29 Jul 2025** by **17:00 Hrs** to **Priyanka Meena Scientist 'E'** (Tele:080-25049461, Fax No.:080-25049123 / 25222326, Email ID: **Priyanka-meena.cabs@gov.in.** along with the names of Personnel attending the Pre Bid Meeting (Only Indian Nationals).

   c) The Confirmation of participation by vendors in Pre Bid Meeting should be forwarded by on or before on **29 Jul 2025 by 17:00 Hrs**

   d) Attending of Pre Bid meeting at CABS, DRDO at the designated time and place is a mandatory requirement.

   e) Video-conferencing not Allowed.

2. **Payment Milestone MATRIX**

   Payment Terms: Milestone payment will be made within 30 days on completion of each phase and acceptance/ completion certificate from user for the work carried out and on production of invoice. (30 days will be counted from the date of receipt of invoice at CABS). The following milestone payments with percentage as follows: -

		Payment Milestone MATRIX		
SL. No.	**Milestone**	**Description (Milestone Wise)**	**% of Payment**	**Project Time Line(T0+Months)**
1	**Milestone-1**	**Submission of High Level Documentation(HLD) and Implementation Plan in AGEST rig** Payment will be made within 30 days of Submission of Bills and confirmation of receipt and acceptance of High Level Documentations (HLD) & Implementation plan by user, whichever is later	**10%**	**T0+01 month**
2	**Milestone-2**	**Critical Design Review (CDR)** Payment will be made within 30 days of Submission of Bills and confirmation of receipt and acceptance of CDR document by user, whichever is later	**20%**	**T0 + 02 months**
3	**Milestone-3**	**FAT at Bidder's premises** Payment will be made within 30 days of Submission of Bills and confirmation of receipt and acceptance of FAT and ATP document by user, whichever is later	**10%**	**T0 + 08 months**
4	**Milestone-4**	**Installation, Integration & Commissioning of Rig at CABS** Payment within 30 days of submitting the bill and receipt of certificate of satisfactory completion of job from acceptance committee, whichever is later	**20%**	**T0 + 10 months**
5	**Milestone-5**	**ATP and Certification of Rig at CABS** Payment within 30 days of submitting the bill and receipt of certificate of satisfactory of ATP and certification of rig from acceptance committee, whichever is later.	**20%**	**T0 + 12 months**
6	**Milestone-6**	Manpower (4 Nos) for Operation, Maintenance and Testing of Aircraft electrical system in the rig after certification of the rig for 1 years	**20%** (with leftover balance if any)	**T0 + 24 months**

Conformance of Acceptability by Bidder	Yes / No	Yes / No
❖	❖ **T0: Indicates date of the GeM Contract**	

2. **Delivery Period:**

   Delivery Period for supply of items/rendering services would be **24 months** from the Effective Date of the Contract. Please note that the Contract can be cancelled unilaterally by the Buyer in case items are not received within the contracted delivery period including installation, Training and acceptance tests etc., as per scope, if any. Extension of contracted delivery period with/ without LD clause will be at the sole discretion of the Buyer.

3. **Intellectual Property Right**:

   CABS shall hold the sole rights for the complete design, software including its source code, etc. developed as part of the project. Selling of this product or its variant to any agency other than CABS shall be with the formal consent of CABS. CABS in such cases, reserves the right to levy appropriate royalty. All development and application software must be delivered in original CDs/Pendrives with source code. Bidder shall also provide the licenses for all the required development platforms on which the entire software has been developed (e.g. LABVIEW). Also the Bidder is required to deliver all the documents, data sheets, design documents, drawings for all the items specifically developed under this contract. In the case of off the shelf items Bidder is required to deliver complete data sheets, and manuals which are delivered as standard as part of the items

4. **Non-disclosure of Contract Documents:** Except with the written consent of the Buyer/ bidder, other party shall not disclose the Contract or any provision, specification, plan, design, pattern, sample or information thereof to any third party.

5. **Installation & Commissioning:** Required.

6. **Installation & Commissioning to be Completed: As per milestone mentioned in SOW**

7. **Training Details** : Required (**For 10 personnel for 15 days in CABS premises**).

8. **Vendor Qualification Criteria:**

8.1. Technical/ Work Experience Criteria:

8.1.1. As the EPGS rig has to work in conjunction with the existing Aircraft Generators & Electrical System Test (AGEST) facility at CABS, it is mandatory for all the bidders to attend the pre-bid meeting in person and visit the existing AGEST facility in order to understand the requirements unambiguously. Attending of Pre Bid meeting at CABS, DRDO at the designated time and place is a mandatory requirement. Bids of only those vendors will be considered for further evaluation, who attend the pre bid meeting at CABS, DRDO. **No video conference is being organized.**

8.1.2. Bidders should have executed jobs involving Design & Development, Integration & Commissioning and Certification of Aircraft Electrical Rigs involving Motor drive controls, Motor-Gearbox drive systems for Generators, Hydraulic power pack for driving Hydraulic Motor Driven Generators, including their Cooling systems, Loading, Control, Data Acquisition, Networking and Remote control/monitoring with associated Software and GUI development. Bidder should have designed the rig for High Power (> 100 kW) and High Speed (> 24000 rpm) Motor-Gearbox drive systems for driving Electrical generators and Hydraulic power pack for Hydraulic Motor Driven Generators (Power up to 5kVA or higher)

8.1.3. Bidder must have executed and commissioned similar rigs/Projects in Defence/Aerospace sectors which are certified by CEMILAC as Type-1 or Type-2 rig. The documentary evidence of executing and installing such systems like Order copy, Project completion letter, Invoice, User references, type certificate from CEMILAC etc., are mandatory and must be included in the Technical Bid. Non enclosure of proof shall be a cause of rejection without any notice.

8.1.4. Bidder should be OEM of the designed system. If the OEM is from outside India, he shall have an authorized technical/commercial understanding with an Indian firm with at least **5-10 years of experience** in managing similar systems/rigs. The OEM should have Support/Service Centre operational in India for at least past 05 years.

8.1.5 The bidder shall have experience in data acquisition with National Instruments (NI) based cards as the existing system in AGEST is based on NI and shall have developed control and monitoring software for aircraft electrical and hydraulic rigs. Appropriate documents, such as Supply orders and corresponding completion certificates for aircraft electrical rigs from government agencies shall be provided by bidder.

8.2. Certifications.

8.2.1. Bidders shall also have familiarity with NABL Accreditation and CEMILAC/DGAQA certification processes. Appropriate proof shall be furnished by bidder for the same, such as, any approved document, certificate, etc.

8.2.2 Bidder shall produce the documentary evidence for certifying the similar projects as Type- 1 or Type-2 rig from CEMILAC/DGAQA.

8.2.3. The bidder shall have demonstrable Quality Systems and should have ISO 9001- 2015/AS9100 or their equivalent Certifications.

8.2.4. Bidder shall have experience and must be well acquainted with various certifications and standards including JSS 55555, CMM, ISO & MIL-STD-704D, ISO, etc.

8.3. Commercial/Financial Criteria:

8.3.1. The bidder shall have adequate business volume of greater than INR **03 Crores** cumulative of preceding 3 years to provide credibility in his commitments and capabilities in terms of skilled manpower and infrastructure for long term support. For assessing the same Audited/certified Balance sheet, Profit/Loss account for the corresponding 3 years' period to be submitted.

8.3.2. Consideration shall be emphasized on the track record of the Bidders and also the reputation of the bidder, his influence and market share of the corresponding units in the market. The bidder shall be selected such that there is scope to support the project for a period of 25 to 30 years.

8.3.3. Exemption of turnover and experience will not be applicable for MSME/Start-ups firms for this procurement.

8.4. Compliance Matrix.

The proposal should contain the compliance report in respect to this document. Point wise compliance statement to be provided clearly stating the specific values of the offered product. Following format is to be submitted along with the technical proposal.

Sl No.	CABS Specification	Vendor Specification	Remarks

Important Note:

- Compliance for the qualification criteria must be provided, with supporting documents and evidences in the technical bid without which the proposal may be rejected. If required, a Technical Evaluation Committee from CABS may visit the facilities of the bidder for fact finding and further consideration of the bid.

- **Bidder should submit proof of all the above along with the Technical Bid, in the absence of which his bid will not be considered further.**

- **Based on the details provided by the bidder against 8.1,8.2,8.3 and 8.4 above, CABS will assess the capability of the bidder to take up this work. The decision of Technical Evaluation committee in this regard will be final and binding.**

8.5. Industry partner should submit IT Returns of last 3 years.

Desirable: - The Industry Partners required to register with SAMAR.GOV.IN and SAMAR rating may be considered.

Advisory: ATMA NIRBHAR BHARAT

Please register as Defence Manufacturer with "INDIA DEFENCE MART" at **URL: https://www.idm.gov.in**

Note: Bidder are required to submit all necessary documents in support of the above-mentioned criteria. Failure to submit documents will result in rejection of the bid.

9. **Product Support after Warranty: N/A**

10. **Clauses regarding 'Restriction on Public Procurement from Certain countries'** : Provisions of Government of India (Min of Finane), Dept of Exp. OM F.No. 6/18/2019-PPD dated 23rd Jul 2020 (as amended time to time) regarding "Restriction on Public Procurement from Certain countries" will be applicable to this tender. Certain important aspects of the letter have been included in this RFP for information and ready reference of the bidders.

11. **Clauses on Restriction on Public Procurement from Certain countries**

11.1 Any bidder from a country which shares a land border with India will be eligible to bid in this tender only if the bidder is registered with the Competent Authority.

11.2 "Bidder" (including the terms 'tenderer' 'consultant' or 'service provider' in certain contexts) means any person or firm or company, including any member of a consortium or joint venture (that is an association of several persons, or firm or companies), every artificial juridical person not falling in any of the descriptions of bidders stated hereinbefore, including any agency branch or office controlled by such person, participating in a procurement process.

11.3 "Bidder from a country which shares a land border with India" for the purpose of this Order means: -

a. An entity incorporated, established or registered in such a country; or
b. A subsidiary of an entity incorporated, established or registered in such a country; or
c. An entity substantially controlled through entities incorporated, established or registered in such a country; or
d. An entity whose beneficial owner is situated in such a country; or
e. An Indian (or other) agent of such an entity; or
f. A natural person who is a citizen of such a country; or
g. A consortium or joint venture where any member of the consortium or joint venture falls under any of the above.

11.4 The beneficial owner for the purpose of (iii) above will be as under:

11.4.1. In case of a company or Limited Liability Partnership, the beneficial owner is natural person(s), who, whether acting alone or together, or through one or more juridical person, has a controlling ownership interest or who exercises control through other means.

Explanation-

a. "Controlling ownership interest" means ownership of or entitlement to more than twenty-five per cent. Of shares or capital or profits of the company;

b. "Control" shall include the right to appoint majority of the directors or to control the management or policy decisions including by virtue of their shareholding or management rights or shareholders agreements or voting agreements;

11.4.2 In case of a partnership firm, the beneficial owner is the natural person(s) who, whether acting alone or together, or through one or more juridical person, has ownership of entitlement to more than fifteen percent of capital or profits of the partnership;

11.4.3 In case of an unincorporated association or body of individuals, the beneficial owner is the natural persons(s), who, whether acting alone or together, or through one or more juridical person, has ownership of or entitlement to more than fifteen percent of the property or capital or profits of such association or body of individuals,

11.4.4 Where no natural person is identified under (1) or (2) or (3) above, the beneficial owner is the relevant natural person who holds the position of senior managing official;

11.4.5. In case of a trust, the identification of beneficial owner(s) shall include identification of the author of the trust, the trustee, the beneficiaries with fifteen percent or more interest in the trust and any other natural person exercising ultimate effective control over the trust through a chain of control or ownership.

11.5 An Agent is a person employed to do any act for another, or to represent another in dealings with third person.

11.6 **[Applicable for Works contracts and Turnkey contracts only]** The successful bidder shall not be allowed to sub-contract works to any contractor from a county which shares a land border with India unless such contractor is registered with the Competent Authority.

11.7 **Certificate to be furnished by the bidders**

"*I have read the clause regarding restrictions on procurement from a bidder of a country which shares a land border with India; I certify that this bidder is not from such a country or, if from such a country, has been registered with the Competent Authority. I hereby certify that this bidder fulfills all requirements in this regard and is eligible to be considered.* ***[Where applicable, evidence of valid registration by the Competent Authority shall be attached.]***"

12. **Free Issue of Material (FIM): N/A**

13. **Inspection Instructions:**

i) Pre Delivery Inspection

ii) Post Delivery inspection on receipt of store.

iii) **Inspection Authority**: The Inspection will be carried out by a representative of the **CABS** duly nominated by the **Director**.

iv) **Inspection Agency: DGAQA.**

SPECIAL TERMS AND CONDITION

Essential Details of Items/Services Required

1. **Schedule of Requirements:** List of items / services required are as follows –

Name/Description of Item(s)/Service(s)	Unit of Measure	Qty required
Design, Development, Installation, Commissioning and Certification of Electrical Power and Generation System(EPGS) Test Rig	Lot	01

2. **Technical Details:**

2.1. SOW

Statement of Work
For Design, Development, Installation, Commissioning and Certification of **Electrical Power & Generation System (EPGS) Test Rig**

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Abbreviations

AC GPPU	AC Ground Power Protection Unit
AC GPU	AC Ground Power Unit
ACBTC	AC Bus Tie Contactor
ACPMU	AC Power Management Unit
APC	Aircraft Power Control
APMU	Aircraft Power Monitoring Unit
APU	Auxiliary Power Unit
BC	Battery Contactor
BMS	Battery Management System
BOM	Bill of Materials
CABS	Centre for Air Borne Systems
CDR	Critical Design Review
CEMILAC	Centre for Military Aircraft Certification
CMM	Component Maintenance Manual
CMP	Configuaration Management Plan
CS	Current Sensor
CSD	Constant Speed Drive
CT	Current Transformer
DAQCS	Data Accquisition and Control System
DC GPPU	DC Ground Power Protection Unit
DC GPU	DC Ground Power Unit
DCBTC	DC Bus Tie Contactor
DCPMU	DC Power Management Unit
DGAQA	Director General for Aircraft Quality Assurance
DRDO	Defence Research and Development Establishment
ECN	Engineering Change Note

1. Introduction:

1.1. In any aircraft development program, rigs are essential to develop to test and qualify the various systems. The electrical rig is one of them and is used to develop and test the electrical LRUs, integrate them in the rig to test, qualify and evaluate the complete electrical system before it is implemented in the actual aircraft. The electrical power system is flight critical, therefore the certification process for a new aircraft requires a complete laboratory analysis of the assembled power system operating in a rig.

1.2. **Electrical Power Generation Systems (EPGS) Test Rig** defined in this document comes under Test Rigs and Tools, Testers & Ground Equipment (TTGE) which will be used for aircraft electrical system testing and evaluation activities during its development. Later the rig can be one of the TTGE of the aircraft and plays a major role in aiding the ground crew during preparation of aircraft for service, upkeep and maintenance in an efficient way. The requirement of the TTGEs can be for any development activities or for entire life cycle of the aircraft after induction into service.

1.3. EPGS test rig shall be designed to test the hybrid (Hydraulic Motor Driven generators and both AC and DC generation) Electrical Power Generation and Distribution System of Aircraft. The rig shall be capable of replicating the redundancy aspects of the power generation and distribution and power flow amongst different buses.

1.4. The basic requirement of the EPGS test rig is that it should be capable of integrating and testing the electrical system of an aircraft which will have on generating side- one AC Generator, one DC Generator and 3 no’s of Hydraulic Motor Driven Generators (HMDGs).To develop the test Rig for this hybrid configuration, the rig shall comprise of two independent Electric Motor-Gear Drive stands to drive AC and DC Generators (available in CABS) and one Hydraulic power pack system to drive the 3 no’s of HMDGs (To be developed). At present, CABS is having the AGEST (Aircraft Generator and Electrical System Test) Facility configured for testing of Twin Engines/Generators plus APU generator up to 120kVA rating. Hence, two nos of Motor-Gear Drive stands in AGEST Facility available at CABS can be used to test one AC Generator and one DC Generator. Therefore, Industry partner has to develop the Hydraulic Power Pack to test the three no’s of HMDGs and also integration of Hydraulic Power Pack System with the Motor-Gear System for testing the power redundancy of electrical architecture of the aircraft having hybrid power generation.

1.5. AC Generator have a closed loop cooling system. To cool the generator oil, an external cooling system needs to be developed, comprising an aircraft grade heat exchanger (FCOC- Fuel Cooled Oil Cooler) and aircraft fuel as the heat dissipater , to reflect the actual aircraft scenario. The development of the Fuel circulation system for the AC generator is within the scope of the Industry Partner.

1.6. Once the rig is equipped and functional with these prime power sources (drive stands and Hydraulic power pack), all other aspects of a typical aircraft electrical system like power conversion, distribution, control, protection and monitoring will be integrated and tested in the rig. Industry Partner shall provide the cable harness as per aircraft standard for aircraft electrical system LRUs integration and testing and validation.

1.7. The rig shall have the capability of testing the Electrical System LRUs of aircraft in standalone mode and in integrated mode which includes control, monitoring and analyzing various design parameters of the electrical power generating system.

1.8. The rig shall be capable of testing all the Electrical LRUs together as a single system replicating the actual aircraft electrical system architecture.

1.9. EPGS test rig shall be capable of identification, recording and display of the fault codes that are set by the individual LRUs of aircraft.

1.10. All the rig parameters such as flow, pressure, speed, temperature, current, voltage, contactor status, fault indications, initiated BIT status etc. shall be brought and displayed in the central console which will be placed in control room.

1.11. The rig shall also have the provision for Measurement Terminal/Patch Panels for manual measurement/monitoring of test and rig parameters.

1.12. This document details the specifications of the major constituents of EPGS to be established in CABS, DRDO, Bangalore. The Industry Partner will be responsible for the Design, Integration, Testing, Commissioning and Certification of the rig. The EPGS will be commissioned in a controlled environment and sheltered within a permanent building

2. **<u>Objectives of the EPGS Test Rig:</u>**

The Electrical Power Generation Systems (EPGS) Test Rig will be established at CABS, DRDO, Bengaluru in order to realize the following objectives after it is commissioned and certified,

(i)Testing and Verification of the functionality including Endurance Testing of Electrical System LRUs of Aircraft such as,

a. Aircraft EGB and AMAGB mounted Alternators/Generators (AC or DC) along with their Generator Control Units (GCUs)
b. Hydraulic Motor Driven Generators along with their Generator Control Units (GCUs/VRUs or CRUs)
c. Transformer Rectifier Unit (TRU) or ATRU or any other Power Conversion units
d. AC and DC Power Management/Distribution
e. AC and DC Ground Power Protection Units
f. Aircraft power control/display panel
g. Battery Management Unit and Batteries

(ii) Integrate, Test and Evaluate the complete Aircraft Electrical System before it is installed in the actual aircraft

a. Prove the complete Electrical System Design
b. Test under various failure conditions
c. Controller functions/BUS transfers
d. External GPU connections
e. BIT functions/Interface with other Aircraft systems
f. Pre-Installation Checks: Verifying the serviceability of LRUs.
g. Periodic exercising and first level defect investigation of LRUs.
h. Built in Self –test for ensuring serviceability of the rig.
i. Defect investigation

3. **<u>EPGS Test Rig Architecture</u>**

a. The EPGS Test Rig architecture is designed to be modular, re-configurable and of sufficient ratings so that it can be adapted to meet the requirements of testing electrical systems of different architectures and aircraft.

b. The architecture of EPGS test rig consists of two Motor-Gear Drive stand, one each for 50kVA(or 40kVA) IDG and 5kW DC Generator, comprising of AC Drives, AC Motors, Step up Gear-boxes, Lubrication Systems for Gear-boxes, Generator cooling system, LRU Test Racks, Terminal Boxes (Patch Panels), Hydraulic Power Pack , 5 kVA HMDG Test Stand, 400W (or 350W) HMDG Test Stands and Data Acquisition and Control System. EPGS architecture is shown as a block diagram in Figure-1 and Figure -2 shows the detail EPGS architecture along with AGEST facility present in CABS.

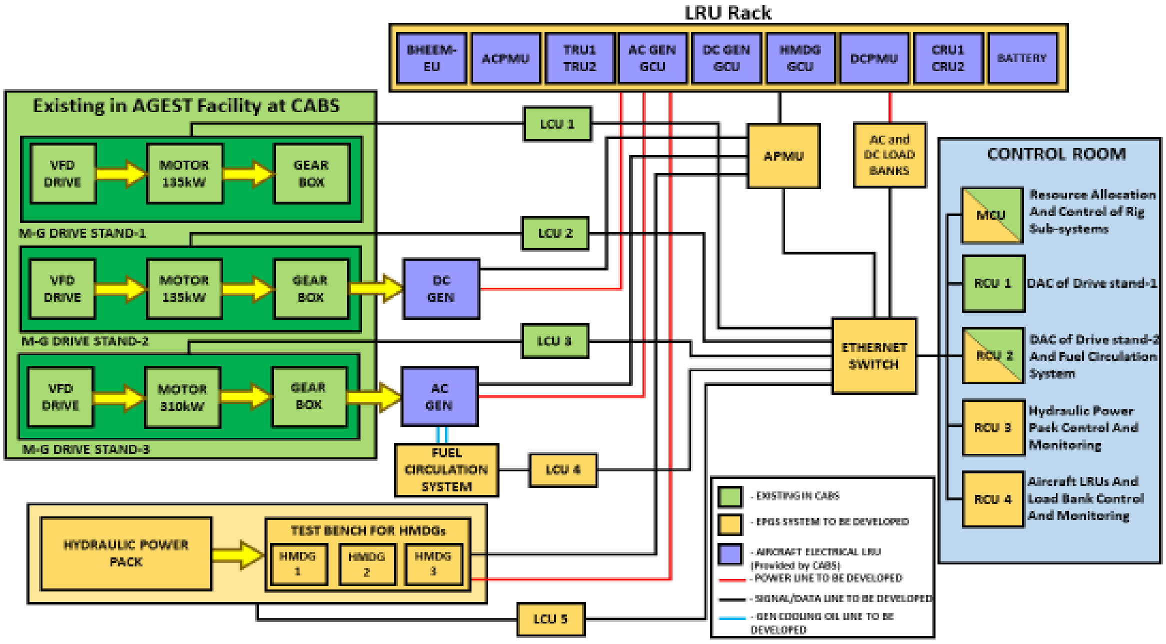


Figure 1:EPGS Test Rig Architecture

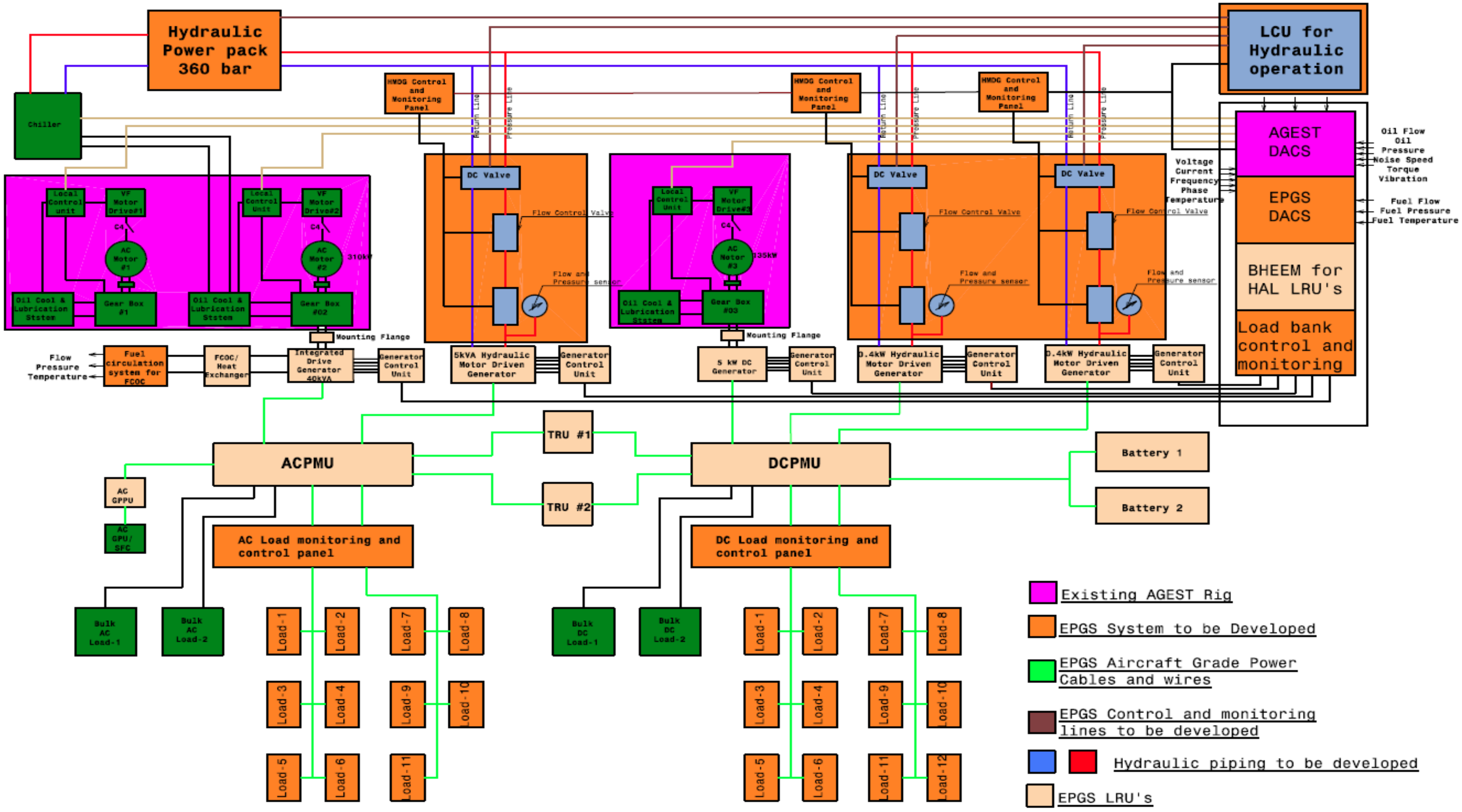


Figure 2 : EPGS Test Rig Architecture along with AGEST system

c. In EPGS Architecture, to test the AC and DC generators two nos of Motor-Drive stands are required. AGEST facility in CABS is already having 3nos of Motor-Drive stands to test the Generators upto 120kVA. Hence, out of these drive stands , any two drive stands can be used for establishing the EPGS Rig. Integration of these drive stand with EPGS system along with control and monitoring will be in the scope of the Industry Partner.

d. The following are the Rig sub systems which are in the scope of the Industry Partner, who has to integrate them and test it as a complete rig.
   i. Hydraulic Power Pack with mounting arrangements and cooling arrangements with required Instrumentation for three HMDGs
   ii. Test stands for 3 nos of HMDGs
   iii. Mounting arrangements for Generators (Both AC & DC Generators)
   iv. Fuel Circulation System with Fuel Cooled Oil Cooler (FCOC) for AC Generator cooling and lubrication
   v. Line Contactors electrical panels for interfacing Generator output to AC Loads and DC Loads
   vi. AC load Banks
   vii. DC load Banks
   viii. Cable looms (along with MIL grade connectors) as per Aircraft standard for complete aircraft electrical systems from generation, conversion, distribution, control, protection and monitoring (as per drawings given by CABS)
   ix. Electrical Panels and Racks for LRUs
   x. Data Acquisition and Control System for the EPGS Test Rig and aircraft electrical LRUs
   xi. Testing and Report generation for all Aircraft Alternators/Generators with their Generator Control Units and electrical LRUs (**Aircraft LRUs -Provided by CABS**)

e. The rig is to be designed for integrated testing of complete electrical system of an Aircraft. LRUs or the EUTs of aircraft electrical system are listed here which will be tested together as a single system in the rig to validate the complete electrical system of an aircraft. The LRUs will be provided by CABS. Data acquisition and control system of electrical LRUs from central console and cable looms along with connectors are in the scope of Industry Partner. The list of LRUs to be tested in the EPGS test rig are:

Table 1: List of Aircraft LRUs

Sl No	Item Description
i.	Integrated Drive Generators (IDGs) along with its Generator Control Units (GCUs)
ii.	Hydraulic Motor Driven Generators (HMDGs)
iii.	DC Generators
iv.	Transformer Rectifier Units or any other Power Conversion Units
v.	AC Power Management Unit
vi.	DC Power Management Unit
vii.	AC/DC Secondary Distribution units
viii.	AC/DC Ground Power Protection units
ix.	AC Ground Power Protection Unit (GPPU)
x.	Batteries and Battery Management units

4. EPGS Physical Layout and Civil Infrastructure Requirements

a) The Aircraft Generator and Electrical System Test (AGEST) Facility had been established at CABS, Bengaluru in the existing Annexe of Flight Test Hangar which is a permanent, RCC sheltered building. The facility is developed for testing the complete Aircraft Electrical Architecture with Twin engines/Generators plus APU generator configuration. The layout diagram of the existing AGEST Facility with dimensions is given in Figure 3 in shaded colour.

b) AGEST facility consists of rig sub systems – 3 nos of Motor-Gear Drive stands, Gear Lubrication units, Local Control Units, AC and DC ground power units, AC and DC Load banks etc. Electrical power with incoming panel of 1000kVA capacity is available for the rig. The entire AGEST rig is controlled and monitored by the Remote Control Units (RCUs) and Master Controlled Units (MCUs) placed in the control room.

c) **Electrical Power Generation Systems(EPGS) Test Rig** will be established in the AGEST facility with additional augmentation to meet the hybrid configuration of the Aircraft Electrical Systems. Aircraft Hybrid electrical system includes, two generators (AC and DC generators) and 3 nos of Hydraulic Motor Driven Generators (HMDGs). For testing, 2 nos of generators (both AC and DC generator), Motor-Gear Drive stands with their respective Gear Lubrication units are already available in the AGEST facility. Hence, it can be used to mount and test the AC and DC generators. Mounting arrangement and testing of these generators are in the scope of Industry Partner. For cooling and lubrication of AC Generator, Fuel Circulation System has to be developed by the Industry Partner. The details about it is given in the subsequent section.

d) The test setup for testing the 3 nos of HMDGs are in the scope of Industry Partner. Industry Partner has to develop the Hydraulic power pack to drive the 3 Nos of HMDGs and a rigid metallic bench to mount these HMDGs. The Figure-3 below shows the existing AGEST facility in shaded color and augmented sub-system of EPGS test rig in bold color.

e) The EPGS Test rig will be controlled and monitored by the RCUs and MCU placed in the control room.

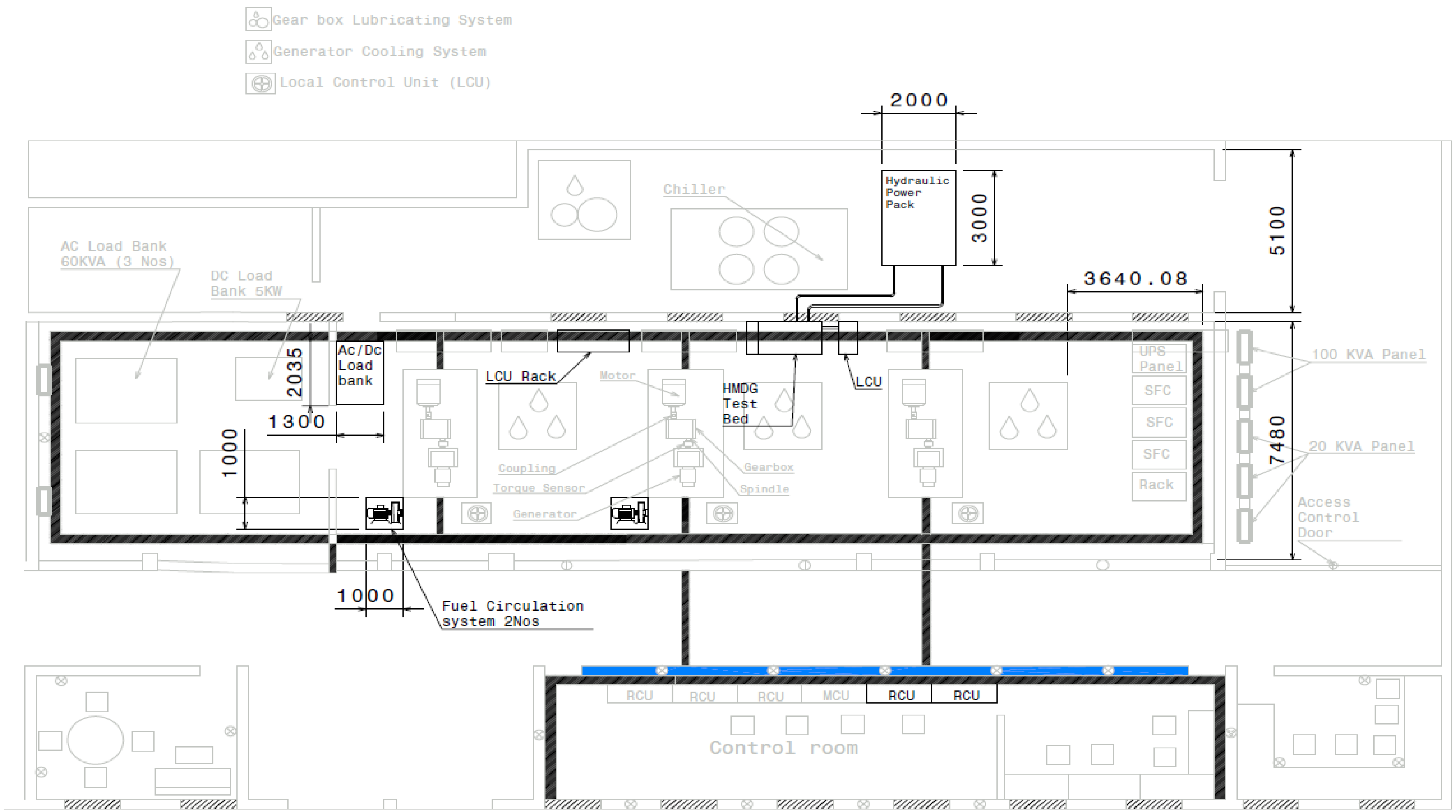


Figure 3 : Layout diagram of Existing AGEST Facility and additional constituents for EPGS Test Rig

5. Constituents of EPGS Test Rig and their Specifications

a. This Chapter gives the brief description and requirements of each sub system that constitute the Rig. The specifications of each of the Rig sub system are given in respective sub-sections as indicated in Table-2 below.

b. The detailed design, fabrication, integration, testing, commissioning and certification of the rig will be the total responsibility of the chosen Industry Partner. The Industry Partner has to deliver and integrate the items mentioned in Table 2 with the existing AGEST Facility and tentative location of the sub-system as per Figure 3 and prove it by testing as per the approved ATP for the Rig. The Rig will have the subsystems as indicated in Figure 1 and Table 2, which will be independent of the aircraft electrical system given in Table-1.

Table 2: Details of Major Constituents to be delivered , Installed and Integrated as Test Rig by the Industry Partner

Sl No.	Rig Equipment	Quantity/Resources
1)	Design and Development of Hydraulic Power Pack with Test Bench	1 Nos
2)	Design and Development of Fuel circulation system with FCOC including Reservoir for IDG oil Heat Exchanger	2 Nos
3)	Design and Development of AC and DC Electrical Distribution Load Bank as per EPGS requirement	1 Set
4)	Data Acquisition and Control System (DAQCS) – Control, Instrumentation, DAQ, analysis and Display/GUI System	1Set
5)	Aircraft grade Cable Looming along with MIL grade connectors	1Set
6)	Mounting arrangements ,Power Panel and Racks for LRUs	1Set
7)	Tool Kit	1Set
8)	Manpower (Software, Electrical and Mechanical) for Operation, Maintenance and Testing of Aircraft electrical system in the rig after certification of the rig for 1 years	4 Nos

5.1 <u>Hydraulic Pack</u>

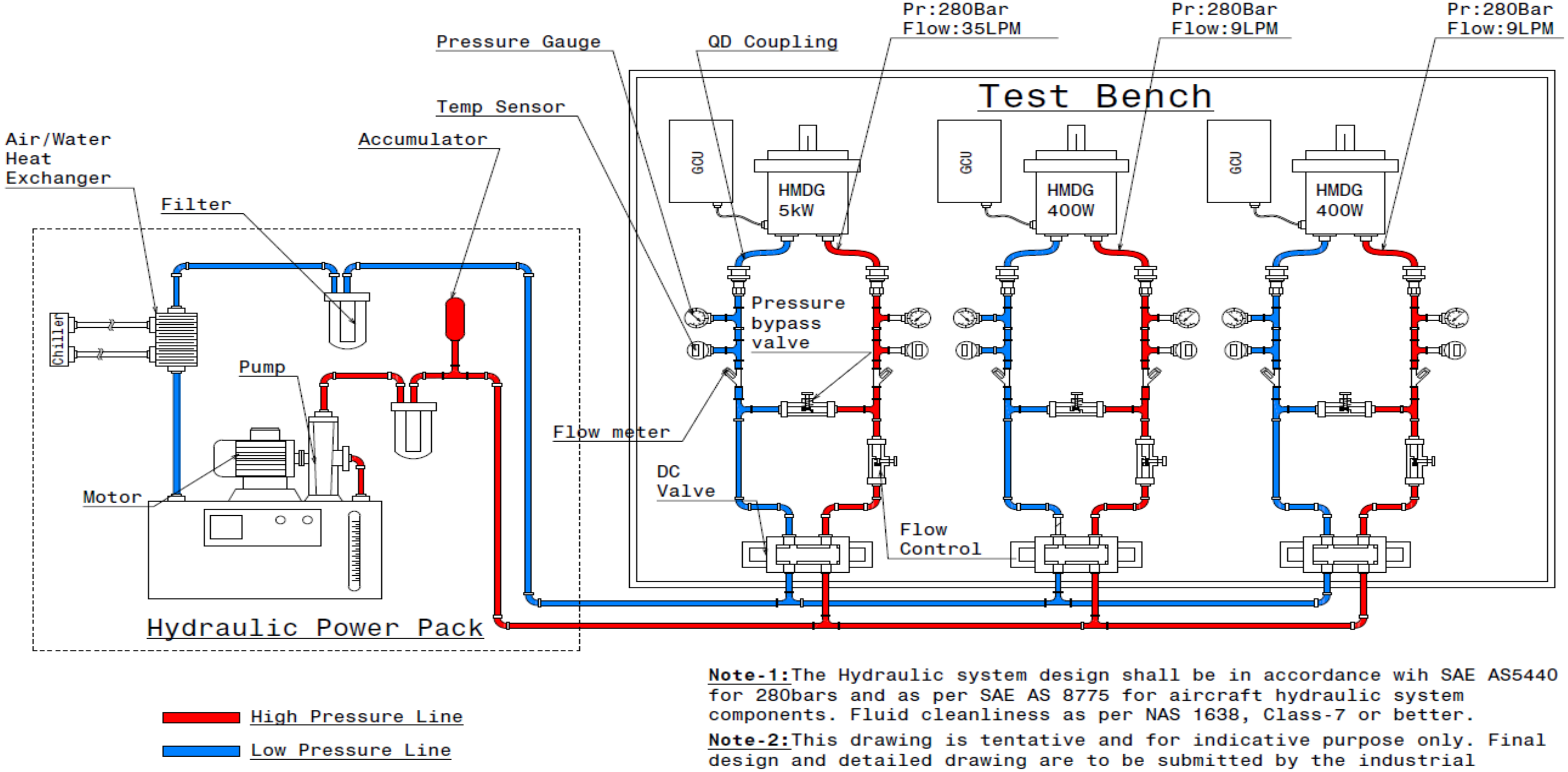
The Hydraulic power pack shall be capable of delivering high pressure aircraft hydraulic fluid as per the requirement specified below in Table 3. It shall consist of a hydraulic pump driven by an electrical motor with suitable hydraulic and electrical control. Test bench with mounting arrangement for 3 nos of HMDGs and all accessories required for testing the HMDGs from installation to control to monitoring are in the scope of Industry Partner. Three Nos of HMDGs will be provided by CABS.

The entire Hydraulic power pack shall be of portable type mounted on a rigid frame work with portable caster wheels, steering handle and removable type safety enclosure. The size of the hydraulic power pack shall be optimized for compactness, ensuring ease of maintenance, adherence to all applicable safety norms and uncompromised operational quality.

For 3 Nos of Hydraulic Driven Generators, provision shall be made to mount the units on their respective test benches with hydraulic pressure and return lines connected. The return line of the hydraulic system shall have the provision for indication of hydraulic fluid flow of respective 3 nos of HMDGs.

Electrical connections required for monitoring all parameters of Hydraulic Pack and 3 nos of HMDGs shall be brought to the Local Control Unit (LCU) and from here to the Central Console (Remote Control Unit). From Control room, provision to be provided for switching controls, monitor the inlet and outlet parameters of hydraulic pack like fluid flow, pressure, temperature and for monitoring the HMDGs output parameters such as Voltage, Frequency, Phase Sequence, Power Factor and Current.

The tentative schematic diagram of hydraulic test bed is shown below in Figure-4 and Indicative diagram with test bench for mounting HMDGs is shown in Figure-5.



Note-1: The Hydraulic system design shall be in accordance wih SAE AS5440 for 280bars and as per SAE AS 8775 for aircraft hydraulic system components. Fluid cleanliness as per NAS 1638, Class-7 or better.

Note-2: This drawing is tentative and for indicative purpose only. Final design and detailed drawing are to be submitted by the industrial partner prior to the commencement of fabrication

Note-3: Pressure and flow should be independently controllable for each individual HMDG line.

Figure 4: Schematic of Hydraulic Test Bed

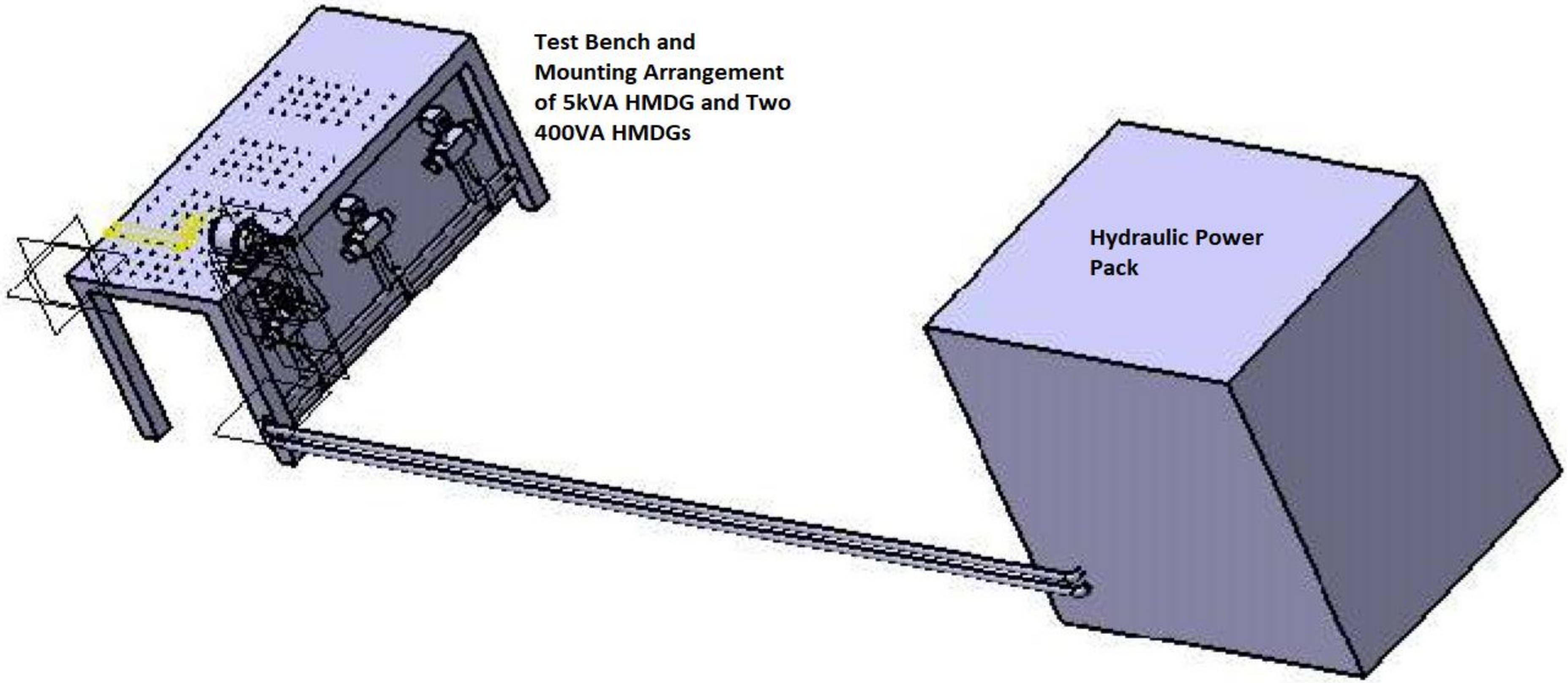


Figure 5 : Test bench for mounting HMDGs

Table 3 :Specification for Hydraulic Power Pack

Sl.No	Parameter / Description	Constraints / Values
1.	Hydraulic pump Type	Variable delivery pump with pressure control
2.	Hydraulic pump Continuous Working Pressure	360 bar
3.	Pump Duty Cycle	8 hours continuous
4.	Rated Flow(pump delivery)	At least 60 LPM
5.	Fluid Operating Temperature	Ambient to 65°C
6.	Control Options	Standard pressure compensator with screw / knob and nut for pump pressure adjustment for the full range of pressure. Pilot pressure to control the pump pressure remotely
7.	Hydraulic Fluid	Aircraft Hydraulic Fluid: MIL-PRF-5606 (Mineral Oil)
8.	Mounting and handling of hydraulic pack	Power pack shall be portable and it shall be mounted on a rigid frame work with caster wheels, steering handle and safety enclosure (removable type).
9.	Reservoir	The reservoir shall have all the standard provisions such as ail filling port with dust cap, air breather, suction strainer, temperature probe rated at 100°C with digital indicator for tank oil temperature indication, oil level indicator, drain cock, inspection cover and 3inch diameter cut-outs (quantity-2) on tank top with cover for inserting flexible hose of oil chilling unit when required.
10.	Reservoir Capacity	500Liters
11.	Reservoir Material	Stainless steel tank with corrosion resistance coating
12.	High Pressure Relief Valve	One high pressure relief valve in the circuit for full range (350 bar) of pressure adjustment shall be provided with knob / screw and lock nut. This shall be used for starting the pump under no load condition.
13.	Safety Valve	One high pressure safety valve settable from 350 to 370 bar shall be provided.
14.	Pressure gauge	One glycerin filled analogue pressure gauge for the specified rated pressure range with 10bar resolution along with isolation valve shall be provided in the pressure line.
15.	Filters	a. One 5-micron pressure line filter with clogging indicator of Non-Bypass type. b. One 10 micron return line filter with clogging indicator c. One 125 micron strainer to prevent large particles from entering the pump.
16.	Accumulator	About 10 –liters capacity accumulator rated at a maximum working pressure of 600 bar in the out let line shall be provided in the pressure line of suppressing pressure fluctuations. The accumulator shall be mounted rigidly on a manifold block with safety block and shut off valve.
17.	Flow Control Valve	One flow control needle valve in the delivery line for adjusting the rated pump flow (50LPM) shall be provided and it shall withstand the rated pressure (350bar).
18.	Heat Exchanger	Suitable capacity air/water-cooled heat exchanger to be provided in the return line for cooling return oil, pump leakage oil and relief valve return oil. This is required for continuous running of the power pack.
19.	Check Valve	Check valve (NRV) in the pump delivery line to avoid back pressure shall be provided.
20.	Electric Drive	Hydraulic pump motor shall be driven by variable frequency drive (VFD). A motor drive control panel to be integrated with power pack and shall have following features like, Mains input indication lights Motor ON/OFF control Switch Motor input parameter display meter, which shall read voltage, current, frequency, power, power factor, etc. For safety, an emergency stop switch will be installed. Faulty status will have indicated by dedicated light and alarms to notify operator. The panel shall support remote/stop functionality and can be equipped with communication capabilities.

		Selection of Motor rating- adequate to deliver the required flow and pressure. Motor protection- IP55 rated for dust and water protection with thermal overload protection.
21.	Electrical Cables	5-core, copper electrical cable to be routed from hydraulic pump motor control panel to mains input power panel as per motor current rating. Length of copper cable to be at least 50 meters.
22.	Test bench	The Output of Hydraulic Pack shall be connected to HMDG at the HMDG Test Bench. Two Small and one bigger rigid Metallic Bench shall be provided to mount the HMDGs for testing. Dimension is to be decided
23.	Design Instructions	1. Hydraulic control valves and pressure gauge should be mounted on a hydraulic control panel (panel mounted type) on the power pack. 2. Distance between the Hydraulic Power pack and the Equipment under testing is approximately 15 meters therefore, the piping should be planned accordingly to facilitate efficient hydraulic control. 3. Delivery and return lines should be terminated with BSP female bulkhead fittings. 4. All the components, plumbing as well as gasket and seals shall be made of materials compatible with the specified hydraulic fluid and should withstand the specified operating pressure. 5. All plumbing items shall be of stainless steel. As per as possible the industry partner shall try to avoid too many joints and fitting in order to minimize the oil leakage during the operational life of power pack. 6. All the major components shall be of reputed make and shall be specified clearly in the quotation with technical catalogues enclosed, particularly for pump, valves, filters, motor etc. Quotations not containing the above supporting documents along with technical compliance report are liable to be rejected. 7. Detailed hydraulic circuit diagram with part list and make to be provided. 8. The power pack shall be painted (Color: Grey/Light blue) with suitable oil resistant paint. 9. All the internal components and electrical wiring shall be identified properly as per the hydraulic circuit diagram. Hydraulic circuit diagram shall be displayed on the front panel of the power pack. 10. Calibration certificates shall be provided for all the instrument/gauges used in the design of the power pack. 11. Hydraulic power pack will include push-to-connect Quick Disconnect Coupling rated upto 400bar (wherever applicable) 12. Hydraulic Power Pack shall be mounted and installed on rigid platform at distant location from HMDG Test Bench at around 15 meters away. The required arrangements for building the platform for mounting the Power pack and routing the pipelines till the Test Bench shall be in the scope of the Industry Partner. 13. The required pressure of minimum 280±3.5 Bar and flow of 38 lpm shall be ensured at the input of 5kVA HMDG and pressure of minimum 280±3.5 Bar and flow of 10lpm at the input of other two 400VA HMDGs. One set of pressure gauge, flow meter and temperature sensor shall be connected in the input line and output line and near to each HMDGs as indicated in the diagram. Nominal operating pressure for 3 HMDGs is 280 bar. Inlet LPM rating: a) 5kVA-24 LPM (at 200% load -38lpm)

		b) 400W-9.5 LPM
24.	Accessories, Consumables and spares	1. First fill of hydraulic oil of about 500 litres and additional 500liters for refilling in containers. (MIL-PRF- 5605) 2. Two sets of filter cartridges (pressure and return lines). 3. Two sets of 5-meters-long High-pressure hydraulic hose for pressure and return lines to withstand rated pressure with BSP male fitting. 4. Two sets of all the replaceable components and sensors. 5. One set of spare sensors and measuring equipment
25.	Control System	PLC(Programmable Logic Control) based for automated testing and precise control of pressure, flow and timing. HMI(Human Machine Interface): User friendly interface for real time monitoring, control and data logging. Capability to monitor and control the Hydraulic power pack remotely and locally via Ethernet or Wireless communication.
26.	Safety Features	a. Electrical Safety: Protection from short circuit or overload. Proper grounding of all electrical component to avoid electrical shocks. Easily accessible Emergency Stop button to cut off power from system in case of emergency. b. Emergency Stop button to be integrated with the Emergency button available in control room of existing system at CABS. c. Overheat Protection: Thermal switch: Shuts down the electrical motor if the hydraulic fluid exceeds the set temperature d. Pressure Gauge: High Pressure Gauge: Calibrated to monitor working pressure uptp 400bar Flow rate indicator: Installed to display the flow rate for real-time monitoring during testing
27.	Documentation	1. Manuals for operating instructions, Maintenance guidelines and troubleshooting. 2. Service manual with list of replaceable components and service intervals. 3. Test certificate
28.	Noise Level	< 70 dB(A) at 1m distance

5.2 Fuel Circulation System with Fuel Cooled Oil Cooler (FCOC) for AC Generator cooling and lubrication

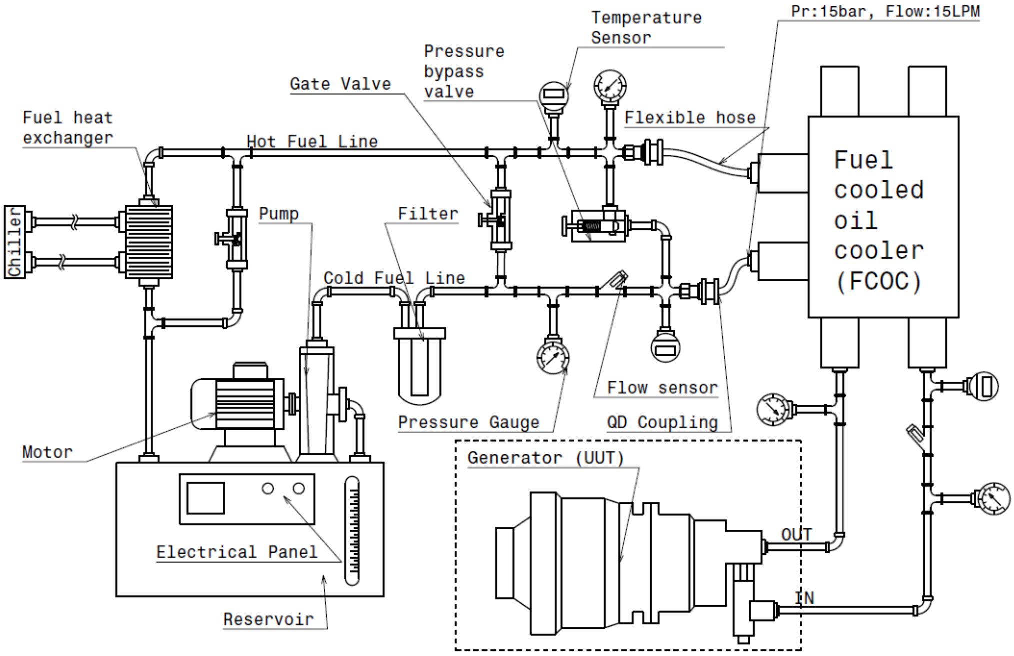
Two independent Fuel Circulation System with Fuel Cooled Oil Cooler (FCOC) is to be designed and developed by the Industry Partner to cool the aircraft generator. FCOC is basically a heat exchanger in the form of Radiator in which the aviation fuel passes through the cooler core to dissipate the heat from the oil. The FCOC serves the function of cooling Generator oil by utilizing aviation fuel as a cooling medium. So, an aircraft generator has closed loop external cooling system comprising of aircraft grade Fuel Cooled Oil Cooler (FCOC) and aviation fuel for extracting heat from FCOC. Hence, the same cooling scheme to be developed in the rig, to replicate the actual scenario operation.

In the EPGS Test rig, Fuel circulation system consist of FCOC, Fuel tank, electrical pump, heat exchanger to cool the aviation fuel and chiller. Both FCOC and fuel tank will have interconnecting pipelines with provision for monitoring fuel flow, fuel pressure and fuel temperature at inlet and outlet of FCOC by ensuring the external oil circuit limits the fuel to 750 ml (<1000 ml). Adequate rated pump shall be used to circulate the fuel between fuel tank and FCOC. The fuel circulation system must have the capability to deliver high flow of aviation fuel to FCOC, meeting the specified requirement. The system should include pump with appropriate electrical control and provision to monitor various parameters of the system like temperature, flow rate, pressure etc. The tentative schematic circuit of fuel circulation system is given below in Figure- 6. The specification of Fuel circulation system and Chiller are given in Table -4 and Table-5.

AC Generator requires external cooling system as mentioned in above para. **2 Nos** of aircraft grade FCOC (**Part No. 41013911, M/s Bharat Heavy Electrical Limited-Heavy Plates and Vessels Plant**) shall be purchased by the Industry Partner to design the compact and light weight cooling system for Generator and to meet the time line of the EPGS rig.AC Aircraft Generator will be provided by CABS. The Specification of FCOC is given in Table-6.

The DC Generator is air cooled generator with integral fan. Hence, no external cooling system for DC Generator.

Mounting of both AC and DC Generators on Motor-Gear Drive stands (available with CABS), integrating with electrical system and testing are in the scope of Industry Partner. All parameters required to monitor for Generators are to be displayed on Central console (LCU & RCU). The details of coupling and interface of the generators shall be provided by CABS during the PDR/CDR.



Note-2:This drawing is tentative and for indicative purpose only. Final design and detailed drawing are to be submitted by the industrial partner prior to the commencement of fabrication

Figure 6 : Fuel Circulation System Schematic

Table 4 : Specification of Fuel Circulation System

Sl.No.	Parameters / Description	Values / Constraints
	Fuel Pump	
1.	Type	Variable Delivery Pump
2.	Duty Cycle	8 hours continuous
3.	Rated flow (Pump Delivery)	20LPM
	FCOC Parameter	
4.	Maximum fuel inlet temperature	< 79 ºC
5.	Maximum continuous inlet operating temperature	120 ºC
6.	Heat dissipation required	23kW
7.	Maximum operating pressure	15 bar
8.	Pressure drop	0.4 bar
9.	Fluid type	Aviation fuel: DERD-2494, F-35, JET A1
10.	Mounting and handling	Fuel Circulation system shall be portable and it shall be mounted on a rigid frame work with caster wheels, steering handle and safety enclosure (removable type).
11.	Piping	Stainless steel piping will be used throughout the system. Seal and gasket will be selected based on temperature and chemical compatibility High quality fuel resistant quick connect coupling or fitting will be used to enable easy

12.	Reservoir	1. Capacity: The reservoir should be sized adequately to effectively dissipate heat from oil of temp 120°C running the system for 8 hours. 2. Tank Material: The reservoir should be constructed from durable material, preferably stainless steel, with an added fire-resistant coating. Additionally, thermal enclosure to be provided. 3. The reservoir shall have all the standard provisions such as filling port with dust cap, air breather, suction strainer, temperature probe rated at 150°C with digital indicator for tank temperature indication, level indicator, drain cock, inspection cover and 3inch diameter cut-outs (quantity-2) on tank top with cover for inserting flexible hose when required.
13.	Gate Valve	Needle type gate valve in the pump delivery line shall be provided.
14.	Filter	A 10-micron filter equipped with a clog indicator shall be installed in return line.
15.	Heat Exchanger	Type: Stainless steel, Plate-type liquid cooled heat exchanger with not less than $0.5m^2$ heat transfer area. Heat Dissipation: 23kW, and should be capable of continuously operation for at least 8 Hrs. Fluid Type: Aviation fuel at 15 LPM Coolant Type: water at 30 LPM (Provided by Chiller Unit)
16.	Sensor	Flow, Temperature (Inlet and outlet) and Pressure sensor with digital display and data-capturing capabilities, must be integrated in the lines.
17.	Electrical Drive and control panel	The fuel circulation should be driven by appropriate rating of electrical motor operating on 415V, 50Hz, 3 phase supply with flexible coupling in between. The motor drive panel shall have the features like over load relay, timers, emergency stop and input parameter display meter.
18.	Electrical cables	5-core, copper electrical cable to be routed from hydraulic pump motor control panel to mains input power panel as per motor current rating. Length of copper cable to be at least 20meters.
19.	Design Instruction	1. Valves and pressure gauge should be mounted on control panel. 2. Distance between the Fuel circulation system and the Heat Exchanger (FCOC) is approximately 5 meters therefore, the piping should be planned accordingly. 3. All the components, plumbing as well as gasket and seals shall be made of materials compatible with the specified fuel and should withstand the specified operating pressure. 4. All plumbing items shall be of stainless steel. As per as possible vendor shall try to avoid too many joints and fitting in order to minimize the leakage during the operation. 5. All the major components shall be of reputed make and shall be specified clearly in the quotation with technical catalogues enclosed, particularly for pump, valves, filters, motor etc. Quotations not containing the above supporting documents along with technical compliance report are liable to be rejected. 6. Detailed circuit diagram with part list and make to be provided. 7. All the internal components and electrical wiring shall be identified properly as per the circuit diagram. Circuit diagram shall be displayed on the front panel. 8. Calibration certificates shall be provided for all the instrument/gauges used in the design.
	Accessories, Consumables and spares	1. Two sets of filter cartridges and Gate Valve. 2. Two sets of hose for pressure and return lines to withstand rated pressure with fitting. 3. First fill aviation fuel approx 200 litres and additional 200 liters for refilling in containers. (DERD-2494, F-35, JET A1) 4. One set of Sensors
	Safety Features	Pressure relief valve will be installed to prevent any pressure build up with in the system. Thermal insulation will be applied to critical components

		An emergency shutoff or a safety interlocks system to be integrated allowing immediate cut-off of fuel flow in case of excessive temperature, pressure.

Table 5 : Specification for Chiller to cool the aviation fuel oil

Sl No	**Parameters / Description**	**Values / Constraints**
1.	Type	Portable type, Water cooled, low noise, compact chiller designed for indoor, equipped with scroll compressor for quite operation
2.	Construction	Corrosion resistant steel and insulated housing to reduce operational noise, quick-disconnect couplings and anti-vibration mounting
3.	Refrigerant	R-410A or Equivalent
4.	Input Voltage	3-phase,415V(L-L),50Hz
5.	Current	20-25A depending on cooling capacity (10TR)
6.	Power Factor	Less than or equal to 0.9 for efficiency
7.	Cooling Capacity	10TR (24 to 28 kW)to handle the required 23Kw load continuously with a safety margin
8.	Water Flow Rate	Designed to deliver a flow of 30LPM, with adjustable control to regulate fuel temperature
9.	Advanced Control and Monitoring Features	**Function:** An automatic controller adjusts the water flow rate in real-time to maintain the fuel temperature with in a specific range. **Temperature Range Control:** Maintains the fuel temperature within the required range set by the operator. Feedback Mechanism: Continuously receives temperature data from the fuel circulation system and adjust water flow to keep temperature stable. Digital Control Panel with Monitoring and Display Display: LCD/LED display on the chiller showing live data, including fuel temperature, coolant outlet temperature, water flow rate and system status. User Interface: User-friendly interface for setting temperature range, flow rate and other operational parameters. Alarm System: High/Low operational data, including temperature fluctuations, water flow adjustment, and system diagnostics for performance tracking and troubleshooting. Data Logging: Logs operational data, including temperature data, including temperature fluctuations, water flow adjustments. Manual Override: Allows for manual control of cooling parameters if needed. Interface: Built in communication port (Eg -Profinet) for interaction with central monitoring systems, allowing remote access to system data and configure system parameters. Chiller Connectors: Vendors should provide the output connector and Stainless steel braided flexible hose that seamlessly connects the chiller outlet to heat exchanger. The vendor shall include Quick-Disconnect coupling on both ends of the flexible hose (Chiller output end and matching inlet connector for Heat exchanger) for easy tool free connection to chiller output and heat exchanger.
10.	General Terms and Conditions	• Two years warranty to be provided from the date of acceptance at CABS • On site servicing/maintenance support should be available till completion of warranty period • Onsite training should be provided • One set of manual to be provided

		• Vendor should have the past experience in the supply/design & development of similar air conditioning units (Proof to be attached) • Number of firms who are using such models/products to be furnished • Proof of any contract for maintenance support is to be attached

Table 6 : Specification of Aircraft Grade FCOC

Sl No	Item Part Number	Part Description	Quantity
1.	Part No: **41013911 (Mfg.: BHEL-HPVP)**	Hydraulic Oil-IDG Oil-Fuel Cooler (HE-1)”	02 Nos
2.	**Hydraulic Oil-IDG Oil-Fuel Cooler (HE-1) requirements:** Integrated Drive Generator (IDG) is required heat exchanger for cooling of MIL Grade hydraulic oil of generator during operation		
3.	Pre-dispatched inspection report to be attached along with Certificate of Conformity (CoC).		
4.	Bharat Heavy Electrical Limited (BHEL), Heavy Plates and Vessels Plant, Visakhapatnam is the sole manufacturer of Hydraulic-IDG Oil –fuel cooler (HE-1), Part No. 41013911		

5.3 AC and DC Electrical Load Banks

AC and DC electrical load banks is being required for loading of different set of AC and DC power sources. EPGS system is having power sources and each power source is supplying power to variety of AC and DC loads. These electrical loads being operated in different loading steps of maximum load current. Hence, these electrical loads to be designed using different ratings of loading elements.

EPGS requires one set of AC Resistive-Inductive load bank and two set of DC Resistive load bank. Each type of load bank to be a standalone unit with caster wheel and hooks.

Both AC and DC load banks to be integrated with the existing (Data Acquisition and Control System) DAQC system in AGEST Facility at CABS.

5.3.1 AC Load Bank

The AC load bank will comprise of 11 buses of different loading capacity as given in Table-7 below. The AC Load power factor shall be varied from 0.6 lagging to 1 unity PF . AC load bank to be combination of resistive & inductive elements and it to be configured in such a manner, desired power factor and active power demand to be achieved. The inductor shall be designed for 400Hz±8Hz operation.

A single inductor effectively in series with the AC load buses shall be designed to achieve the variable PF. The inductor shall be designed for 400Hz±8Hz operation.

The load banks should be able to load individual current feeder lines for different AC buses and also in steps as specified in the specification, along with feature of acquiring the voltage and current parameters for each output loads. Each AC load bus shall be switched by independent AC Switch Contactor/Circuit Breakers compatible with 400 Hz operation. A pictorial representation that provides information of AC Load Bus branching is given below in Figure- 7 . The specification for AC Loads are given in Table-7.

AC LOAD	
BUS-1	**BUS-2**
BUS-3	**BUS-4**
BUS-5	**BUS-6**
BUS-7	**BUS-8**
BUS-9	**BUS-10**
BUS-11	

Figure 7- : Pictorial representation of AC Buses

The load bank shall have integral cooling without any external means. The switching of the various load modules shall be controlled in both mode i.e. remote and in manual mode. Load bank shall be designed for continuous operation of 8 hours.

The neutral point of individual 3 phase loads shall be connected together externally to form a common neutral.

EPGS electrical system requires resistive and inductive (R-L) load bank for loading at 3-phase, 115V, 400Hz AC power. R-L load bank shall have 03 numbers of independent channels for each phase loading and can be connected together to form a common neutral externally. The R-L load bank to be designed for 3-phase, 115V, 400Hz AC power and following loading steps as given in Table -7.

Table 7 :Requirements for Resistive and Inductive load bank of 3-phase, 115V, 400Hz AC power

Sl. No.	Input Voltage (Volt) and frequency (Hz)	Total Apparent Power rating (VA)	Load Bank Buses current rating (Amps)	Incremental load current steps	Lagging Power factor required at each incremental loading steps	Quantity required
1	3-phase, 115V, 400Hz, 6-Wires	12075	35	5,10,10,2,8	0.60, 0.70, 0.75, 0.80, 0.85, 0.90, 1.00	01
2	3-phase, 115V, 400Hz, 6-Wires	25875	75	2,8,20,20,25	0.60, 0.70, 0.75, 0.80, 0.85, 0.90, 1.00	01
3	3-phase, 115V, 400Hz, 6-Wires	12075	35	5,10,10,2,8	0.60, 0.70, 0.75, 0.80, 0.85, 0.90, 1.00	01
4	3-phase, 115V, 400Hz, 6-Wires	12075	35	5,10,10,2,8	0.60, 0.70, 0.75, 0.80, 0.85, 0.90, 1.00	01
5	3-phase, 115V, 400Hz, 6-Wires	12075	35	5,10,10,2,8	0.60, 0.70, 0.75, 0.80, 0.85, 0.90, 1.00	01
6	3-phase, 115V, 400Hz, 6-Wires	5175	15	5,5,5	0.60, 0.70, 0.75, 0.80, 0.85, 0.90, 1.00	01
7	3-phase, 115V, 400Hz, 6-Wires	2760	8	5,3	0.60, 0.70, 0.75, 0.80, 0.85, 0.90, 1.00	01
8	3-phase, 115V, 400Hz, 6-Wires	5175	15	5,5,5	0.60, 0.70, 0.75, 0.80, 0.85, 0.90, 1.00	01

9	3-phase, 115V, 400Hz, 6-Wires	1725	5	5	0.60, 0.70, 0.75, 0.80, 0.85, 0.90, 1.00	01
10	3-phase, 115V, 400Hz, 6-Wires	1725	5	5	0.60, 0.70, 0.75, 0.80, 0.85, 0.90, 1.00	01
11	3-phase, 115V, 400Hz, 6-Wires	1725	5	5	0.60, 0.70, 0.75, 0.80, 0.85, 0.90, 1.00	01

Table 8: Technical specification of Resistive and Inductive Load bank of 3-phase, 115V, 400Hz

Sl.No.	Description/Parameters	Constraints/Value
1.	Load Input Voltage	3-Phase, Nominal: 115V (Line-Neutral); Range: 108 to 118V
2.	Load Input Frequency	Nominal: 400Hz; Range: 392 to 408Hz
3.	Load Input Phase Connection	3 nos. of independent phases, can be connected in STAR or DELTA configuration.
4.	Load Power Factor (pf)	Lagging power factor steps 0.60, 0.70, 0.75, 0.80, 0.85, 0.90, 1.00; at each loading current.
5.	Loading elements	Resistive-Inductive (R-L) elements; Load elements to be designed for 400Hz operations.
6.	Operation mode	Balanced and unbalanced 3-phase operation, Single phase loss operation (two phase operation), Independent phase loading operation. Only Resistive loading operation without inductors.
7.	Manual operation switches	Load bank front panel shall have input ON/OFF switch, Emergency stop switch, Remote/Local switch, Step load switches, Cooling fan/blower switch, Resistive and inductive element switches, etc.
8.	Loading operation	Loading operation to be carried out on the basis of kVA power demand and applicable power factor (pf). Apparent power (kVA) depends on the applied voltage and load current only. Apparent power to be constant at each power factor step. Active and reactive power demand will be changing as per defined power factor.
9.	Parallel operation	Load bank shall have provision to paralleling multiple loads, when powered by single source.
10.	Input terminals	Terminal block/Connector to be provided for the input cable connections and it's rating to be more than 2 times of rated resistive element current capacity.
11.	Load input contactor	Load input contactor to be provided for ON/OFF connected loads. The input contactor rating shall be more than 2 times of rated resistive element current capacity.
12.	Cooling arrangement	Heavy duty fans to be provided for heat removal. Fan shall have independent ON/OFF control and powered by external 50Hz AC power. Unit shall have perforation for better air-circulation and perforation to be provided on top, bottom and side plates.
13.	Ingress protection	Unit shall have ingress protection of IP23 or better.
14.	Operating area temperature	The load bank shall meet its performance requirements in an operating environment of 0 degC to 50 degC.
15.	Operation control	The unit shall have relays, contactors and programmable logic controller (PLC) to achieve desired loading under manual mode and remote mode of operations.
16.	Measurement display meter	Unit shall have front panel display to monitor the load parameters of each line at 400Hz like, voltage, current, frequency, active power, reactive power, apparent power, power factor, phase to phase displacement, etc.

17.	Mode of operation	Unit shall have provisions to operate in manual mode via front panel switches and remote mode via NI LabView based GUI. Programmable Logic Controller (PLC) to be used for operation, control and remote interfaces.
18.	Remote operation and control	Remote operation shall be possible by Ethernet and RS232/RS485 interface. Remote control shall have complete functionality of load bank with suitable GUI. Remote control GUI to be developed in NI LabView software by supplier.
19.	Protection	Fan failure protection, Over-temperature protection, Over-current protections, short-circuit protection to be integrated in the unit. Fuses to be used for protection of load elements. Under any type of fault, unit shall have to withdraw the loading operations.
20.	Marking	All front panel display, CB, terminals, switches etc. are marked with their nomenclature/notations.
21.	Load bank modularity and handling	Caster wheels, hooks and handles to be provided in load bank to move the complete assembly.
22.	Ratings of wires, elements and components	The current rating of wires, resistive element and inductive elements to be more than 2 times of full load current. Contactors and relays to be rated for at-least1.8 times of full load current. Protection fuses and MCBs to be rated according to I^2t curve. These components to be best quality available in the market
23.	Accessories	Resistive-Inductive load bank to be connected with 3-phase, 115V, 400Hz AC power source and it requires 50meters of 5-core, 16-sq.mm, copper cable; 5-channel, 125Amps distribution box integrated with 4-pole 125 MCCB and 12-way terminal block. For data transfer interface, an 8-channel ethernet switch and RS232/RS422/RS485 to ethernet converter with I/O cables and power adaptor to be supplied. All necessary accessories, which are essential for operation of Load bank to be supplied.
24.	Passive loading elements	Resistors: Resistive element to be constructed using resistance wire (Nichrome 80/20) embedded in compacted magnesium oxide powder with the outer sheath constructed from stainless steel or Inconel to provide corrosion resistance. Resistive load element to be totally weatherproof with an electrically grounded outer sheath to prevent electrical short circuits by an external foreign object and to protect personnel against accidental electrical shock. Elements to be individually serviceable in the field without disturbance to other elements. The change in resistance due to temperature to be minimized by maintaining conservative watt densities. Open wire type elements where the live conductors are exposed and which can be short circuited to each other or to ground by foreign objects or by the breakage of an element or an element support shall not be permitted. Element chambers to be surrounded by aluminized steel heat shields reducing heat gains to the outer surface. Inductors: All inductors to be iron cored and vacuum impregnated with insulation varnish and coated externally with corrosion resistant paint. The inductors shall have a minimum thermal classification of – Class H (180degC). Fuse failure of one phase leaving the inductor energized by two phases shall not cause any damage to the load inductor.

		Inductors shall be capable of continuous operation, when absorbing power with the voltage harmonics generated by a typical AC generator complying with current standard. Larger inductors shall be fitted with a discharge resistor to prolong the life of the switchgears.
25.	Enclosure construction and finish	IP 43 or Higher Load bank shall be assembled in metal casing and live parts to be separated by insulators. Enclosure shall have earth/chassis ground point. Enclosure body shall be protected against the line to chassis fault. Complete body of load bank shall be shock hazard proof. The container structure shall be manufactured from mild steel/carbon steel with thickness of 1.5mm. Hinged door and bolted panels to be used for access of control, switchgear and load element compartment. The frame contains fork lift pockets for maneuverability and transportation. A durable powder coated finish is used on all panels. Standard color is grey RAL70XX.
26.	Safety features	An emergency stop/disconnect switch to be provided for full isolation of load, fan and control supply. Fan/blower shall be protected with fuse and thermal overload. Thermal detectors to be fitted to protect against overheating of passive elements. Each element group associated contactors are protected by an HRC fuses. Internal access to be restricted by bolted panels. Polycarbonate screens to be used behind the doors to prevent accidental contact with live parts.
27.	Operation & Maintenance Manual	Two Set

5.3.2 DC Load Bank

Aircraft electrical system requires multiple resistive load bank for operations at 28V DC power. Resistive load bank to be designed for incremental loading steps for 28V DC power.

The DC load banks should be able to load individual current feeder lines for different DC buses and also in steps as specified in the Table- 9, along with feature of acquiring the voltage and current parameters for each output loads.

Each load bank shall have independent front panel control & monitoring and remote control interface shall be configured in unit. Load bank shall have built in protection for over-current, short-circuit, over-temperature, etc. Each load bank shall have additional load line drop compensation step to over-come lengthy line voltage drop.

The load bank shall have integral cooling without any external means. The switching of the various load modules shall be controlled in both mode i.e. remote and in manual mode.

DC Bus is rated at nominal voltage of 28V and would range between 16V – 30V. A pictorial representation that provides information of DC Load Bus branching is given below in Figure- 8 . The specification for DC Loads are given in Table- 9.

DC LOAD - 1		**DC LOAD - 2**	
BUS -1	**BUS -2**	**BUS -1**	**BUS -2**
BUS -3	**BUS -4**	**BUS -3**	**BUS -4**
BUS -5	**BUS -6**		
BUS -7	**BUS -8**		

Figure 8 : Pictorial representation of AC Buses

Table 9: Requirement for Resistive load bank of 28V DC power

Sl. No.	Input Voltage (Volt)	Total Power rating (Watts)	Load Bank current rating (Amps)	Incremental load current steps	Load line drop compensation load step (Amps)	Quantity required
1	28V DC, 2-wires	4480	160	50,25,25, 20,20,10	10	01
2	28V DC, 2-wires	3080	110	25,25,20, 20,10	10	01
3	28V DC, 2-wires	2240	80	25,25,10, 10,5	5	01
4	28V DC, 2-wires	644	23	10,5,5	3	01
5	28V DC, 2-wires	1540	55	20,10,8, 5,5,2	5	01
6	28V DC, 2-wires	1540	55	20,10,8, 5,5,2	5	01
7	28V DC, 2-wires	1540	55	20,10,8, 5,5,2	5	01
8	28V DC, 2-wires	1120	40	10,10,8, 5,2	5	01
9	28V DC, 2-wires	1120	40	10,10,8, 5,2	5	01
10	28V DC, 2-wires	1120	40	10,10,8, 5,2	5	01
11	28V DC, 2-wires	1120	40	10,10,8, 5,2	5	01
12	28V DC, 2-wires	112	4	3	1	01

Table 10: Technical specification of Resistive load bank for 28V DC power

Sl.No.	Description/Parameters	Constraints/Value
1.	Load Input Voltage	DC, Nominal: 28V; Range: 22 to 29V
2.	Loading elements	Resistive element
3.	Operation mode	Resistive loading operations
4.	Manual operation switches	Load bank front panel shall have input ON/OFF switch, Emergency stop switch, Remote/Local switch, Step load switches, Cooling fan/blower switch, Resistive switches, etc.
5.	Loading operation	Loading to be carried out on behalf of load current.
6.	Parallel operation	Load bank shall have provision to paralleling multiple loads, when powered by single source.

7.	Input terminals	Terminal block to be provided for the input cable connections and it's rating to be more than 2 times of rated resistive element current capacity.
8.	Load input contactor	Load input contactor to be provided for ON/OFF connected loads. The input contactor rating shall be more than 2 times of rated resistive element current capacity.
9.	Cooling arrangement	Heavy duty fans to be provided for heat removal. Fan shall have independent ON/OFF control and powered by external 50Hz AC power. Unit shall have perforation for better air-circulation and perforation to be provided on top, bottom and side plates.
10.	Ingress protection	Unit shall have ingress protection of IP23 or better.
11.	Operating area temperature	The load bank shall meet its performance requirements in an operating environment of 0 degC to 50 degC.
12.	Operation control	The unit shall have relays, contactors and programmable logic controller (PLC) to achieve desired loading under manual mode and remote mode of operations.
13.	Measurement display meter	Unit shall have front panel display to monitor the load parameters like, voltage, current, power, etc.
14.	Mode of operation	Unit shall have provisions to operate in manual mode via front panel switches and remote mode via NI LabView based GUI. Programmable Logic Controller (PLC) to be used for operation, control and remote interfaces.
15.	Remote operation and control	Remote operation shall be possible by Ethernet and RS232/RS485 interface. Remote control shall have complete functionality of load bank with suitable GUI. Remote control GUI to be developed in NI LabView software by supplier.
16.	Protection	Fan failure protection, Over-temperature protection, Over-current protections, short-circuit protection to be integrated in the unit. Fuses to be used for protection of load elements. Under any type of fault, unit shall have to withdraw the loading operations.
17.	Marking	All front panel display, CB, terminals, switches etc. are marked with their nomenclature/notations.
18.	Load bank modularity and handling	Caster wheels, hooks and handles to be provided in load bank to move the complete assembly.
19.	Ratings of wires, elements and components	The current rating of wires, resistive element and inductive elements to be more than 2 times of full load current. Contactors and relays to be rated for at-least1.8 times of full load current. Protection fuses and MCBs to be rated according to I^2t curve. These components to be best quality available in the market
20.	Accessories	Resistive load bank to be connected with 28V DC power source and it requires 50meters of 2-core, 16-sq.mm, copper cable; 5-channel, 125Amps distribution box integrated with 4-pole 125 MCCB and 12-way terminal block. For data transfer interface, an 8-channel ethernet switch and RS232/RS422/RS485 to ethernet converter with I/O cables and power adaptor to be supplied. All necessary accessories, which are essential for operation of Load bank to be supplied.
21.	Passive loading elements	Resistors: Resistive element to be constructed using resistance wire (Nichrome 80/20) embedded in compacted magnesium oxide powder with the outer sheath constructed from stainless steel or Inconel to provide corrosion resistance. Resistive load element to be totally weatherproof with an electrically grounded outer sheath to prevent electrical short circuits by an external foreign object and to protect personnel against accidental electrical shock.

		Elements to be individually serviceable in the field without disturbance to other elements. The change in resistance due to temperature to be minimized by maintaining conservative watt densities. Open wire type elements where the live conductors are exposed and which can be short circuited to each other or to ground by foreign objects or by the breakage of an element or an element support shall not be permitted. Element chambers to be surrounded by aluminized steel heat shields reducing heat gains to the outer surface.
22.	Enclosure construction and finish	Load bank shall be assembled in metal casing and live parts to be separated by insulators. Enclosure shall have earth/chassis ground point. Enclosure body shall be protected against the line to chassis fault. Complete body of load bank shall be shock hazard proof. The container structure shall be manufactured from mild steel/carbon steel with thickness of 1.5mm. Hinged door and bolted panels to be used for access of control, switchgear and load element compartment. The frame contains fork lift pockets for maneuverability and transportation. A durable powder coated finish is used on all panels. Standard color is grey RAL70XX.
23.	Safety features	An emergency stop/disconnect switch to be provided for full isolation of load, fan and control supply. Fan/blower shall be protected with fuse and thermal overload. Thermal detectors to be fitted to protect against overheating of passive elements. Each element group associated contactors are protected by an HRC fuses. Internal access to be restricted by bolted panels. Polycarbonate screens to be used behind the doors to prevent accidental contact with live parts.
24.	Operation & Maintenance Manual	Two Set for each

5.4 Data Acquisition and Control System – (DAQC)

Data Acquisition and Control (DAQC) System is the heart of the Rig and acts as an interface between the Operator and the System. The functions of DAQC system includes Status monitoring, System Control, Data recording, Data analysis, Calibration, Simulation, States management and Report generation. The DAQC system shall be capable of controlling every aspect of the Electrical Rig System like Motor-Gear Drive Stand, Gearbox cooling system, Hydraulic Power Pack System, Aircraft Generators, Generator cooling system, HMDGs, AC & DC Load banks and Aircraft LRUs.

The Data Acquisition & Control (DAQC) System is used to control the EPGS Test rig, based on the commands received from the operator under normal and faulty operating conditions. The DAQC system is also responsible for monitoring various parameters of the subsystems in the rig. This system receives the instrumented values of various parameters from different sensors placed at different locations in the rig. The received parameter information is analysed to generate the proper control, status and fault signals for the rig operation. The DAQC system also stores the received parameter information during the entire test for future analysis. The monitored parameters shall also be displayed properly on GUI for this module. This GUI is able to display the required data in the time and frequency domain.

The DAQC system will be designed under a distributed and modular architecture concept. The DAQC system will be comprised of Local Control Unit (LCU), Remote Control Unit (RCU), Master Control Unit (MCU), Aircraft Power Monitoring Unit (APMU) and Accessories like Ethernet switches, NAS, sensors, interfaces etc.

The DAQC includes software capabilities and provides an intuitive GUI for operators to interact with the rig and Aircraft electrical systems for performing control and monitoring operations. The LCU allows the operator to perform the data acquisition function along with pre-test start up and required maintenance operations. The RCU displays data and control functions and provides all interface communications between the operator and the MCU Unit. MCU will be used for resource allocation of LCUs and load banks to RCUs. The APMU performs the data acquisition, monitoring and control of the Aircraft electrical subsystems like Generator GCUs, TRUs, ACPMU, DCPMU, ACGPU, DCGPU etc through different interfaces like Ethernet, ARINIC,MIL-1553,RS- etc.

The general DAQC system interfaces with the EPGS rig subsystem and Aircraft LRUs under test is described in Figure- 9.

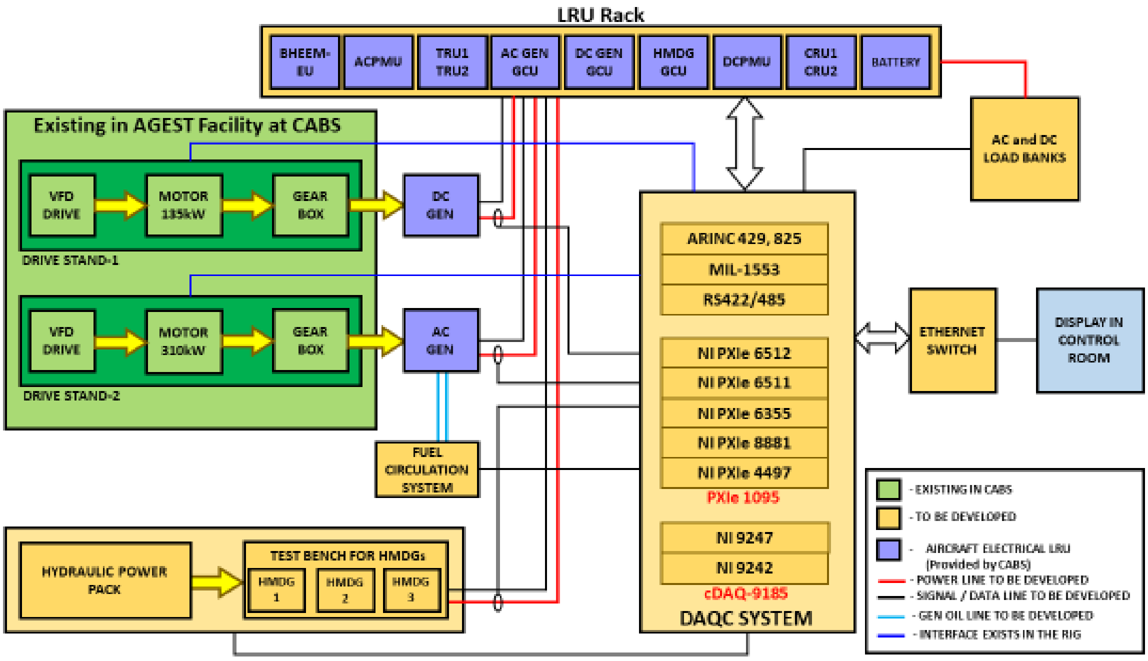


Figure 9 : DAQC System Interface with EPGS Rig and LRUs under test

The DAQC system for two nos of Moto-Gear Drive stand as shown in the Figure-9 are already established in CABS and are controlled and monitor on RCU through their respective LCUs.

The Industry partner has to establish the LCU and RCU for data acquisition and control of Hydraulic power pack system and for Fuel circulation system for AC generator. The existing DAQC system of CABS in AGEST Facility is to be integrated with the DAQC system of Hydraulic power pack and Fuel circulation system and is under the scope of Industry Partner.

AC and DC Load Banks will be used to test the generators and electrical LRUs of aircraft. To simulate the electrical LRUs, 11nos of AC Load and 12 nos of DC loads will be used. These loads will be monitored and control through RCU in control room. Industry Partner has to develop these AC and DC loads and integrate it with DAQC system through suitable interface.

An Aircraft Power Monitoring Unit (APMU) shall be provided by the Industry partner for monitoring the Aircraft LRUs from generation to distribution like GCUs, ACPMU, TRUs, DCPMU etc. This APMU shall have different interfaces like discrete, ARNIC, MIL-1553,RS-422/485 etc. All the aircraft LRUs shall be controlled and monitored though these interfaces. An APMU shall have all hardware, software and user friendly GUI to control and monitor the different electrical aircraft LRUs. The data acquired by APMU shall be displayed on RCU in control room. The design and development of APMU, integration with Aircraft LRUs, data acquisition, control and monitoring from RCU and testing in integrated mode are in the scope of Industry Partner.

5.4.1. Hardware Considerations of DAQC System

The DAQC system of the rig comprises of the following sub-system components:

(ii) Master Control Unit (MCU) – 1 Number

(iii) Remote Control Unit (RCU) – 4 Numbers

(iv) Local Control Unit (LCU) – 2 Numbers

(v) Aircraft Power Monitoring Unit (APMU) - 1 Number

(vi) Network Attached Storage (NAS) - 1 Number

(vii) Ethernet Switch (ES) - As required

(viii) Colour Printer - 1 Number

The functionality of each of the above subsystems of the DAQC system and their technical specifications are given in subsequent sections.

5.4.1.1 Master Control Unit (MCU)

The Master Control Unit (MCU) which is already available in CABS will be used for EPGS Test rig resource management and controls the LCU and RCU. Industry Partner shall have to integrate the LCU and RCU of Hydraulic power pack with the existing MCU in AGEST Facility. The specification of MCU is given in the Table-11 below.

Table 11: Specification of MCU available in CABS

Sl. No.	Parameters	Compliance
1	Form Factor	Workstation
2	Processor Make	Intel
3	Number of Processors	2
4	Processor Configuration	Intel Xeon silver 4210 (10 Core,3.2 GHz, 13.75 MB
5	Number of PCIe Slots Gen 3 (x 1, x 4,x 8 ,X16) (Number)	(2) PCIe x16, (2) additional x16 slots with 2nd CPU (1) PCIe x8 open ended, (1) x16 wired as x 4, (1) x16 wired as x1
6	RAM	DDR4
7	Operating Frequency	2933MHz
8	Display (antiglare, LED-backlit)	Monitor
9	Display Resolution (Full HD or better) (Pixel)	1920X1080

a. MCU is primarily used as Health Monitoring station and Time server. It allots the functionality like motor control, load control, parameter monitoring, etc, to different RCUs. It is a workstation PC with GUI software and is be located in the control room.

b. All the subsystems of the AGEST Facility are time synchronized using the time provided by the Time server at the MCU. The Time Server sends the Network Time Protocol (NTP) time as a UDP network packet at every 64 secs.

All the subsystems of EPGS Test rig also use this time for their internal operation and as a time stamp in the network packets.

c. The Health monitoring of the subsystems of the rig including RCU & LCU is carried out through Power ON Built-In Test (PBIT), Maintenance Built-In Test (MBIT) and Continuous Built-In Test (CBIT) tests. The PBIT, also called Power ON Self-Test (POST) is initiated during the Boot time. All the subsystems check the health status of their internal components and sends the results as PBIT to MCU. The MBIT is initiated by the operator from the MCU for a particular subsystem to perform certain functionality and know its health status. Each subsystem sends CBIT results at every 3 Minutes to MCU to ensure the subsystem is in working state throughout the test duration. Any faults or deviations observed are indicated through Alerts on MCU in the form of message Alerts and Audio Alerts.

5.4.1.2 Remote Control Unit (RCU)

The Remote Control Unit (RCU) is used for controlling the LCUs and monitoring the parameters of the subsystems in the rig. The functionality of the RCU to control certain functionality of the LCUs is defined by the MCU. RCU sends the commands related to Start/STOP of the subsystems, parameter references etc. to LCU. The measured parameters of the subsystems in the EPGS Test rig received from the LCUs are displayed on the RCU. The RCU is basically a PC with GUI software. The GUI software is planned to be user friendly and allow the operators to handle the operations with less efforts. RCUs will be located in the control room about 8 m from the LCUs.

The specifications for the RCU are given in Table-12 below .

Table 12 : Specification of Remote Control Unit (RCU)

Sl. No.	**Parameters**	**Values**
1	FormFactor	Tower Model
	1. PROCESSOR	
2	Processor Make	Intel
3	Number of Processors	1
4	ProcessorGeneration	8th
5	ProcessorConfiguration	IntelXeonGold 5222 (4CoreHT,3.8GHz,8MBL3Cache)or Equivalent/Better
	2**. CHIPSET**	
6	Chipset	Intel C621
7	Number of PCIe Slots	4xPCIeslots:1x8Gen3(x16Connector)FH/HL1x8Gen3(x8Connector)FH/HL1x4Gen3(x8Connector)FH/HL1x1Gen3(x1Connector)FH/HL
	3. **GRAPHICS**	
8	Type of Graphics	Discrete
9	GraphicCard	nVIDIAQuadroP50006GBorbetter
	4. **MEMORY**	

10	RAM	DDR4
11	OperatingFrequency	2666MT/s
12	RAM Type	ECC
13	RAM Size (GB)	64
14	RAMExpandability(GB)	64
	5. **STORAGE**	
15	TypeofHardDiskDrive -I	SATA
16	SizeofHardDiskDrive -I(GB)	1000
17	NoofHardDiskDrives-I	1
18	TypeofHardDiskDrive -II	SSD
19	SizeofHardDiskDrive -II(GB)	256
20	NoofHardDiskDrive -II	1
	6. **OPERATING SYS**TEM	
21	OperatingSystem	Windows 10 Pro / Higher Version
22	OS Certifications	Windows,Ubuntu
	7. **APPLICATION & CERTIFICATIONS**	
23	Safety Certifications	YES
24	ROHS Compliance	Yes
	8. **CONNECTIVITY**	
25	NetworkingInterface	DualGigabitNIC10/100/1000
	9. **PORTS**	
26	VGA Port	1
27	HDMI Port	2
28	Display Port	4
29	Serial Port	1
30	Number of USB 2 Ports	2
31	No. of USB 3.0 Port (No.)	6
	10. **DISPLAY**	
32	Display(antiglare,LED-backlit)	Monitor
33	Display Size (cm)	60.47
34	Panel Type	IPS
35	DisplayResolution(FullHDorbetter)(Pixel)	1920X1080
	11. **POWER**	
36	MaximumPower(Watt)	1400 max
37	PowerEfficiency	Minimum 80%
38	Power Management Unit	YES
39	Keyboard	Standard Keyboard
40	Mouse	OpticalScrollMouse
41	Optical Drive	DVD R/W
42	No. of DIMM Slots (No.)	4
43	On Site OEM Warranty (Year)	2

5.4.1.3. Local Control Unit (LCU)

a. The LCU comprises of the embedded hardware, software and sensors for carrying out the data acquisition and instrumentation function. The Scope of LCU is limited to Hydraulic Power Pack and Fuel circulation system of EPGS test rig. Since, Hydraulic power pack along with the test setup for 3 Nos of HMDGs is an independent sub-system of rig, it will have an independent LCU. For fuel circulation system for AC generator, one of the LCUs of the Motor –Gear Drive stand which is already available in AGEST Facility shall be upgraded with additional hardware/instrumentation and can be used for Fuel circulation system or separate new LCU to be designed. The Industry Partner can do the feasibility study to accommodate the additional augmentation in the existing LCUs required for Fuel circulation system or can provide the new LCU for Fuel circulation system.

b. The LCU has provisions to be controlled through RCU and MCU. LCU receives the commands from the RCUs to control the subsystems of the electrical rig. LCU also receives the instrumented data of the parameters of the electrical rig subsystems from different sensors. These parameter data are used by the LCU to generate necessary control and fault signals for proper and safe operation of the rig. The data from the LCU is transferred to RCU/MCU and NAS for further monitoring and control through Ethernet (1G/10G).

c. The LCUs of EPGS sub-system can also be operated locally from a GUI replicating the functionality of the RCU and running in a System with a Display/Key board/Mouse in standalone mode for testing and debugging a particular sub-system. LCU is the system which turns OFF the rig automatically in case of emergency apart from the emergency stop from the operator through RCU.

d. The software of LCU can be realized using LabVIEW Model-based development framework or any Equivalent GUI development framework. The Hardware is preferably a PXI based system with necessary signal conditioning or equivalent system capable of meeting all the requirements specified in this document. The hardware of the LCU should be fully modular using COTS units and should have the provision for future expansion.

e. The LCU should control and monitor all the subsystems of the rig as given in Table 13. The LCU shall have the capability to acquire the signals, perform signal conditioning required and feed the data to LCU controller for engineering conversion and displaying the results in GUI.

Table 13 : Specification for the parameters to be monitored By LCU in the rig

Sl.No.	Parameter	Measurement Range	Accuracy	Resolution	Sensor Type	Interface/Signal Type	Remarks
1	**Hydraulic Power Pack**						
1.1	All the Parameters as given by the Hydraulic Power Pack	--	--	--	--	Standard Interface Provided by the Hydraulic Power Pack (Ethernet, Profinet, ModBus, etc.)	To LCU from Hydraulic Power Pack
1.2	VFD Health Status	0-Not Healthy 1-Healthy	--	--	--		
1.3	Emergency STOP	--	--	--	--	Discrete from LCU, RCU and MCU through which the rig can be stopped apart from the digital interface.	
2							
2.1	Inlet Oil Flow rate	0-100LPM	±0.5% of FS	8-bit	Screw Type PD Flow Meter	Analog	Between LCU & Hydraulic Power Pack; These parameters can be transmitted & received on the communication bus (Ethernet, ModBus etc.)
2.2	Inlet Oil Pressure	0-40000kPa	±0.5% of FS	10-bit	Piezo Resistive	Analog	
2.3	Inlet Oil Temperature	0-200^{0}C	±0.5% of FS	8-bit	PTD/RTD	Analog	
2.4	Outlet Oil Flow rate	0-100LPM	±0.5% of FS	8-bit	Screw Type PD Flow Meter	Analog	
2.5	Outlet Oil Pressure	0-40000kPa	±0.5% of FS	10-bit	Piezo Resistive	Analog	
2.6	Outlet Oil Temperature	0-200^{0}C	±0.5% of FS	8-bit	PTD/RTD	Analog	
2.7	Reservoir Oil Temperature	0-200^{0}C	±0.5% of FS	8-bit	PTD/RTD	Analog	
2.8	Reservoir Oil Level	0-500L	±0.25% of FS	10-bit	Level Sensor/Sight Gauge	Analog	

2.9	Filter Clog by Delta Pressure	0-750kPa/100PSIG	--	--	Pressure Differential	Digital 0-No Clog 1-Clog	
2.10	Oil Flow Control	0-100LPM	1LPM	1LPM	--	Analog through potentiometer / as a Ethernet Packet	
2.11	Heat Exchanger Flow Switch Control	0-OFF 1-ON	--	--	--	Digital	
2.12	Heat Exchanger Valve Control	0-OFF 1-ON	--	--	--	Digital	
2.13	Flow Pump ON/OFF	0-OFF 1-ON	--	--	--	Digital	
2.14	Pump Health Status	0-Not Healthy 1-Healthy	--	--	--	Digital	
3	**Sound/Noise Measurement**						
3.1	Sound/ Noise Level	0-100dBA	±1% of FS	8-bit	Microphone based sound sensor	Analog	To LCU; The sensor shall be placed at appropriate place near test bench for HMDGs
4	**Hydraulic Motor Driven Generator (HMDG)**						
4.1	Vibration-3 Axis –(5 nos)	10 MILS	±2% of FS	8-bit	Piezo Electric	Analog	To LCU

4.2	Case Temperature (5 nos)	0-200^{0}C	±0.25% of FS	8-bit	PTD/RTD	Analog	
4.3	Ambient Temperature (5 nos)	0-70^{0}C	±1% of FS	8-bit	Thermocouple	Analog	
	Fuel Circulation System for cooling of Generator						
4.4	Inlet Oil Flow	0-100LPM	±0.5% of FS	8-bit	Screw Type PD Flow Meter	Analog	Between LCU & Fuel cooling System; These parameters can be transmitted & received on the communication bus (Ethernet, ModBus etc.) Any other parameters and Controls should be included as per the design by Industry Partner
4.5	Inlet Oil Pressure	0-4200kPa	±0.5% of FS	10-bit	Piezo Resistive	Analog	
4.6	Inlet Oil Temperature	0-200^{0}C	±0.5% of FS	8-bit	PT200/RTD	Analog	
4.7	Outlet Oil Flow	0-100LPM	±0.5% of FS	8-bit	Screw Type PD Flow Meter	Analog	
4.8	Outlet Oil Pressure	0-4200kPa	±0.5% of FS	10-bit	Piezo Resistive	Analog	
4.9	Outlet Oil Temperature	0-150^{0}C	±0.5% of FS	8-bit	PTD/RTD	Analog	
4.10	Reservoir Oil Temperature	0-200^{0}C	±0.5% of FS	8-bit	PTD/RTD	Analog	
4.11	Reservoir Oil Level	0-100L	±0.25% of FS	10-bit	Level Sensor/Sight Gauge	Analog	
4.12	Filter Clog by Delta Pressure	0-750kPa	--	--	Pressure Differential	Digital 0-No Clog 1-Clog	
4.13	Oil Flow Control	0-100LPM	1LPM	1LPM	--	Analog through potentiometer / as a Ethernet Packet	
4.14	Pump ON/OFF	0-OFF 1-ON	--	--	--	Digital	
4.15	Pump Health Status	0-Not Healthy	--	--	--	Digital	

		1-Healthy					
5	**AC Loads**						
5.1	All the parameters as specified for the Load Banks		--	--	--	Ethernet	Between LCU & AC Load Bank;
5.2	Load Enable	0-Disable 1-Enable	--	--	--		These parameters can be transmitted and
5.3	Emergency STOP					Discrete from LCU, RCU and MCU through which the rig can be stopped part from the digital interface	received on the gigabit Ethernet between LCU & AC Load Bank.
6	**DC Load Bank**						
6.1	All the parameters as specified for the Load Banks		--	--	--	Ethernet	Between LCU & DC Load Bank;
6.2	Load Enable	0-Disable 1-Enable	--	--	--		These parameters can be transmitted and
6.3	Emergency STOP	--	--	--	--	Discrete from LCU, RCU and MCU through which the rig can be stopped apart from the digital interface	received on the gigabit Ethernet between LCU & DC Load Bank.
7	**Spare Channels**						
7.1	Analog Channels	20 Nos.					

7.2	Digital Channels	40 Nos.

f. LCU shall also contain Reference Signal Generation Source/Module and Switch Matrix used for internal calibration of the Data Acquisition and Control System. The Reference Signal Generation Source generates all the internal reference signals equivalent to the different sensors outputs. These internal reference signals along with the actual sensor outputs shall be connected to the Switch Matrix. The calibration or normal operating mode can be selected through the command received by LCU from RCU. Based the command received, the Switch Matrix in LCU routes either known internal reference signals from Reference Signals Generation Source or the actual sensor outputs to all channels of the data acquisition modules. During an internal calibration, each data acquisition module digitizes the known internal reference signals, compares the reading to the known value, and updates and stores new correction factors in the module's onboard calibration memory.

The technical specification of LCU hardware are given in Table 14,15,16 and 17

5.4.1.3.1. Sampling Rate for Parameters Monitoring

a. The parameters of the electrical rig LRUs have to be measured with the adequate sampling rate. The sampling rate for the different parameters acquisition is selected based the requirement to capture the transients and distortion on the signals, criticality of the signals etc.
b. The proposed minimum sampling rates for the different parameters acquisition in the EPGS Test Rig are in the Table 14 below. The hardware of LCU consists of data acquisition and processing modules and controller assembled in the chassis. The details of these LCU subsystems are given below.

Table 14 : Sampling Rate

Sl.No.	Parameter	Sampling Rate
01.	Voltage	10kSamples/Sec
02.	Current	10kSamples/Sec
03.	Temperature(at different locations of the rig)	Upto50Samples/Sec,Programmable
04.	Speed	Upto2kSamples/Sec,Programmable
05.	Vibration	10kSamples/Sec
06.	Oil Flow Rate	Upto50Samples/Sec,Programmable
07.	Oil Flow Pressure	Upto50Samples/Sec,Programmable
08.	Oil Temperature	Upto50Samples/Sec,Programmable
09.	Sound/Noise Level	Upto50Samples/Sec,Programmable
10.	Any other Analog Signals	10kSamples/Sec

5.4.1.3.2 Controller of LCU

a. The embedded controller in the LCU is used to control the data acquisition modules. It receives the measured data of the parameters of rig LRUs, analyses and generates the control and status signals for controlling the LRUs for safe operation of the rig. It also communicates with the RCU for receiving the commands related to rig operation and sends the status parameters of the rig LRUs to RCU at regular intervals. It also sends the status parameters of the rig LRUs to NAS at regular intervals for storage and further analysis.

b. The embedded controller is a high bandwidth PXI Express compatible or equivalent system controller. The controller integrates standard I/O features in a single unit. The standard I/O on each module includes one Display Port 1.2, video port, high-speed USB ports, a PCI-based GPIB controller, two Gigabit Ethernet connectors, a reset button and PXI Express triggers.

The specifications of the LCU controller are given in Table 15 below,

Table 15 : Specification of the LCU Controller

Sl. No.	Feature	Specification
01.	CPU	Intel® Xeon W-2245 or better
02.	CPU Frequency	2.3 GHz (base), 3.4 GHz (single-core Turbo mode)
03.	Intel Smart Cache	20 MB
04.	Triple-Channel DDR4 RAM	8 GB Standard 24 GB Maximum
05.	Solid-State Drive	240 GB or larger Serial ATA
06.	Ethernet	10/100/1000 BaseT, 2 ports
07.	Hi-Speed USB (2.0) Ports	Yes
08.	Hi-Speed USB (3.0) Ports	Yes
09.	PS/2 Keyboard/Mouse Connector	No
10.	PXI Express 4 Link Configuration	x4, x4, x4, x4
11.	PXI Express 2 Link Configuration	x8, x16
12.	GPIB (IEEE 488 Controller)	Yes
13.	Installed Operating System	Windows 10 Professional (64-Bit)/ Higher Version

5.4.1.3.3 . <u>Chassis</u>

a. All the data acquisition modules and the embedded controller shall be assembled in a chassis. The chassis combines a high-performance 18 slot PXI Express backplane or equivalent with a high-output power supply. The bandwidth of the chassis shall be 12GB/S or better. Extra slots should be provided as spares for further expansion and shall have multi chassis support. The spare channels/slots are used for acquisition of the parameters identified during the operation of the rig. The chassis shall generate the different DC voltages like 3.3V, 5V, ±12V etc. required for the data acquisition modules. The load and line regulation of these DC generation modules shall be less than or equal to 1%. The chassis design shall be modular to ensure a high level of maintainability, resulting in a very low mean time to repair (MTTR).

b. The chassis along with other LCU related LRUs/components shall be integrated in a rack.The specifications of the chassis are given in Table-16 below.

Table 16 : Specification of the Chassis

Sl. No.	**Parameter**	**Specification**
1.	**AC Input**	
1.1.	Input voltage range	100 VAC to 240 VAC
1.2.	Operating voltage range	90 VAC to 264 VAC
1.3.	Nominal Input frequency	50 Hz
1.4.	Operating frequency range	47 Hz to 63 Hz
1.5.	Input current rating	12 A to 6 A
1.6.	Over-current protection	15 A circuit breaker
1.7.	Line regulation for DC Outputs 3.3 V 5 V ±12 V	<±0.2% <±0.1% <±0.1%
1.8.	Efficiency	70% typical
2.	Module cooling system	Forced air circulation
3.	Ambient temperature range	0 to 55 °C

5.4.1.4 Aircraft Power Monitoring Unit (APMU)

The Aircraft Power Monitoring Unit (APMU) is used to monitor the parameters of the aircraft electrical LRUs that are being functionally evaluated in the electrical rig. The Industry Partner shall make the APMU and interface it with the aircraft LRUs for their control and monitoring. The Industry Partner will test the loading of aircraft generators and HMDGs by enabling the GCUs through the AMPU.

The parameters measured by the APMU shall be sent to LCU/RCU/MCU and NAS through gigabit Ethernet (1G/10G) for further monitoring, control and display in GUI. The critical parameters of the aircraft LRUs measured by APMU shall also be displayed on the GUI replicating the Multi-Functional Display (MFD) of the aircraft. The tentative GUI is shown in the Figure- 10 below. The interface between the APMU and aircraft LRUs is Discrete, Ethernet, MIL-STD-1553B multiplexed bus & RS-422/232 serial interface, ARNIC-429 etc..

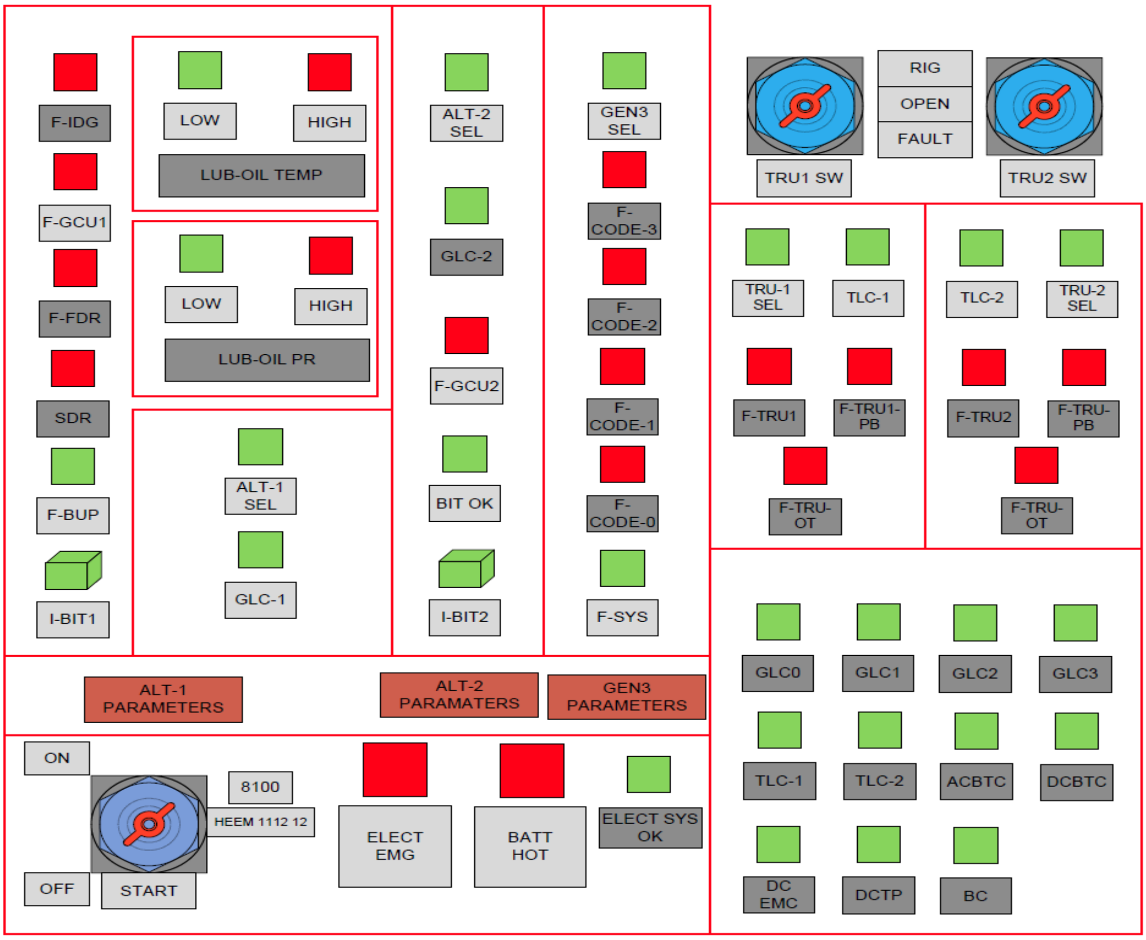


Figure 10 :GUI for APMU

12 number of toggle switches should be provided in the APMU. 5 toggle switches are used to enable or disable the five Generator Control Units (GCUs) and two nos for TRUs. APMU should monitor and display the status of these switches. Extra five switches are used as spares.

APMU also sends the status of all the aircraft LRUs to the RCU/LCU for display in GUI and for necessary action towards safe operation of the rig in case of failure conditions.
The software of APMU can be realized using LabVIEW Model-based development framework or any Equivalent GUI development framework. The Hardware is generally a PXI based with necessary signal conditioning or any equivalent system. The hardware of the LCU should be fully modular using COTS units and should have the provision for future expansion.

The hardware of APMU consists of data acquisition and processing modules, Oscillographic Recorder, Power Analyzer and controller assembled in the chassis. The specifications for the controller and chassis are same as LCU controller and chassis. The sampling rate for voltage and currents to be monitored by APMU is 2 Msamples/sec. The sampling rate for measurement of other parameters in APMU is same as given for LCU.

The detail of the Power Analyzer is given below.

5.4.1.4.1 Power Analyzer

The main objectives of the computer based Power Analyser are to capture power quality parameters and disturbance events and the storage of the acquired data for post-monitoring analysis and corrections. The Power analyzer allows for measurement and analysis of system parameters such as Power (active, reactive, apparent and instantaneous), power factor, Total Harmonic Distortion (THD) (%), energy and Signal-to-noise and distortion ratio (SINAD) (in decibels). The calculated power quality parameters are sent to RCU/MCU over Gigabit Ethernet connection. It is implemented as a virtual instrument (VI) in MCU, whereby, programming and user interface is developed using LabVIEW or equivalent GUI development software. The voltage and current probes supplied by the OEM of the power analyser should be compatible for power quality measurement of the electrical rig LRU.

Table 17 : Specification of Power Analyser

Sr. No	Parameters	Specifications
	Make	M/s Yokogawa (WT333E Series) or Equivalent
	Quantity	**2 Nos**
1.	Power measuring elements	03Nos. (03 voltage and 03 Current for carrying out 3-phase, 4-wire measurement and also provision to perform 1-Phase,3-wire & 3-phase,3-wire measurement)
2.	Power Measurement frequency Range	DC and 0.1Hz to 100kHz
3.	Power measurement Accuracy at 50Hz	≤ 0.1% of reading + 0.05% of range
4.	Voltage Measurement range	15V/30V/60V/150V/300V/600V
5.	Current Measurement range	0.5A/1A/2A/5A/10A/20A (Direct), upto 400 A with external Current Clamp
	Instantaneous Maximum allowable input (1 period, for 20ms)	
6.	Voltage	Peak value of 2.8kV or RMS value 2.0kV
7.	Current	Peak value of 450A or RMS value 300A
	Instantaneous Maximum allowable input (for 1s)	
8.	Voltage	Peak value of 2.0kV or RMS value 1.5kV
9.	Current	Peak value of 150A or RMS value 40A
	Continuous Maximum allowable input	
10.	Voltage	Peak value of 1.5kV or RMS value 1.0kV
11.	Current	Peak value of 100A or RMS value 30A
12.	External current sensor input Channel	02 nos.
	Voltage, Current and Power Measurement	
13.	Measurement Method	Digital Sampling method
14.	Crest Factor	3 or 6
15.	Wiring System	Single-phase, Three-wire, Three-phase, Three-wire Three-phase, Four-wire
16.	Range Selection	Auto or Manual
17.	Peak Measurement	Measure the peak(max, min) of voltage, current and power from instantaneous values.
18.	Zero-level Compensation	Remove internal offset
	Frequency Measurement	

19.	Method	Reciprocal method
20.	Measurement range	0.1Hz to 100kHz
21.	Accuracy	± 0.06% of reading
	Harmonic Measurement	
22.	Method	PLL Synchronization
23.	PLL source frequency range	10Hz to 1.2kHz
	Measurement	
24.	Measurement and calculation	Voltage, current, active power, reactive power, apparent power, power factor, phase angle, frequency, peak voltage, peak current, crest factor, integration (Wh, Ah, varh, Vah), time, %
25.	MAX hold Measurement	Voltage RMS, MEAN simultaneous measurement Average active power (user defined function) Active power integration (WP) (Wh) Apparent power integration (WS) (VAh) Reactive power integration (WQ) (varh)
26.	Voltage & Current Accuracy	± (0.1% of reading + 0.3% of range) up to 10kHz and ± (0.5% of reading + 0.5% of range) ± [{0.04 x (f-10)}]% of reading, beyond 10kHz
27.	Communication	Serial (RS-232) Interface, USB PC Interface, Ethernet Interface
	Display	
28.	Display type	7 Segment LED
29.	Simultaneous display	4 items
30.	Display range (Voltage, Current, Power(P, Q & S))	Displayed in 5 digits (00000 to 99999)
31.	Power factor display range	1.0000 to -1.0000
32.	Response Time	At maximum, 2 times the data update rate
	Internal Memory	
33.	Measured Data	Save and recall the stored measured data. Store interval: Data update interval or in the range of 1sec to 9hrs 59min 59sec.
34.	Setup information	Saves/loads four patterns of setup information.
	D/A Output	
35.	Output Voltage	±5V FS (approx. ±7.5V FS Maximum)
36.	Number of output channel	12 Outputs for product
37.	Output Items	Set for each channel. U, I, P, S, Q, fU, fI, Upk, Ipk, WP, WP±, q, q± and MATH

38.	Accuracy	±(accuracy of each measurement item + 0.2% of FS)(FS = 5 V)
39.	D/A Conversion Resolution	16bits
40.	Remote control input signal	EXT HOLD, EXT TRIG, EXT START, EXT STOP, EXT RESET
	General Specification	
41.	Operating Environment	Temp: 5°C to 40°C Humidity: 20% RH to 80% RH (No Condensation)
42.	Rated Supply voltage	100VAC to 240VAC
43.	Rated Supply Frequency	50Hz (48Hz to 63Hz)
44.	Battery Backup	Setup parameters are backed up with a lithium battery
45.	Safety Standard	EN 61010-1, EN61010-2-030
46.	Emission and Immunity Standard	EN 61326-1
	Accessories	
47.	Measurement lead	0.75 m safety terminal cable with 2 leads (red and black) in a set
48.	Small Alligator clip adapter	Safety terminal-alligator clip adapter, containing 2 pieces (red and black) in a set
49.	Fork Terminal Adapter	Safety terminal-fork terminal adapter, containing 2 pieces (red and black) in a set
50.	Conversion Adapter	BNC binding post adaptor
51.	Safety Terminal Adapter	Screw-in type 2 adapters (red and black) in a set
52.	External Sensor Cable	For connection the external input of the instrument to current sensor. Length: 0.5 m
53.	Connection Cable	26-pin cable for D/A output
54.	Current Probe	Necessary current measurement clamp-on probes to be provided
55.	Calibration Certificate	Calibration certificates to be provided alongwith instrument

The schematic drawings giving the details of APMU must be included in the Technical bid.

The list of LRUs to be tested and their parameter to be measured, analyzed and displayed is given in the Table- 18.

Table 18 :LRUs wise –Aircraft LRUs Parameters to be monitored by APMU (To be developed and integrated in the EPGS Test Rig by the Industry Partner)

NOMENCLATURE	SIGNAL	TYPE	RANGE
50kVA IDG Testing Parameters			
50kVA integrated drive generator (50kVA IDG)	Ac current Ø1	Analog	0-300A RMS
	Ac current Ø2	Analog	0-300A RMS

	Ac current Ø3	Analog	0-300A RMS
	Ac voltage Ø1- Ø2	Analog	0-250V RMS
	Ac voltage Ø2- Ø3	Analog	0-250V RMS
	Ac voltage Ø3- Ø1	Analog	0-250V RMS
	Frequency Ø1	Analog	500 Hz
	Frequency Ø2	Analog	500 Hz
	Frequency Ø3	Analog	500 Hz
	Torque	Analog	0-200 Nm
	Speed	Analog	0-10000 RPM
	temperature	Analog	0-200°C
	Phase sequence	--	Ø1- Ø2- Ø3
IDG GCU	FIDG	discrete	GND/OPEN
	FGCU	discrete	GND/OPEN
	FFDR	discrete	GND/OPEN
	FSDR	discrete	GND/OPEN
Fuel cooled oil cooler (FCOC)	FCOC ON/OFF	discrete	GND/OPEN
	Oil inlet pressure	Analog	0-500 psi
	Differential pressure	Analog	0-100 psi
	Inlet oil flow	Analog	0-50 LPM
	Oil inlet temperature	Analog	0-200°C
	Oil outlet temperature	Analog	0-200°C
5kVA HMDG Testing Parameters			
5 kVA hydraulic motor driven generator (5kVA HMDG)	Ac current Ø1	Analog	0-50A RMS
	Ac current Ø2	Analog	0-50A RMS
	Ac current Ø3	Analog	0-50A RMS
	Ac voltage Ø1- Ø2	Analog	0-250V RMS
	Ac voltage Ø2- Ø3	Analog	0-250V RMS
	Ac voltage Ø3- Ø1	Analog	0-250V RMS
	Frequency Ø1	Analog	500 Hz
	Frequency Ø2	Analog	500 Hz
	Frequency Ø3	Analog	500 Hz
	Phase sequence	--	Ø1- Ø2- Ø3
	Oil inlet pressure	Analog	0-300 bar
	Oil inlet flow rate	Analog	0-50 LPM

	Oil inlet temperature	Analog	0-200°C
	Oil outlet temperature	Analog	0-200°C
HMDG GCU	System shutdown	Discrete	GND/OPEN
	Bit status	Discrete	GND/OPEN
5kW DC Gen Testing Parameters			
5 KW DC generator	DC current	Analog	0-300A
	DC output voltage	Analog	0-50A
	Input shaft torque	Analog	0-50Nm
	speed	Analog	0-15000 RPM
5 kW DC generator GCU	ARINC 429	Data	
5 kW DC generator GCU(by CVDRE)	Gen status	Discrete	GND/OPEN
	BIT STATUS-1	Discrete	GND/OPEN
	BIT STATUS-2	Discrete	GND/OPEN
	BIT STATUS-3	Discrete	GND/OPEN
	BIT STATUS-4	Discrete	GND/OPEN
400W HMDG Testing Parameters			
400W hydraulic motor driven Generator-1(0.4 KVA HMDG-1)	Ac current Ø1	Analog	0-20A RMS
	Ac current Ø2	Analog	0-20A RMS
	Ac current Ø3	Analog	0-20A RMS
	Ac voltage Ø1- Ø2	Analog	0-400V RMS
	Ac voltage Ø2- Ø3	Analog	0-400V RMS
	Ac voltage Ø3- Ø1	Analog	0-400V RMS
	Frequency Ø1	Analog	600 Hz
	Frequency Ø2	Analog	600 Hz
	Frequency Ø3	Analog	600 Hz
	Phase sequence	--	Ø1- Ø2- Ø3
	Oil inlet pressure	Analog	0-300 bar
	Oil inlet flow rate	Analog	0-50 LPM
	CB status	Discrete	GND/OPEN
CRU-1	HMDG-1 ON	Discrete	28V DC/NIL
	DC VOLTAGE	Analog	0-50 V DC
	DC VOLTAGE	Analog	0-50 V DC
	DC VOLTAGE	Analog	0-50 V DC
	DC VOLTAGE	Analog	0-50 V DC

400W hydraulic motor driven Generator-2(0.4 KVA HMDG-2)	Ac current Ø1	Analog	0-20A RMS
	Ac current Ø2	Analog	0-20A RMS
	Ac current Ø3	Analog	0-20A RMS
	Ac voltage Ø1- Ø2	Analog	0-400V RMS
	Ac voltage Ø2- Ø3	Analog	0-400V RMS
	Ac voltage Ø3- Ø1	Analog	0-400V RMS
	Frequency Ø1	Analog	600 Hz
	Frequency Ø2	Analog	600 Hz
	Frequency Ø3	Analog	600 Hz
	Phase sequence	--	Ø1- Ø2- Ø3
	Oil inlet pressure	Analog	0-300 bar
	Oil inlet flow rate	Analog	0-50 LPM
	CB status	Discrete	GND/OPEN
CRU-2	HMDG-2 ON	Discrete	28V DC/NIL
	DC VOLTAGE	Analog	0-50 V DC
	DC VOLTAGE	Analog	0-50 V DC
	DC VOLTAGE	Analog	0-50 V DC
	DC VOLTAGE	Analog	0-50 V DC
Common LRU for Integrated Testing			
DC power management unit (DCPMU)	GLC 3 close	Discrete	GND/OPEN
	DCBTC CLOSE	Discrete	GND/OPEN
	DCEMC CLOSE	Discrete	GND/OPEN
	BC CLOSE	Discrete	GND/OPEN
	DCTP CLOSE	Discrete	GND/OPEN
	GPC-2 CLOSE	Discrete	GND/OPEN
	DC GPU ENGAGE	Discrete	GND/OPEN
	EMDP CC	Discrete	GND/OPEN
	RCCB STATUS	Discrete	GND/OPEN
	RCCB STATUS	Discrete	GND/OPEN
	RCCB STATUS	Discrete	GND/OPEN
	RCCB STATUS	Discrete	GND/OPEN
	RCCB STATUS	Discrete	GND/OPEN
	RCCB STATUS	Discrete	GND/OPEN
	DC current sensor	Analog	0-10V

	TLC-1 CLOSE	Discrete	GND/OPEN
	TLC-2 CLOSE	Discrete	GND/OPEN
	TRU-1 FAIL	Discrete	GND/OPEN
	TRU-2 FAIL	Discrete	GND/OPEN
	TRU-1 current	Discrete	GND/OPEN
	TRU-2 current	Discrete	GND/OPEN
	A/C BAT HOT	Discrete	GND/OPEN
	A/C BAT OK	Discrete	GND/OPEN
	EMDP BAT HOT	Discrete	GND/OPEN
	EMDP BAT OK	Discrete	GND/OPEN
	A/C BAT MAIN	Discrete	GND/OPEN
	EMDP BAT MAIN	Discrete	GND/OPEN
	A/C BAT thermal switch	Discrete	GND/OPEN
	EMDP BAT thermal switch	Discrete	GND/OPEN
	RCCB control	Discrete	28V DC/NIL
	Dual redundant RS 422		
AC power management unit (ACPMU)	GLC 1 close	Discrete	GND/OPEN
	GLC 2 close	Discrete	GND/OPEN
	ACBTC close	Discrete	GND/OPEN
	GPC-1 close	Discrete	GND/OPEN
	AC GPU engage	Discrete	28V DC/NIL
	AC voltage	Analog	0-115V
	AC frequency	Analog	0-500Hz
	ALT-2 current sensor	Analog	0-10 V DC
	Dual redundant RS 422		
Battery	DC output current	Analog	0-300A
	Dual redundant ARINC 429		
BHEEM	ELECT EMERGENCY	Discrete	28V DC/NIL
	5KVA ALT OVER CURRENT	Discrete	28V DC/NIL
	5KW DC GEN OVER CURRENT	Discrete	28V DC/NIL
	BACKUP POWER	Discrete	28V DC/NIL
	DC VOLTAGE	Analog	0-50 V DC

Transformer rectifier unit-1(TRU-1)	OUTPUT RIPPLE	Analog	0-5 Vpp
	TRU-1 HOT	DISCRETE	GND/OPEN
Transformer rectifier unit-2 (TRU-2)	DC VOLTAGE	Analog	0-50 V DC
	OUTPUT RIPPLE	Analog	0-5 Vpp
	TRU-2 HOT	DISCRETE	GND/OPEN
Central console	A/C BAT IBIT	Discrete	28V DC/NIL
	EMDP BAT IBIT	Discrete	28V DC/NIL
	Power SW selection	Discrete	28V DC/NIL
	AC SW selection	Discrete	28V DC/NIL
	DC SW selection	Discrete	28V DC/NIL
	Rig power selection	Discrete	28V DC/NIL
DC GPU	DC current	Analog	0-500A
	DC voltage	Analog	0-50V
AC GPU	AC current Ø1	Analog	0-300A RMS
	AC current Ø2	Analog	0-300A RMS
	AC current Ø3	Analog	0-300A RMS
	AC voltage Ø1- Ø2	Analog	0-250V RMS
	AC voltage Ø2- Ø3	Analog	0-250V RMS
	AC voltage Ø3- Ø1	Analog	0-250V RMS
	Frequency Ø1	Analog	500 Hz
	Frequency Ø2	Analog	500 Hz
	Frequency Ø3	Analog	500 Hz
	Phase sequence		Ø1- Ø2- Ø3

5.4.1.5. Network Attached Storage (NAS)

The Network Attached Storage (NAS) is used as a central Repository. To Ensure safety of the data, the RAID based configuration is chosen. The LCU, RCU and MCU are connected to NAS and continuously store the data related to their functional activities in respective partition of the NAS. Each of the subsystem will be having restricted access to the other subsystem based on the memory management design. Overall the Test report and status is recorded along with the screen recording video of LCU, RCU and MCU. The Operator can playback the recordings on MCU and perform debrief and analysis of the Tests. It is also envisaged to have a centralized record and playback hardware at NAS to facilitate the storage of screen video and the data.

Table 19 : Specification of Network Attached Storage (NAS)

Sl NO	Parameters	values
1.	Storage System Type	Unified Storage Group 2
2.	Storage Capacity (TB)	24 Terabyte
3.	Hardware Form Factor of Storage System (RU)	5
4.	Disk Type	SSD
5.	Speed of dual ported disk drive in Gbps	6
6.	Total Numbers of Drive Slots	4
7.	Number of Drive Slots Populated with drives of different type in the Storage System	4
8.	Total no of drive slot populated with SSD	0
9.	Total no of drive slot populated with Flash Drive	0
10.	Total no of drive slot populated with NL-SAS	0
11.	Drive Type Wise Storage Capacity of System in GB	SATA - 16000 GB
12.	Automated Storage tiering feature across the populated Drives types(in case of multiple drive system)	Yes
13.	Automated Storage tiering Software License Included	Yes
14.	Hot Spares	No
15.	Number of Populated Disks per Hot Spare	0
16.	Provision for additional Capacity	No
17.	Type of Front end Ports	USB 3.0
18.	Number of Front end Ports	1 USB 3.0
19.	Speed of front end Ports in Gbps	5
20.	Type of Back end Ports	USB 3.0, eSATA, Gigabit LAN Ports
21.	Number of Back end Ports	5
22.	Speed of Back end ports in Gbps	1
23.	Number of Remote Replication Ports(FC/Ethernet)	2
24.	Number of Controllers/VSD/ Node available in the storage System on common back plane without using external switch/device OR without common backplane using switches	1

25.	RAID Level Support	0,1,5,6,10
26.	Active-Active Controllers Configured in HA	No
27.	Active Stand by Controller Configured in HA	No
28.	Cache Availability Type	NA
29.	Total Configurable Cache (GB)	8
30.	Wide Stripping or equivalent feature	Yes
31.	No of Snapshot Copies Per Volume	255
32.	License for Snapshot included	Yes
33.	Remote Replication	Yes
34.	Remote Replication license included	Yes
35.	Synchronized Replication Support	Yes
36.	Synchronous Replication license included	Yes
37.	Asynchronous Replication Support	Yes
38.	Asynchronous Replication license included	Yes
39.	3-DC Zero Data Loss Support	No
40.	Type of Data Compression	S/W Level
41.	Type of Data Deduplication	NA
42.	No Single point of Failure with Non-Disruptive replacement of Hardware	Yes
43.	The Storage provide Non-disruptive Firmware /Microcode upgrade	Yes
44.	Firmware upgrade without any controller reboot	Yes
45.	Encryption Mode	
46.	Encryption Required	Yes
47.	Type of Data at Rest Encryption	SW Based
48.	Storage management software for configuration and multi-pathing(part of the supply).	YES

49.	Multi-pathing and load balancing and fail over software (part of supply) with license for windows/Linux servers or shall support native multi pathing of OS.	YES
50.	Protocols Supported by the storage system from day one	CIFS/SMB, AFP 3.3, NFS v3, HTTP, HTTPS, FTP, SSH, WebDAV, iSCSI, rsync, TCP/IP, IPv4, IPv6, IEEE 802.3ad link aggregation LACP, SNMP v2/v3, NTP
51.	Operating System Platform and Clustering Supported by the Storage from day one	ReadyNAS OS 6.4.0 or later, Linux 4.x, Internal File System: BTRFS, External File System: EXT3, EXT4, NTFS, FAT32, HFS+
52.	Operating Temp Range	0°-40° C
53.	Storage Temp Range	0°-40° C
54.	Operating Humidity (Rh)	5-95%
55.	Storage Humidity (Rh)	5-95%
56.	BIS Registration under CRS of MeiTY	Yes
57.	If Yes, BIS Registration No under CRS of MeiTY otherwise indicate N/A	R-41086126
58.	BIS Registration Certificate shall be furnished when demanded by the buyer	Yes
59.	Storage System is compliant with IPv6	Yes
60.	Standby power consumption of System in Watt	90
61.	Scope of Supply	It includes installation, commissioning & integration together with all necessary software to make the system fully functional as intended.
62.	On Site OEM Warranty	2 Years

5.4.1.6. Ethernet Switch (ES)

All the subsystems of the electrical rig connected to each other via Gigabit Ethernet. Towards this, a 24 port Ethernet switch is used. The Ethernet switch should have the ability to add more switches as needed and to manage these stacked as a single switch.

The Specifications for the Ethernet Switch (ES) are given in Table-20.

Table 20 :Specification of Ethernet Switch (ES)

Sl No	Parameters	Values
1.	Type of Switch	Managed
2.	Technology	Non PoE
3.	Number of 1G Copper Ports	24
4.	Number of 10G Copper Ports	0
5.	No. of 1 G SFP Port (Uplink)	4
6.	No. of 10 G SFP+ Port (Uplink)	0
7.	Multi-Gigabit Support	Yes
8.	Redundant Power supply (from day one)	Available
9.	Console Port	Not Available
10.	Switching Capacity Non Blocking (Gbps)	128
11.	Throughput (MPPS)	95.8
12.	Operating System	available
13.	Dedicated Stacking Port/Slot (from day one)	available
14.	Dedicated Stacking Port/Slot (from day one)	available
15.	Basic Layer-3 Protocol	Static Routing DHCP Relay* IGMP Snooping V1/V2/V3 802.3ad LACP (Up to 14 aggregation groups, containing 8 ports per group) Spanning Tree STP/RSTP/MSTP BPDU Filtering/Guard TC/Root Protect Loopback detection 802.3x Flow Control LLDP, LLDP-MED
16.	Security Feature	Access Control Lists (ACLs) L2/L3/L4 (MAC, IPv4, IPv6, TCP, UDP) Protocol-based ACLs ACL over VLANs Dynamic ACLs IEEE 802.1x Radius Port Access Authentication 802.1x MAB Address Authentication Bypass (MAB) Port Security IP Source Guard DHCP Snooping Dynamic ARP Inspection MAC Filtering Port MAC Locking Private Edge VLAN Private VLANs
17.	Management Protocol	Password management Configurable Management VLAN Auto Install (BOOTP and DHCP options 66, 67, 150 and 55, 125) Admin access control via Radius and TACACS+ Industry standard CLI (ISCLI) CLI commands logged to a Syslog server Web based graphical user interface (GUI) Telnet IPv6 management Dual Software (firmware) image Dual Configuration file IS-CLI Scripting Port descriptions SNTP client over UDP port 123 XMODEM SNMP v1/v2 SNMP v3 with RMON 1,2,3,9 Port Mirroring SSH V1/V2 Syslog (RFC 3164)

18.	QoS	ACL Support (include L3 IP and L4 TCP/UDP) MAC ACL Support Inbound direction, Supports LAG interfaces, Multiple ACLs per interface (inbound), Mixed-type ACLs per interface (inbound), Mixed L2/IPv4 ACLs per interface (inbound) Supports LAG interfaces COS Mapping Config IP DSCP Mapping per interface Queue Parms configurable per interface Drop Parms configurable per interface Traffic Shaping (for whole egress interface) Minimum Bandwidth Weighted Deficit Round Robin (WDRR) Support 127 Maximum Queue Weight WRED Support
19.	Operating Temperature Range (Degree C)	-5 to 50
20.	Operating Humidity (RH)(%)	0 to 90
21.	IPv6 Ready from day one and dully certified	Yes
22.	PoE Power Budget (Watt)	0
23.	On Site OEM Warranty	2 years

5.4.1.7. Laser JET Printer

The Laser JET printer is used for printing the test reports and other documents related to rig. The printer shall be integrated on the rig network over Ethernet to get the prints from any test PC.

The specifications of the Laser JET Printer are given Table 21.

Table 21 : Specification for Printer

Sl NO	Parameters	values
1.	Print quality, black	Up to 600 x 600 dpi
2.	Print quality, colour	Up to 600 x 600 dpi
3.	Resolution technology	HP ImageREt 2400, Ultraprecise toner, HP edge-enhancement
4.	Language	Host-based
5.	Print speed: Letter-size and A4-size paper	Black and white: up to 12 pages per minute Color: up to 8 pages per minute
6.	First page out	As fast as 24 seconds As fast as 30 seconds
7.	Monthly duty cycle	Up to 25,000 sheets of paper
8.	Recommended monthly print volume	250 to 1000 sheets of paper
9.	Duplex printing (printing on both sides of paper)	Manual
10.	Power requirements	100 to 127 V (+/-10%) ,50 to 60 Hz (+/-2 Hz)

The schematic drawings showing the complete DAQC system should be included in the Technical bid.

5.4.2. Software Requirements of the Electrical Rig

The electrical rig shall be controlled and monitored by the application software of the DAQC system through the GUI commands received from the operator. The DAQC system software is also responsible for monitoring the various parameters of the rig LRUs and aircraft LRUs that are being tested in the rig. The DAQC system software generates the necessary control and status signals based on the measured parameters of the LRUs for proper and safe operation of the rig.

The software development process shall be as per the IMTAR-21, Version 2. (Ground/Test equipment software and Test Rig Software). The software certification shall follow the guidelines charted out as per IMTAR-21, Version 2., Subpart C6. The above reference clearly defines the software certification processes that are required for the Ground/Test equipment software and test rig software.

The functionalities of the DAQC System are:

i. Status Monitoring of the Rig and Aircraft LRUs.

ii. Generation of Control and Status signals based on the measured parameters of the Rig and Aircraft LRUs.

iii. Controlling the Rig subsystems based on the commands received from the operator under normal operating conditions.

iv. Controlling of the rig subsystems under failure conditions of Rig and Aircraft LRUs

v. Analysis and Display of the various parameters of the Rig and Aircraft LRUs.

vi. Recording of the parameter data of the Rig and Aircraft LRUs during the test for offline analysis.

vii. States management for internal calibration and normal operation of the rig.

viii. AC and DC Electrical Power sources and Load banks Management

ix. Test procedure setting and Test Report Generation

The functionalities of the electrical rig assigned to the DAQC system software are divided into five Computer Software Components Items (CSC) namely,

i. LCU CSC
ii. RCU CSC
iii. MCU CSC
iv. APMU CSC
v. NAS CSC

The brief functionalities of each CSC are given below. The details functionalities of each CSC with associated CSUs shall be given in the Software Requirement Specification (SRS) document of the electrical rig.

i. <u>LCU CSCI</u>

The following functionalities shall be implemented in Local Control Unit

LCU-CSC:

a) Monitoring the various parameters of the Rig subsystems.

b) Controlling the Rig subsystem like AC motor through VFD, AC loads, DC loads and, Gear and Generator Oil, Hydraulic Power Pack & Lubrication Systems based on operator commands and under faulty operating conditions.

c) Generation of control and status signals based on rig subsystem parameters monitored.

d) Sends the rig subsystem parameter information, control, fault and status signal information to RCU and NAS at regular intervals.

e) Sends the health status of rig subsystems and the individual modules in LCU to MCU at regular intervals. Aslo sends the critical parameter information of rig subsystems to MCU, as necessitated during development phase.

ii. <u>RCU CSC</u>

The following functionalities shall be implemented in Remote Control Unit RCU-CSC:

a) Sends the commands set by the operator to LCU for Controlling the Rig subsystems.

b) Displays the parameter information of rig subsystems received from the LCU on the GUI.

c) Sends the rig subsystems parameter information, control, fault and status signal information to MCU at regular intervals.

d) Sends the health status of RCUs to MCU at regular intervals.

iii. <u>MCU CSC</u>

The following functionalities shall be implemented in Master Control Unit MCU-CSC:

a) Resource allocation and management between LCU and RCU.

b) NTP time generation and distribution for synchronous operation of rig subsystems.

c) Health monitoring of the rig sunsystems

d) Analysis and Display of critical parameters of the rig subsystems after receiving from NAS.

e) Test report Generation.

iv. <u>APMU CSC</u>

The following functionalities shall be implemented in Aircraft Power Monitoring Unit APMU-CSC

a) Monitoring the various parameters of the aircraft LRUs.

b) Generation of control and status signals based on aircraft LRU parameters monitored.

c) Sends the aircraft LRU parameter information, control, fault and status signal information to LCU and NAS at regular intervals.

d) Sends the health status of aircraft LRUs and the individual modules in APMU to MCU at regular intervals.

v. <u>NAS CSC</u>

The functionalities assigned to NAS CSC are,

a) Storing the parameters information received from rig and aircraft LRUs.

b) Storing the command level transactions between rig and aircraft LRUs.
c) Sending the parameter and command information to MCU for analysis and display.

5.4.3. <u>Calibration of Data Acquisition and Control System</u>

To ensure measurement accuracy, all the modules in Data Acquisition and Control System requires periodic calibration. The Factors such as time in service, temperature, humidity, environmental exposure, and misuse of modules can affect the accuracy in the measurements. Calibration quantifies measurement uncertainty by comparing the actual measurements to a known value and adjusts the device's measurement capability until it meets the published specifications.

Two types of calibration are used for Data Acquisition and Control System of the electrical rig — **Internal Calibration and External Calibration**.

5.4.3.1 <u>Internal Calibration</u>

a. Internal Calibration can be done without any external connections. The main purpose of an internal calibration is to compensate for changes in the operating environment such as temperature variances and other factors that may affect the measurement modules making measurements. The internal calibration of data acquisition system shall be performed after initial installation, whenever the environmental conditions change, or when the user/certification agency/manufacturer specifies.

b. This procedure involves routing the known internal reference signals from an internal reference signals generation source/module to all channels of the data acquisition modules. During an internal calibration, each data acquisition module digitizes the high accuracy onboard source, compares the reading to the source's known value, and updates and stores new correction factors in the module's onboard calibration memory.

c. A software function calls from the RCU GUI typically initiates either an internal calibration or normal operation of data acquisition system without changing the connections, and a software-controlled internal calibration technique can lead to reduced test times and maximum availability of the rig.

d. The accuracy of Internal calibration depends on the accuracy of the on-board reference signals generation source. Hence the reference signals generation source shall be externally calibrated at regular and frequent intervals compared to data acquisition modules.

5.4.3.2. External Calibration

The data acquisition modules including measurement sensors shall be sent for external calibration to NABL accredited firms, that maintain traceable standards, at regular intervals. The intervals of calibration will be decided based on the detailed design and specifications of the individual modules. During external calibration, the device's measurement capabilities shall be compared to its published specifications to quantify how well the device is measuring. If the device is not measuring within specs, new calibration constants and correction factors shall be updated in the module's onboard calibration memory. To maintain traceability, all verifications and adjustments should be performed with standards that are traceable to national or international standards.

5.5 Cable Looms with Connectors as per Aircraft Standards

The EPGS Test Rig will be used for testing the Electrical System LRUs in integrated modes which includes power generation, conversion and distribution. The cable harness along with connectors for electrical LRUs i.e. from generation to conversion to distribution and finally to load banks are in the scope of Industry Partner.

The cable loom for Aircraft LRUs shall be designed and fabricated as per the interface details given in the Table -22 and 23 below. Aero-grade cables shall be used for connecting A/c LRUs with the test rig. Length of cable loom should be kept as actual to the Aircraft. The part numbers of standard Aero grade cables are given in Table- 24.

Industry Partner shall follow the airborne standard for manufacturing the Cable looms for Aircraft LRUs and shall submit the relevant process documents during inspection by DGAQA/CEMILAC.

Table 22 : Interface details of AC Electrical LRUs

SL NO	FROM CONNECTOR	FROM LRU NAME	TO CONNECTOR	TO LRU NAME	LENGTH BETWEEN LRU	GUAGE SIZE	NO. OF CABLES
1	9X		3Xc(J3) (D38999/26WF35PN)		10300	22 AWG	1
2	3Xc (D38999/26WF35PN)		72Tb		7820	22 AWG	7
3	3Xc (D38999/26WF35PN)		GND		1400	22 AWG	2
4	3Xb(J2) (97B4106F2222P)		7Pa (D38999/26FE06SN)		2200	8/12 AWG	4
5	3Xa(J1) (97B4106F32B13P)		6P		1100	10 AWG	4
6	1Xa (Terminal block)		TERMINAL BLOCK-2		4500	2 AWG	1
7	4Xb (D38999/26FC98PN)		SGRP		3000	12 AWG	1
8	2Xa(J1) (MS27467T23B53S)		1Xb(J2) (MS27467E15B97S)		5900	20 AWG 22 AWG	5 2
9	2Xa(J1) (MS27467T23B53S)		1Xa (MS27467E13B98S)		4500	20 AWG 22 AWG	4 4
10	2Xa(J1) (MS27467T23B53S)		9X		8600	22 AWG	1
11	2Xa(J1) (MS27467T23B53S)		3Xc (D38999/26WF35PN)		3300	20 AWG 22 AWG	2 7
12	5Xa (D38999/26FE26SN)		4Xd(C4) (D38999/26FA98SN)		2250	22 AWG	2
13	5Xa (D38999/26FE26SN)		4Xc(C3) (M83723/95K8986)		2250	22 AWG	2
14	5Xa (D38999/26FE26SN)		4Xb(C2) (D38999/26FC98PN)		2250	20 AWG	5
15	5Xa		40Da		2980	22 AWG	1

	(D38999/26FE26SN)		(D38999/26KA35SN)				
16	5Xa (D38999/26FE26SN)		3Xc (D38999/26WF35PN)		1400	22 AWG	5
17	5Xa (D38999/26FE26SN)		72TB		8800	22 AWG	3
18	6X (MS90362)		TERMINAL BLOCK-1		2550	2 AWG	4
19	7Xa (D38999/26FC35SN)		3Xc(J3) (D38999/26WF35PN)		3600	22 AWG	1
20	7Xa (D38999/26FC35SN)		5α		1100	22 AWG	1
21	7Xa (D38999/26FC35SN)		37P		2550	22 AWG	1
22	40Da (D38999/26KA35SN)		3Xc(J3)		3030	22 AWG	1

Table 23 : Interface details of DC Power systems

SL NO	FROM CONNECTOR	FROM LRU NAME	TO CONNECTOR	TO LRU NAME	LENGTH BETWEEN LRU	GAUGE SIZE	NO. OF CABLES
1	15Pb (MS3126F14-8S)		72TE (NSXN3T369S00)		7600	22 AWG	5
2	3Pa (D38999/26FC98PN)		5Pa (D38999/26MC98SN)		5570	20 AWG	1
3	3Pa (D38999/26FC98PN)		5Pb (D38999/26ME26SN)		5570	20 AWG	2
4	72TE (NSXN3T369S00)		5Pb (D38999/26ME26SN)		7400	22 AWG	2
5	TERMINAL BLOCK-8P		51P (MS3506)		2470	0	0
6	8Pa (D38999/26WH54PN)		10P (210PS01B1A)		2470	20 AWG	1
7	8Pb		1Wb		10360	22 AWG	3

	(D38999/26WG35PN)		(D38999/26FF35SN(REF))				
8	8Pb (D38999/26WG35PN)		72TE (NSXN3T369S00)		8100	22 AWG	19
9	8Pb (D38999/26WG35PN)		5Pa (D38999/26MC98SN)		2270	22 AWG 20 AWG	2 2
10	8Pb (D38999/26WG35PN)		72TE (NSXN3T369S00)		8100	22 AWG	6
11	TERMINAL BLOCK-8P		TERMINAL BLOCK-3P		6820	0	0
12	TERMINAL BLOCK-8P		15Pa (MS25182-2)		5020	0	0
13	TERMINAL BLOCK-8P		TERMINAL BLOCK-6P		1050	0	0
14	TERMINAL BLOCK-8P		TERMINAL BLOCK-7P		2220	0	0
15	8Pb ((D38999/26WG35PN))		6Pb (D38999/26FB35SN)		1050	22 AWG	1
16	8Pb (D38999/26WG35PN)		7Pb (D38999/26FB35SN)		2220	22 AWG	1
17	5Pb (D38999/26ME26PN)		1306DEL_M		7800	22 AWG	1
18	TERMINAL BLOCK-6P		SGRP		3000	12 AWG	1
19	TERMINAL BLOCK-7P		SGRP		3000	12 AWG	1
20	15Pa (MS25182-2)		SGRP			16 AWG	1
21	TERMINAL BLOCK-3P		SGRP			12 AWG	1

Table 24: Standard Aero grade Cables/wires part number

SL NO	COMPONENT TYPE	PART NUMBER	REMARKS
1	WIRE	M22759/86-12-9	12 AWG
2	WIRE	M22759/86-20-10	20 AWG
3	WIRE	M22759/86-22-11	22 AWG
4	CABLE	M27500A-01-WJ-2-U06	0 AWG T/P
5	CABLE	M27500A-02-WJ-2-U06	00 AWG T/P
6	CABLE	M27500A-10-WJ-4-S06	10 AWG T/Q SH
7	CABLE	M27500A-12-WP-1-S24	12 AWG S/C SH-LIGHT WEIGHT

8	CABLE	M27500A-12-WP-2-U24	12 AWG T/P -LIGHT WEIGHT
9	CABLE	M27500A-12-WP-4-S24	12 AWG T/Q SH-LIGHT WEIGHT
10	CABLE	M27500A-16-WP-1-S24	16 AWG S/C SH-LIGHT WEIGHT
11	CABLE	M27500A-16-WP-2-U24	16 AWG T/P -LIGHT WEIGHT
12	CABLE	M27500A-16-WP-4-S24	16 AWG T/P SH-LIGHT WEIGHT
13	CABLE	M27500A-20-WP-2-S24	20 AWG T/P-LIGHT WEIGHT
14	CABLE	M27500A-20-WP-2-U27	20 AWG T/P SH-LIGHT WEIGHT
15	CABLE	M27500A-20-WP-3-S24	20 AWG T/T SH-LIGHT WEIGHT
16	CABLE	M27500A-20-WP-4-S24	20 AWG T/Q SH-LIGHT WEIGHT
17	CABLE	M27500A-20-WP-4-U24	20 AWG T/Q-LIGHT WEIGHT
18	CABLE	M27500A-22-WP-1-S24	22 AWG S/C SH-LIGHT WEIGHT
19	CABLE	M27500A-22-WP-2-S24	22 AWG T/P SH-LIGHT WEIGHT
20	CABLE	M27500A-22-WP-2-U24	22 AWG T/P-LIGHT WEIGHT
21	CABLE	M27500A-22-WP-4-S24	22 AWG T/Q SH-LIGHT WEIGHT
22	CABLE	M27500A-22-WP-4-U24	22 AWG T/Q-LIGHT WEIGHT
23	CABLE	M27500A-2-WJ-4-S24	2 AWG T/Q SH
24	CABLE	M27500A-4-WJ-2-U24	4 AWG T/P
25	CABLE	M27500A-8-WJ-2-U24	8 AWG T/P
26	CABLE	M27500A-8-WJ-4-S24	8 AWG T/Q SH

5.6 Mounting Arrangements, Power Panels and Racks for LRUs

The mounting and installation of AC and DC generators and 3 Nos of HMDGs along with their GCUs shall be developed as per drawings provided by CABS, are in the scope of Industry Partner.

Drawing pertaining to the IDG mounting profile and the gearbox interface adaptor profile will be provided during the development stage. After analysis of the mounting profile of both the IDG and gearbox, suitable adaptor shall be designed and manufactured. The adaptor will facilitate the mounting of the IDG to Gearbox for both 40/50kVA and 5KW IDGs.

Quill shaft, featuring splines, shall also be manufactured. The shaft is instrumental in transferring motion from the gearbox to the IDG. Detailed drawing pertaining to the gearbox’s internal spline and external spline of the IDG will be shared during the development stages to ensure precise manufacturing and integration.

Reports to be provided during fabrication.

1. Raw material test report
2. Dimensional report
3. Heat treatment report
4. Hardness test report
5. Magnetic particle test report
6. Balancing report
7. Plating/painting report
8. Respective IS/EN/ISO/ANSI/ASME standard document

The test bench for 3 nos of HMDGs with sensors like pressure, temperature, flow rate etc. and piping are in the scope of Industry partner. All the port size of individual HMDGs are tabulated in the below table.

Table 25 : Hydraulic Port Connectivity Details for HMDGs

400W HMDG 2Nos (Left and Right)		
Port	**Size**	**Standard**
Inlet, High Pressure	MJ12X1.25	ISO 7321-6 type-1
Outlet, Low Pressure	MJ14X1.5	ISO 7320-8
Seep Drain		ISO 7320-6
5kVA AC HMDG		
Inlet, High Pressure	MJ16X1.5	ISO 7320-10
Outlet, Low Pressure	MJ18X1.5	ISO 7320-12
Seep Drain	MJ8X1.1	ISO 7320-4
Case Flow Return	MJ12X1.25	ISO 7320-6

5.6.1. Switches and Breakers

The control panels, switchgears, junctions, breakout etc. between electrical LRUs wherever required are to be provided by the Industry Partner. These switchgears shall be of Aero grade. The list of some switches and circuit breakers are mentioned in the Table- 25 below.

All the indicators, controls and measurement shall be provisioned on the Panel wherever required.

Table 26 : List of switches and Breakers

Sl. No.	Type	Part No.	Function	Quantity
Switches				
1	Pressure Switch	211C223-490	Start command switch	2
2	8PDT Switch	940UN01A4BE5P	AC power reset switch	2
3	SPDT Switch	210PS01B1A	Reset switch	2
4	8PDT Switch (RH)	940UN01A4BE5P	Reset switch	2
5	DPDT Switch	MS27785-23	Switch	2
6	4PDT Switch	975UN01B4AA6P	Battery switch	2
7	SPDT Switch	210PS01B1A	Reset switch	2
8	DPDT Switch	915UN01B4AL5P	Switch	2
9	Reset Switch	584H91A1B4C1D1G28L564N1(H)	Reset switch	2
10	SPDT Switch	MS27785-23	Switch	3
12	Switch (With AUX Contacts)	MS14154-2.5V	Switch	3
13	Relay	M210D4NCER	Relay switch	6
14	DC contactor	LRE-M306-1	DC contactor	2
Circuit Breakers				
1	CB (1A, 2-Pole)	MS3320-E1V	To provide protection for 28V	3
2	CB (2.5A, 2-Pole)	MS3320-E2.5V	To read battery status and provide protection for 28V	2
3	AC-CB (1A, 2-Pole)	-	To provide protection for 28V	2
4	DC-CB (2.5A, 2-Pole)	-	To provide protection for 28V	2
5	DC-CB (7.5A, 2-Pole)	-	To provide protection for 28V	3

6	CB (25A, 2-Pole)	-	To provide protection for 28V	3
7	CB	-	To provide protection for 28V	2

5.6.2. Racks for Aircraft LRUs

All the equipment other than the generators of the test rig shall be Placed/Installed in the racks with easy accessibility and cable routing. For this purpose, three numbers of Multi-tier open racks for keeping and installing A/c Electrical LRUs except Alternators/generators with proper connection and routing shall be developed.

The electrical LRUs of the aircraft may be installed on vertical (preferably) or horizontal rack. The installation rack shall be designed in such a way that the accessibility for installation and removal of LRUs and its associated connections/wiring is carried by the technicians at ease (with minimum effort). The general specification of rack is given in Table-26 below.

Table 27 :Specification for Racks

Parameter	**Constraints / Description**
Rack Quantity	03 Nos.
Technical Specification for each floor mount enclosure	
Rack Type	Floor Mount Steel Enclosure
Rack Standard	Conforms to DIN 41494 or equivalent standard
Construction	CKD (Completely Knocked Down)
Basic Frame	Steel four pillars with steel end frame
Front Door	Lockable toughened glass / perforate, single door
Rear Door	Lockable vented / perforated single / dual with 63% perforation, dual door
Equipment mounting	DIN standard 10mm square slots
19” mounting angle	Formed steel 2mm thick fully accessible, L Section with 10mm square slots at both edges.
Standard Finish	Powder coating
Standard Color	RAL 90XX (Black)
Top Tray	4nos of fans with exhaust vents and cable entry cut-outs with rubber grommet.
Bottom Tray	Cable entry cut-outs with rubber grommet.
Standard Mounting	Caster Wheels (2 front wheels with brake and rear wheel without brake) with levelers / base frame
Static Load	≥ 300 kgs (Recommended to distribute the load along with U space in the rack)
Total Height	1798 mm (36U usable)
Total Width	600 mm (19” usable)
Total Depth	1000 mm (920 mm usable)
Cable organizer	Horizontal 32U Cable organizer front and back
Equipment Support Angles	06 Nos. Cantilever Sliding Shelves and support plates, 100kg loading capacity
Cantilever Sliding Tray	02 Nos.
Side Panel	Slam Latch Panels with vents
Door Lock	Single point lock with swing handle
Storage Space	02 nos. >=4U height and >=700mm depth at bottom, Each storage shall have locking mechanism with keys, Toppling to be avoided.
AC mains power distribution channels	1) Vertical Rackmount power distribution unit indian round pin (Type D) 12nos., 5A/15A, 230V AC with input CB and Fuse. 2) Vertical Rackmount power distribution unit 12nos., IEC C13 type 10Amps 230Volt AC with input CB and Fuse. 3) 4-Core, 10 sq-mm power cable 40 meters. 4) Rack ground kit (Copper Rod with brass nut-bolts 06 Nos.)
Blanking panels	19”, 10nos. 1U height and 06nos. 2U height
Accessories	Nut-bolts for equipment fixing in rack to be supplied, Rack Grounding kit, Copper strip of 10mter (20mm X 2mm), Circuit breaker 3-pole, 63Amps-04nos along with mount arrangements in rack.

Other requirements	Rack shall be topple proof and weight balancing to be done. Complete rack systems shall be protected against any lizard/reptiles entry.

5.7 Timelines for EPGS Test Rig

The Project execution plan:

Project Stage	**Description**	**PDC**
1.	Placement of PO by CABS	T_0
2.	Submission of High Level Documentation (HLD) and Implementation Plan AGEST Facility	T_0 + 1month
3.	CDR	T_0 + 2 months
4.	Preparation of all the documents and obtaining approval from CEMILAC/DGAQA	T_0 + 5 months
5.	Readiness of the Hardware of the Rig at Bidder's Premises	T_0 + 6months
6.	Finalization of FAT & ATP Documents	T_0 + 7months
7.	FAT at Bidder's Premises	T_0 + 8months
8.	Completion of installation of the rig at CABS	T_0 + 10months
9.	Rig Acceptance with Certification	T_0 + 12 months
10.	Testing of Aircraft Electrical LRUs in EPGS Rig (Manpower for Operation, Maintenance, software development for monitoring and testing of Aircraft electrical LRUs)	T_0 + 24Months (1 years from the date of Rig Acceptance)

5.8 <u>Man Power for EPGS Test Rig</u>

Considering the Time line of the project minimum of four numbers of manpower shall be deputed to CABS for the period of 24 months. Man power consist of one mechanical, one Electrical and two Computer background.

After commissioning and certification of the rig, the manpower shall be utilized for operation and maintenance of the full rig and they are responsible for the testing activities of electrical LRUs of aircraft and preparation of test reports. Their primary role is to ensure that the EPGS rig is functioning properly and any issues that arises are resolved as quickly as possible.

The personnel deputed by the Industry Partner for EPGS rig should have experience in operating and maintenance of similar machines/equipment stated above.

Two personnel from computer background should have experience in LabVIEW coding, software documentation preparation etc.

The technical manpower shall have following skill sets and qualification:

Sl No	Academic Qualification	Experience & Skill Set	Minimum Experience	No. of Person
1	BE/B.Tech (Electronics/Computer Engg /CS/IT)	Skills: LabVIEW coding, VC/C++, GUI development, MySQL Database Mgmt. Software Documentation Testing Skills: Software Integration & Testing	3 Years	02
2	Diploma/BE/B.Tech (Mechanical/Aerospace / Hydraulic Technology)	Hydraulic System assembly, installation, testing, troubleshooting, operation and maintenance. Knowledge of hydraulic safety and fluid handling, mechanical drawings in CAD software	3 Years	01
3	Diploma/BE/B.Tech (Electrical/EE/ECE)	Harness routing, testing of electrical system, troubleshooting, operation and maintenance of electrical system in electrical rig, knowledge of aviation wiring standards. Electrical documentation	3 Years	01

Manpower shall report to CABS on all working days and shall be adhering to CABS safety and security rules and regulations. The manpower shall be in the payroll of the industry partner and the industry partner shall be responsible for any issue raised related to manpower.

Industry Partner should ensure the safety of their manpower when they are working on site. It is the responsibility of Industry Partner to provide necessary personal safety equipment to their manpower deputed to CABS.

Regular inspection and preventive maintenance should be conducted by the manpower to ensure the optimal functioning of the equipment and repair to be carried out if required.

6. General Aspects of the EPGS Rig - Integration, Assembly, Accessories, Panels, Grounding, Wiring etc.

Industry Partner will ensure that all the items including the items provided by CABS as listed in Table-1 are integrated and tested as a complete test system. The details of Accessories required, Panels, Wiring and other general aspects of Integration are given in below section.

6.1 Build Quality, Reliability, Calibration, Software

a. The construction of the Test Rig shall be of rugged construction with a life of over 30 years. The design life of the rig sub systems shall not be less than 30,000 hrs of operation. This works out to 8 hrs of operation per day for 260 working days per year. In case of failure, the failed components of the Rig should be replacable within 24 hours except for the major components like Motor, Gear box, equipment in Hydraulic Power Pack.

b. Modularity in the design of the rig should be ensured so that any faulty component can be easily repaired or replaced.

c. The rig will have to work in the hot tropical dry conditions, hot & humid sea level condition. Care shall be exercised while selecting the materials to comply with fault free operation of the equipment under the above climatic conditions. The electrical drive and controls shall be of the latest industry standards

d. Importance shall be given to the aesthetics and the ergonomic aspects shall be considered while developing the rig. There shall be sufficient space for accessing each equipment/component/sensor for its visual inspection and replacement.

e. Noise level of Hydraulic Power Pack should be less than 88 dB when all the three HMDGs are operating together.

f. Each item of the rig must be labeled as to its identification, function, or operating limitations, or any applicable combination of these factors. Suitable part numbers with future revision requirement to be mentioned in the label. All components/fabricated parts/units/systems and console shall be numbered as per part numbering scheme to be defined by CABS

g. The integrated system should be tested with the system operating, to include connected displays, sensors, actuators, and other equipment. Place the system in various operating modes and test sequence to ensure the integrated system is tested when operating to its maximum load.
h. Expected Reliability of the components should be mentioned and system reliability calculations should be provided.
i. It should be ensured that the calibration of all the components and the system should be valid up to the warranty period.
j. All Hardware and Software should be protected against Obsolescence
k. Industry Partner shall submit the technical documents provided by the OEMs of all machines, subsystems, Items / components / materials, including Drivers/BSP's/Libraries/ Programming Manuals etc and traceability certificate of the items.
l. The Industry Partner developed software and also the COTS software being used such as OS, Application specific software, etc., will be delivered to CABS with perpetual original license. Also original media to be delivered.

6.2 Grounding and Bonding

a. The EPGS rig will have provisions for two grounds, namely Rig Ground and Aircraft Ground. All the Rig sub systems will be connected to the Rig ground which is same as the AGEST facility supply ground. The chassis of all the rig LRUs will be connected to this rig ground for personnel safety purposes. All the Aircraft LRUs including the electrical load banks will have a separate ground which will be isolated from the Rig ground to simulate the conditions of aircraft in flight.

b. Dedicated neutral earthing shall be provided for aircraft power generators and provision shall be made for creating the floating neutral of aircraft generators. Separate copper strip/bus-bar shall be routed for connecting all aircraft LRUs/sub-systems to aircraft generator neutral. This aircraft generator neutral shall be used to create a common ground reference plane similar to aircraft ground plane. Aircraft generator neutral copper strip/bus-bar shall be separated from the rig grounding by using insulators. Each aircraft LRU shall have chassis ground point and it shall be bonded to this isolated copper strip/bus-bar. To create the common ground reference system, all the LRUs/sub-systems chassis is bonded to ground reference system via suitable thickness copper wire or copper braid. Bonding resistance is to be kept below the 2.5 mΩ. If interconnecting cables are having shielding, it shall be bonded/terminated to interfacing LRU/sub-system chassis. The bonding of aircraft LRUs will be as per MIL-B-5087. The copper strip/bus-bar shall handle the full rated current of the generators.

c. In essence, the rig shall be capable of replicating the aircraft grounding scheme for integrated testing of the complete electrical system.

6.3 Panels, Wiring and Accessories in EPGS Test Rig

a. An incoming Panel of 1000 kVA rating is installed by CABS in the rig area. The independent distribution panels with appropriate switch gear for Hydraulic Power Pack, AC and DC load banks, Fuel Circulation System, UPS and all other rig LRUs supplied by the Industry Partner including the power cabling from incoming panel to these distribution panels shall be done by the Industry Partner.

b. The wires will be as per IEC STD 60502. All the cables and wires shall be appropriately de-rated with respect to their current-carrying capacity (AWG size) and voltage rating. The current rating of wire must not be less than the current-rating of the circuit breaker used in the line. The Wire and Cable Harness assemblies will meet the requirements of IPC standard IPC/WHMA-A-620.

c. Where the circuits work at different voltages, the conductors shall be separated by suitable barriers (separate conduit) or all the wires shall be rated for the highest voltage to which the conductor may be subjected.

d. All the wires used and other associated components for wiring, e.g., terminal blocks, circuit breakers, contactors, relays, DIN rails, connector blocks, screws, plugs, sockets etc. shall be of industrial or higher grade.

e. Analog, digital, electrical load and power supply cables shall be bunched separately and routed to maintain separation between power and signal cables. Cable or wire markers shall be used for identification purpose, especially in multiway cable or wire harnesses. Both ends shall be marked with the same number. The markers shall be placed such that number is read from the joint/termination. Number of digit shall be same for each wire marker. Terminals and terminal blocks shall be clearly marked and identified to correspond to the markings in the wiring diagram.

f. All the Cables will be routed in the Cable trenches provided in the rig floor area. Mains and other power wiring shall be formed into a separate cable form from all other wires.

g. There shall be enough wire length at the ends so that a strain relief loop can be provided if needed. There shall be no strain on any wire at junctions or breakouts. When laying the cable form into the equipment, the following shall be avoided: heat dissipating parts, sharp edges on the hardware, moving parts, covering parts that need access.

h. The cables fabricated for the Electrical test rig is to be qualified for continuity checks and Megger checks (>20 MΩ at 500V DC).

i. All cables used for special purposes may be accepted provided they are manufactured and tested in accordance with the relevant acceptable standards.

j. When supply is taken from DC Power supplies, polarized connectors or protective devices shall be provided against damage due to accidental connection of reverse polarity.

7. Safety Requirements

a. All the equipment in the rig shall be incorporated with safety features required to safeguard personnel and equipment during operation, maintenance, repair and calibration and Industry safety guide lines to be followed for safety. Safety features will be implemented in Software to operate in safe mode whenever a parameter exceeds the set threshold values.

b. CABS shall perform a Risk Assessment on the Hydraulic Power pack system and Fuel circulation system design. Safety requirements shall be discussed at PDR. The Hydraulic Power pack system and Fuel circulation system design will be reviewed at CDR and required changes are to be incorporated to meet safety requirements. The Industry Partner shall agree to change the hardware/software design for improved safety in response to CABS review. A final safety review will be conducted to verify if changes were addressed properly. Final acceptance shall not occur until all safety related changes identified at CDR have been implemented to CABS satisfaction.

c. Emergency Stop pushbuttons shall be provided throughout the system, particularly at Hydraulic Power pack system and Oil Supply. Emergency Stop button for Hydraulic power pack shall be integrated with the existing safety system of the Rig in CABS. Provisions shall be included in the equipment design to allow for installation of additional devices (additional Emergency Stop pushbuttons, door switches, etc.) A Safety Controller shall be used to manage the Emergency Stop and other Safety inputs.

d. Each of individual module and the total integrated system will incorporate all the necessary safeties to ensure the Test operator safety, UUT safety and Test stand hardware safety. This includes but not limited to Over speed, UUT mounting detection, UUT signature identification, Over temp for Oil reservoir, UUT inlet oil, UUT outlet oil, UUT oil delta, UUT temp rise rate, UUT vibration, Oil level, Water pressure, Shop Air pressure, UUT oil scavenge Pressure/Flow switch, UUT /IDG charge pressure, IDG case pressure, IDG oil flow, Generator oil flow, load bank air flow, load bank over temp, load bank under frequency, over voltage, over frequency and all other parameters essential for safety.

e. Personnel protection from hazards will be provided by use of guards, enclosures or shields. Metal shields shall be provided around the rotating parts of the drive stands. Applicable PPEs for at least four personnel should be provided by the Industry Partner.

f. Suitable warning signs indicating Hot surfaces, Rotating parts, Electric shock hazards, etc should be installed at all appropriate places in the rig.

8. Maintainability and Serviceability of the EPGS Test Rig:

a. Periodic maintenance of Hydraulic Power Pack, Fuel Circulation system, Motor-Gear drives and other interfaces will be ensured for uninterrupted availability of the Rig.
b. Necessary fault repair/product support activities including supply of spares, consumables, training of personnel for operation and maintenance are critical requirements which has to provisioned by the Industry Partner. The time required for repairing/replacing minor components in the rig should be less than 24 hrs. Recommended spare parts list for quick replacement towards maintaining the continuity of operation shall be submitted.
c. The rig shall be so designed that it should be possible to upgrade it with minimum effort and down time. In case the facility is used for other programmes the changes are to be made as minimum and modularity to be ensured for the respective programme. Any such change will be taken up with the approval of IQA and DGAQA.
d. The Industry Partner shall provide training to around ten personnel designated by CABS divided into three phases: Theory, Operation and Maintenance of both hardware and software aspects over a minimum period of one week exclusively dedicated for this after commissioning of the rig at CABS. Well designed and detailed training manuals and material (Slides, Videos, Notes) should be provided for future training requirements. Operational, maintenance and software manual shall be provided in hard copy.
e. One set of standard tools, Safety kits, Mating Connectors, data and signal converters and additional set of necessary power & data cables shall be provided by the contractor which is required to maintain and operate the Rig.

f. Consumables (like coolant liquids, greaseeing compounds etc) required for the operation and maintenance of the facility upto the warranty period should be provided by the Industry Partner.

8.1. Maintainability Requirements

a. As far as possible conventional available tools shall be used for maintenance activities.
b. The electrical system design shall consider preventive maintenance program as part of design & development activity wherever applicable and should be well-documented.
c. A sufficient number of accessible test points shall be provided so it is not necessary to remove assemblies or subassemblies to accomplish testing.
d. Performance monitoring and fault detection and isolation at a point shall be available from a single sensor.
e. The system shall provide an alert signal as well as alarm in case of failure of a vital part; and also provide information in case of a failure of a non-vital component or element, that requires certain corrective action.
f. The system shall also provide data logging of parameters of electrical rig for rectification, modification and further analysis of the system/sub-system.
g. The calibration of system should always be checked. Periodicity and detailed calibration procedure should be provided by the Industry Partner.
h. The Industry partner shall support the software for period of 2 years which shall include Upgrade and update to the software in terms of new test and measurement functionalities for the new electrical aircraft LRUs, fixing of bugs etc.
i. The Industry Partner shall ensure the software development is carried out at CABS premises.
j. The industry partner shall create proper software configuration control methodology and establish the same at CABS for software maintenance and upgradation
k. All development and application software must be delivered in original CD/pen drive along with source code.
l. Proper software version maintenance must be defined in documentation.
m. The Industry partner shall provide the licenses for all the required development platform on which the entire software has been developed (eg. LABVIEW)
n. All the source code developed shall be the proprietary of the CABS/DRDO and as such shall be delivered to CABS.

9. Acceptance and Certification

a. This Chapter brings out the process that needs to be followed for the Acceptance and Certification of the EPGS Test Rig which will be used in the testing and validation of the aircraft LRUs. Certification is essential for facilitating in maintaining the airworthiness of the Air systems and Airborne stores.

9.1. Classification of Rig

EPGS Test Rig is considered as one of the TTGEs and for the purpose of Certification, it is classified as,

a. Based on the procurement and development process, EPGS is classified as TTGE to be developed and used for aircraft electrical system development activities.
b. Based on location/deployment environment, EPGS test rig is classified as Class-5 test rig used in permanent building and is categorized as Type-1/Type-2 test rig used during development phase.

9.2. General Description and Process for Certification

The detail requirement of the documents and the process for the certification are given below. The process flow for acceptance testing and certification of EPGS test rig for aircraft is shown in the form of flow chart in the Figure- 11 .

9.2.1. Flow Chart for Acceptance and Certification of EPGS

The acceptance of the EPGS Test Rig will be based on an Acceptance Test Plan and procedure which will be jointly agreed. The acceptance shall involve verification of equipment, calibration, functionality, software, test result etc.

The industry partner shall take complete responsibility to integrate and test the complete functionality of the EPGS Test Rig at CABS as part of ATP and Clearance of them. Industry Partner is also responsible in testing and report generation of Aircraft LRUs for 01 Years after certification of the rig.

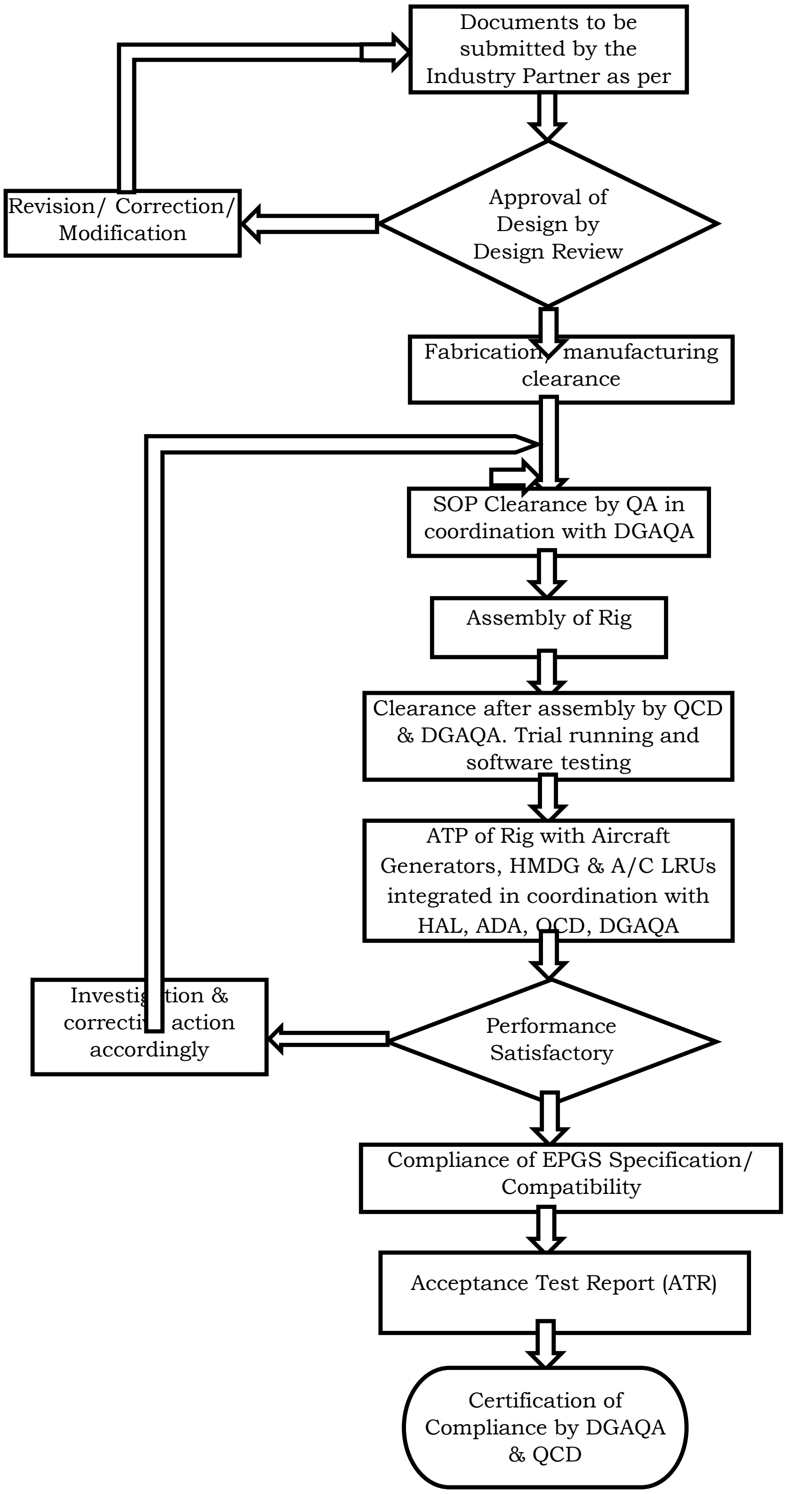


Figure 11 - : Flow Chart for the Process of Acceptance and Certification

9.2.2. Approval of System Requirement Description Document

a. Based on the top level requirement, System requirements have been prepared and presented in this document. As the requirements are generated at the beginning of the project, it need not fully describe the designer’s intention. Considering broad technical requirements and the finalized realization approach for the rig, the detailed Technical Specifications of the complete rig shall be prepared by the Industry Partner and it shall be reviewed and approved by CEMILAC/DGAQA. For the software portion of the rig, Software Requirements Specification (SRS) document shall be prepared by the Industry Partner and provided to CEMILAC for approval.

b. Technical requirements of the facility are only specified in this document. During Acceptance, if any clarification in specification is required, it shall be through change control process as per CABS requirement through ECN, ECP, etc.

9.3. Design Review

a. Post placement of order with the Industry Partner, design reviews (PDR and CDR) shall be conducted by the committee constituting members from CABS, ADA, DGAQA, HAL and RCMA / CEMILAC. PDR demonstrates that the preliminary design meets all the system requirements with acceptable risk and within the cost and schedule constraints. The Industry Partner will show that the correct design options have been selected, interfaces have been properly allocated, and verification methods have been described.

b. The detail design consists of refining the block diagram and drawing the schematics based on the technical specification, bill of material, and interface control documents (ICD). The block diagram, schematics and board layout will be reviewed with the Industry Partner to ensure that the hardware design meets the requirements and follows the electrical design standards.

9.4. Documentation

a. All the necessary documents will be prepared by the Industry Partner as per CABS requirements and Formats. The documents are to be vetted & coordinated by the CABS and submitted to DGAQA for document approval and acceptance of TTGE through respective QA/QC.

b. For certification of the EPGS Test Rig, following documents will be provided for Hardware and Software,

9.4.1. Hardware Documents

i. Technical Specification of the AGEST Facility (TS)
ii. Quality Assurance Plan
iii. Design documents (Electrical and Mechanical)
iv. SOP with Test Rig Drawings (Installation, Mechanical & Electrical) including MDI and BOM, Certificate of Conformance
v. Acceptance Test Procedure (ATP)
vi. Acceptance Test Report (ATR)
vii. User Manual
viii. Installation Manual
ix. Operating Manual
x. Maintenance manual
xi. Fault Analysis and Troubleshooting Manual
xii. Calibration requirements
xiii. List of Spare Parts.
xiv. List of Tools

9.4.2. Software Documents

i. Software Requirement Specification (SRS)
ii. Software Design Document shall define the software design flow, interface and control details etc (SDD)
iii. Software Quality Assurance Plan (SQA)
iv. Software Test Plan (STP)
v. Software Configuration Management Document
vi. Independent Verification and Validation (IV&V) Plan
vii. Software Test Report (STR)
viii. Version Definition Document (VDD)

9.5. Factory Acceptance Testing (FAT) or Site Acceptance Testing (SAT)

a. The Industry Partner shall perform FAT prior to shipment. The Industry Partner shall prepare and submit to CABS the plan for verifying the compliance with the specifications of the individual items of the rig. Compliance with the specifications shall be verified by one of the following methods, Similarity, Analysis, Inspection, Plan, Procedure and Design reviews and Testing.

9.6. Physical Inspection at CABS

a. KOP clearance will be taken from DGAQA through CABS QCD before assembly of rig. All manufactured equipment/LRUs required for rig assembly and their corresponding documents shall be cleared by DGAQA through CABS QCD.
b. During the physical Inspection the following aspects will be checked from the safety point of view and specifications of rig:
   i. Safety features like Personnel safety, Grounding, Thermal hazards,
   ii. Equipment safety, Electrical safety, etc.,
   iii. Good manufacturing practices in respect of Components, Parts, Connectors and other Materials.
   iv. Design and Construction should be as per class, performance, usage and other requirements mentioned in the detailed specifications of the equipment.
   v. Calibration details (with maintained validity) of the Test and Measurement equipment.

10. Assembly of Rig at CABS

a. The various sub-systems shall be integrated during this phase. The system integration includes both hardware and software integration. The rig shall run on trial to test the integrity of the system/sub-system. Tests are carried out in accordance with the test procedures defined and approved by DGAQA.

11. Acceptance Testing

a. Acceptance testing of the EPGS Test Rig, shall be done with the aircraft generators and HMDGs as per the approved Acceptance Testing Procedure (ATP) for the facility. CABS will provide AC and DC generators and HMDGs. Towards this, the Industry Partner shall prepare all the required documents and is solely responsible for the completion of ATP.
b. The Industry Partner shall submit the ATP and on approval of it by DGAQA, perform acceptance testing after installing the complete rig sub-system at CABS in the presence of DGAQA reps.
c. Based on approved Technical Specifications document, Drawings, MDI and BOM, Evaluation and demonstration shall be carried out in coordination with CABS & DGAQA. The facility shall be qualified in coordination with DGAQA as per approved ATP. The objective of acceptance testing is to ascertain the facility meeting the desired specifications of airborne electrical system, thereby mitigating the risk of defects or other inadequacies throughout the operational life of the Test Rig.
d. The EPGS Test Rig with aircraft generators integrated should be tested with the system operating, to include connected displays, sensors and other equipment in various operating modes and test sequences. It must be ensured that the integrated system is tested when operating to its maximum load.

e. ATP identifies the aspects of the Test Rig to be tested, the level of manning required to operate the Test Rig and the anticipated duration of testing. In ATP testing of all functionalities related to the rig to be mentioned. The ATP will mention the test schedule which contains the following:
    - Details of aircraft generators and HMDGs to be tested
    - Objective and aim of the test
    - Details of the test equipment used.
    - Calibration of test equipment and instruments and their records
    - Connectivity of the equipment with other components of the rig to be tested
    - Test Procedures/Test order
    - Measurements to be taken and instrumentation required
    - Pass / Fail criterion

f. After conducting ATP tests, the following should be ensured for necessary clearance:
    - Completion of Acceptance Test Report (ATR) and related certifications.
    - Calibration Instructions.
    - User Manual.
    - Limitations/Deviations/Amendments are authenticated and documented.

g. ATR will be generated which contains test details and the outcomes of test activity. The result of each test case should be highlighted by severity. The report indicates whether the rig is compliant and if it is not compliant, make recommendations for change.

h. The EPGS Test Rig will be certified by CEMILAC/DGAQA in coordination with internal CABS QCD. The Industry Partner shall be solely responsible for certification of the EPGS Test Rig with the CABS supplied generators, HMDGs and Aircraft LRUs.

12. Post Development Warranty and Support

A two-years post development support by Industry Partner shall include all Equipment, hardware and software. The post development support shall commence from the day of completing the delivery and installation of the EPGS rig.

During the support the Industry partner will be responsible:

a) For maintenance of the system including all the equipment for third party OEMs
b) Calibration of equipment as per OEM recommendations
c) Bug fixes on the software
d) Getting the software updated on the equipment from the OEMs
e) Development of additional packages for test and evaluation of electrical aircraft LRUs (Assume 5% to 10% additional development)
f) Optimization of existing processes and procedures
g) Improvement in HMI and reporting's etc.

13. Responsibilities of CABS

a. CABS shall provide required power source and site for installation
b. Input for technical specification for Electrical Rig and facilitation of approval process by CEMILAC/DGAQA/QCD
c. Participation in Acceptance tests of all hardware and software
d. Participation in testing of Electrical aircraft LRUs and report generation

14. Onsite Installation and Commissioning

14.1. The site preparation and associated electrical cable work is the responsibility of industry partner.

14.2. Extreme care shall be taken with regards to the existing infrastructure/AGEST rig. If any damage happens to anything during the site preparation, it has to be replaced by industry partner without any additional cost.

14.3. Epoxy flooring shall be taken care by the industry partner wherever it is required in the rig area, to create a good environment ambience.

14.4. The industry partner shall take care of tapping of electric power from CABS raw power panel and its distribution for EPGS rig.

14.5. All the electrical wiring, fitment, measuring equipment, hydraulic requirement to be catered as per the SOW. All the fitment inside the AGEST Facility to make the EPGS Test Rig functional will be carried out by the Industry Partner at CABS.

14.6. The Industry Partner shall install and demonstrate the functionality of the EPGS Test Rig subsystems along with the automated software at CABS

14.7. The man power, tools etc required for the job shall be arranged by the Industry Partner.

14.8. Any system level deficiency/inadequacy observed during the tests shall be addressed by the Industry Partner. Industry Partner shall absorb all cost and efforts associated with additional requirement for the EPGS rig, ensuring smooth operation and testing process of aircraft electrical system during warranty period.

14.9. The Industry Partner shall be responsible for the operation and maintenance of EPGS rig including the existing AGEST rig as EPGS rig is getting integrated with the existing rig.

14.10. Industry Partner shall provide a master list of parts used in the system

14.11. Industry Partner has to make minor software corrections during the warranty period

14.12. Industry Partner shall provide Labour support for the rig.

14.13. Training shall be given to familiarize the CABS team with the functionality of each sub-system, its usage, operation, setting of parameters etc. Training also shall be imparted to maintain all the equipment and subsystems up to the level of localizing the faulty modules/boards/ components and the use of software for troubleshooting and fault diagnosis.

15. References

Table 28 : References

Sl. No.	Standard/References	Title
1.	JSS 55555: 2000; Revision No. 2	Environmental Test Methods for Electronic and Electrical Equipment
2.	MIL-PRF-21480B	General Specification for Generator System, Electric Power, 400Hz, Alternating Current, Aircraft; performance specification
3.	MIL-STD-704D	Aircraft Electrical Power Characteristics; interface standard
4.	MIL-C-2212G	Contactors and Controllers, Electrical Motor AC or DC, and Associated Switching Devices
5.	MIL-W-16878	General Specification for Electrical Insulated Wire
6.	MIL-STD-7080	Selection and Installation of Aircraft Electric Equipment
7.	MIL-STD-1839C	Standard Practice for Calibration and Measurement Requirements

8.	MIL-STD-202G	Test Method Standard for Electronic and Electrical Component Parts
9.	MIL-HDBK-274A	Electrical Grounding for Aircraft Safety
10.	MIL-STD-461G	Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment
11.	MIL-STD-464C	Electromagnetic Environmental Effects Requirements for Systems
12.	MIL-HDBK-1760E	Aircraft/Store Electrical Interconnection System
13.	MIL-STD-882E	System Safety
14.	MIL-HDBK-217F	Reliability Prediction of Electronic Equipment
15.	MIL-HDBK-561C	Airworthiness Certification Criteria
16.	MIL-C-5015	Connectors, Electrical, Circular Threaded, AN Type, General Specification for
17.	MIL-B-5087	Bonding, Electrical, and Lightning Protection, for Aerospace Systems
18.	MIL-W-5088	Wiring, Aerospace Vehicles
19.	MIL-C-5809	Circuit Breakers, Trip-Free, Aircraft, General Specification for
20.	MIL-R-6106	Relays, Electromagnetic, General Specification for
21.	MIL-F-15160	Fuses: Instruments, Power, and Telephone
22.	MIL-E-25499	Electrical Systems, Aircraft, Design and Installation of, General Specification for
23.	MIL-E-87219	Electrical Power Systems, Aircraft
24.	MIL-C-83413	Connectors and Assemblies, Electrical, Aircraft Grounding, General Specification for
25.	MIL-DTL-38999L	Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect (Bayonet, Threaded, and Breech Coupling), Environment Resistant, Removal Crimp and Hermetic Solder Contacts, General Specification for
26.	MIL-L-23699	Military Specification - Lubricating Oil, Aircraft Turbine Engine
27.	MIL-PRF-7808	Performance Specification-Lubricating Oil, Aircraft Turbine Engine
28.	RTCA-DO-160E	Environmental Conditions and Test Procedures for Airborne Equipment
29.	RTCA-DO-178B/C	Software Considerations in Airborne Systems and Equipment Certification

30.	SIMOTICS M-1PH8 Motors	Configuration Manual for 1PH8 series Asynchronous Motors from M/s SIEMENS
31.	Component Maintenance Manual for IDG	Component Maintenance Manual for IDG (Part No. 752157B, C, and D) from M/s UTC Aerospace Systems
32.	IPC/WHMA-A-620	Requirements and Acceptance for Cable and Wire Harness Assemblies
33.	ADA-Document ADA/AWS/ELEC/AMCA/TR/2018	Technical Specification of Integrated Electrical Test Rig For Advanced Medium Combat Aircraft (Preliminary)
34.	CEMILAC Document No CEMILAC/6003, Issue No 1, DDPMAS-2002	Design, Development and Production of Military Aircraft and Airborne Stores.
35.	ADA-Document No ADA/QASEG/3734/Maint/TTGE/628/2015	Guideline for Certification of Tools, Test Equipment and Ground Equipment (TTGE)
36.	AQA Directive 02/01/2015 Issue-02 – January 2015).	Guidelines for Qualification Test Procedure & Acceptance Test Procedure of Ground / Test Equipment (JIGS) for Airborne Items (Electrical & Electronics)
37.	CABS QAP Document No CABS/QMS/GEN/SQAP/246/2018, May 2018	Standadization of Quality Assurance Plan
38.	ISO 1940-1, Second Edition 2003-08-15	Mechanical Vibration-Balance quality requirements for Rotors.

2.2 Special Terms and Conditions for the procurement of "Design, Development, Installation, Commissioning & Certification of Electrical Power Generation System (EPGS) Test Rig"

2.2.1. Documentation & Responsibilities:

2.2.1.1 In response to this RFP, the bidder shall submit the technical compliance matrix covering all the requirements specified (point-by-point), clearly stating how the technical specification requirement is being met or exceeding by the proposed design/equipment. Technical offer must be accompanied with product catalogues /data sheets, of individual components from respective OEM, additional explanation/analysis clarifying the offer, as supporting documents and the same shall be cross referred in the compliance matrix. Merely stating that the specification requirement is "complied" is not acceptable.

2.2.1.2 All the deliverables are interconnected and interdependent to realize the set objective of the Rig and therefore only single order will be placed on a single Industry Partner.

2.2.1.3. After the placement of Purchase Order, the Bidder shall submit the design documents of the Rig to CABS. PDR/CDR meetings shall be conducted, either at CABS or at Bidder's premises and the design of the Rig and its constituents& Data Acquisition and Control System shall be finalized mutually.

2.2.1.4. Bidder shall be responsible for the preparation of the technical documents like Technical Specification (TS), Software Requirement Specification (SRS), Master Drawing Index (MDI), Bill of Materials (BOM), etc. as listed in the specifications document and submit them to CABS. They shall be approved by CEMILAC/DGAQA. CABS will coordinate with these agencies. However, it is the Bidder's responsibility to submitting additional inputs/clarifications asked by CEMILAC/DGAQA and get the documents approved by them.

2.2.1.5. Quality Assurance directives issued by CEMILAC/DGAQA and CABS-QCD shall be followed for qualification and acceptance of ground based test system. Software Certification Plan for EPGS rig and existing system in CABS and its software shall be prepared by the Industry Partner. The plan shall detail the review process and stage of involvement of certification agencies. Software requirements, software design/development, preparation of software test plan and other software related documents shall be prepared by Industry Partner and approved by CEMILAC/DGAQA.

2.2.1.6. All the hardware and software documents shall be prepared as per the guidelines in IMTAR-21 document. The same shall be thoroughly scrutinized, verified, approved and certified by the Quality department of the Industry Partner prior to submitting to CABS.

2.2.1.7. FAT and/or Pre-Dispatch Inspection shall be conducted at the Bidder's premises for all the individual sub-systems.

2.2.1.8. Documents corresponding to the Factory Acceptance Test (FAT) procedure on-site and Acceptance Test Procedure (ATP) shall be prepared & submitted by the Bidder, which shall be scrutinized and approved by CEMILAC/DGAQA. The documents to be submitted to CABS at least 30 working days before the scheduled FAT.

2.2.1.9. Bidder shall be responsible to integrate and test the complete functionality of the EPGS Rig along with the existing AGEST facility of CABS.

2.2.1.10 Aircraft grade power and signal cable fabrication, routing, and installation as per the ICD (Table-22,23, and 24 of SOW) provided by CABS for aircraft electrical LRUs in the Racks and between rig, loads and aircraft LRUs shall be the responsibility of the Bidder. The testing of aircraft LRUs in standalone and integrated mode shall be under the scope of Bidder. Bidder shall deliver 2 sets of cable looms with connectors for aircraft electrical LRUs and for rig LRUs.

2.2.1.11 Bidder shall be responsible for addressing and fulfilling any minor additional requirements that may arise during the integration and testing of the aircraft electrical system.

2.2.1.12. Bidder shall be responsible for the operation and maintenance of EPGS rig including the existing AGEST rig.

2.2.1.13 Bidder shall absorb all costs and efforts associated with additional requirement for the EPGS rig, ensuring smooth operation and testing process of aircraft electrical system during warranty period.

2.2.2 <u>Other Terms and Conditions</u>

2.2.2.1 Components and materials traceability.

2.2.2.2. Provision for upgrading power ratings, software interfaces and additional cooling system unit.

2.2.2.3 Consider modular architecture to allow for future modification and expansion without major redesign.

2.2.2.4. Maximum usage of standard and proven COTS components (e.g. VFD, Motor, Load banks, Data Acquisition and Control system modules, etc.).

2.2.2.5 Bidder shall submit appropriate data sheet, CoC/Test reports and traceability certificate for these items.

2.2.2.6 Modularity in the design of the rig should be ensured so that any faulty component can be easily repaired or replaced.

2.2.2.7 Expected Reliability of the components should be mentioned.

2.2.2.8 All Hardware and Software should be protected against Obsolescence.

2.2.2.9 Bidder shall describe their after sale service/support strategy during and after Warranty period.

2.2.2.10 Bidder shall submit the technical documents provided by the OEMs of all machines, subsystems, Items / components / materials, including Drivers/BSP's/Libraries/ Programming Manuals etc.

2.2.2.11 When COTS items are used, Bidder shall provide the source of availability of these items.

2.2.2.12. Whenever COTS software is being used such as OS, Application specific software, etc; Bidder shall deliver such software with perpetual original license. Also original media to be delivered.

2.2.2.13 Bidder shall deliver/transfer the OEM's warranty as it is to CABS for all items he is procuring from 3rd party.

2.2.2.14 Bidder shall follow all aerospace processes during all phases of design, manufacturing, testing, certification etc.

2.2.2.15. Part numbering: All components/fabricated parts/units/systems and console shall be numbered as per part numbering scheme to be defined by CABS.

2.2.2.16 Bidder shall submit a Reliability Document for the Rig, as per the requirements of the Certification agencies.

2.2.3. <u>Design Clearance and Inspection</u>

2.2.3.1 The design clearance and Inspection agencies will be Center for Military Airworthiness and Certification (CEMILAC) and Directorate General of Aeronautical Quality Assurance (DGAQA).

2.2.4. <u>Warranty & Calibration</u>

2.2.4.1 Compulsory Comprehensive Warranty to be provided for 2 years from the date of Certification & Acceptance of the rig. Required spares and consumables to operate and maintain the facility during the Warranty period should be provided by the Bidder.

2.2.4.2 The Bidder should have dedicated hardware and software team with sufficient experience related to similar projects and provide onsite support on daily basis with the specified number of manpower for testing of aircraft electrical systems for 1 years after certification of the rig.

2.2.4.3 The spare list and maintenance /trouble shooting manual service bulletins should be supplied along with the unit.

2.2.4.4 The maximum permissible breakdown period of the facility during the warranty period shall be 7 days. If this limit is exceeded, PBG will be forfeited.

2.2.4.5 Bidder shall ensure that the calibration validity of the sensors and instruments is valid during the Warranty period. Bidder shall also indicate the calibration periodicity of the sensors and instruments.

2.2.4.6 To ensure the safety and security of the instruments/items sent out of the CABS for calibration or any other purpose, it is mandatory, the insurance will be taken by the industry partner in name of CABS.

2.2.4.7 A calibration philosophy for the Rig shall be prepared & submitted by the Bidder, which needs to be accepted by the Certification agencies.

2.2.5. CAMC

The Bidder shall explicitly mention in the bid their acceptance of providing CAMC (Comprehensive Annual Maintenance Contract) services for the complete EPGS Test Rig.

CABS may exercise the option for placement of AMC after warranty. The scope of work of AMC shall be as per the user manual supplied by the Bidder. The required spares shall be maintained by the Bidder to avoid delays in making the rig operable.

2.2.6 Cost Breakup:

2.2.6.1. The price bid furnished by the Contactor shall have the following major heads,

2.2.6.2 Design, Installation, Commissioning and Certification of EPGS rig + Applicable Taxes & Duties, if any

Sl No	Description	Unit/Quantity	Price in Figure (INR)	Price in words
1.	Design, Development, Installation, Commissioning, and Certification of EPGS Test Rig including 2 years of operation, Maintenance and Testing with manpower and warranty post commissioning	1 lot		
2	Taxes and Duties			
	Total	-		

2.2.7. Proposed Payment Terms:

Milestone payment will be made on completion of each phase and on production of invoice and completion certificate from user. The following Milestone payment will be made (As per deliverables, Specification & SOW)

Sl No	Milestone	Description (Milestone Wise)	% of Payment	Project Time Line(T0+Months)
1.	M-1	**Submission of High Level Documentation(HLD) and Implementation Plan in AGEST rig** Payment will be made within 30 days of Submission of Bills and confirmation of receipt and acceptance of High Level Documentations (HLD) & Implementation plan by user, whichever is later	10%	T0 + 01
2.	M-2	**Critical Design Review (CDR)** Payment will be made within 30 days of Submission of Bills and confirmation of receipt and acceptance of CDR document by user, whichever is later	20%	T0 + 02
3.	M-3	**FAT at Bidder's premises** Payment will be made within 30 days of Submission of Bills and confirmation of receipt and acceptance of FAT and ATP document by user, whichever is later	10%	T0 + 08

4.	M-4	**Installation, Integration & Commissioning of Rig at CABS** Payment within 30 days of submitting the bill and receipt of certificate of satisfactory completion of job from acceptance committee, whichever is later	20%	T0 + 10
5.	M-5	**ATP and Certification of Rig at CABS** Payment within 30 days of submitting the bill and receipt of certificate of satisfactory of ATP and certification of rig from acceptance committee, whichever is later.	20%	T0 + 12
6.	M-6	Manpower (4 Nos) for Operation, Maintenance and Testing of Aircraft electrical system in the rig after certification of the rig for 1 years	20%	T0 + 24

Note: T0: Purchase Order /Supply Order Date

2.2.8. <u>Delivery schedules</u>

2.2.8.1 The following delivery schedule is envisaged:

2.2.8.2 Placement of PO by CABS: The Bidder shall submit a Security Deposit as mandated during negotiations within 15 days from the receipt of the Acceptance of Tender (AOT). Date of issue of PO shall be To.

2.2.8.3 Submission of High Level Documentation (HLD) and Implementation Plan

in AGEST rig: To + 1 months

2.2.8.4 CDR: To + 2 months

2.2.8.5 Preparation of all the documents and obtaining approval from CEMILAC / DGAQA: T0 + 5 months

2.2.8.6 Readiness of the Hardware of the Rig at Bidder's Premises: To + 6 months

2.2.8.7 Finalization of FAT & ATP Documents: To + 7 months

2.2.8.8 FAT at Bidder's premises: To + 8 months

2.2.8.9 Completion of installation of the Rig components at CABS: To + 10 months.

2.2.8.10 ATP & Certification of the Rig: To + 12 months

2.2.8.11 Operation, Maintenance, and Testing of Aircraft electrical system in the rig (Manpower of 4 Nos): To + 24 months

Sl No.	Delivery Period	Months
1.	**Completion of installation of the Rig components at CABS and ATP and Certification of the Rig (Sl No. 1 to 7 of BoM)**	**10 months +02 months for certification**
2.	**Manpower (4 Nos) for Operation, Maintenance and Testing of Aircraft electrical system in the rig after certification of the rig for 1 year (Sl No. 8 of BoM)**	**1 Years from the date of Acceptance and Certification of the rig.**

2.2.9 Bill of Material (BoM)

Sl No.	Rig Equipment	Quantity/Resources
1	Design and Development of Hydraulic Power Pack with Test Bench	1Nos
2	Design and Development of Fuel circulation system with FCOC including Reservoir for IDG oil Heat Exchanger	2Nos
3	Design and Development of AC and DC Electrical Distribution Load Bank as per EPGS requirement	1Set
4	Data Acquisition and Control System (DAQCS) – Control, Instrumentation, DAQ, analysis and Display/GUI System	1Set
5	Aircraft grade Cable Looming along with MIL grade connectors	1Set
6	Mounting arrangements ,Power Panel and Racks for LRUs	1Set
7	Tool Kit	1Set
8	Manpower (Software, Electrical and Mechanical) for Operation, Maintenance and Testing of Aircraft electrical system in the rig after certification of the rig for 1 years	4 Nos

COMMERCIAL TERMS AT A GLANCE

Kindly fill up and forward the below given Commercial Terms & Conditions compliance from along with your Techno- Commercial Bid. (If CABS terms are strictly adhered to, bids can be evaluated quickly).

CABS Tender Reference No : ____

Your Quotation No : ____

Bid Security Declaration details : ____

Registration details : ____

1. **Payment Milestone MATRIX**

Payment Terms: Milestone payment will be made within 30 days on completion of each phase and acceptance/ completion certificate from user for the work carried out and on production of invoice. (30 days will be counted from the date of receipt of invoice at CABS). The following milestone payments with percentage as follows: -

		Payment Milestone MATRIX		
SL. No.	**Milestone**	**Description (Milestone Wise)**	**% of Payment**	**Project Time Line(T0+Months)**
1	**Milestone-1**	**Submission of High Level Documentation(HLD) and Implementation Plan in AGEST rig** Payment will be made within 30 days of Submission of Bills and confirmation of receipt and acceptance of High Level Documentations (HLD) & Implementation plan by user, whichever is later	**10%**	**T0+01 month**
2	**Milestone-2**	**Critical Design Review (CDR)** Payment will be made within 30 days of Submission of Bills and confirmation of receipt and acceptance of CDR document by user, whichever is later	**20%**	**T0 + 02 months**
3	**Milestone-3**	**FAT at Bidder's premises** Payment will be made within 30 days of Submission of Bills and confirmation of receipt and acceptance of FAT and ATP document by user, whichever is later	**10%**	**T0 + 08 months**
4	**Milestone-4**	**Installation, Integration & Commissioning of Rig at CABS** Payment within 30 days of submitting the bill and receipt of certificate of satisfactory completion of job from acceptance committee, whichever is later	**20%**	**T0 + 10 months**
5	**Milestone-5**	**ATP and Certification of Rig at CABS** Payment within 30 days of submitting the bill and receipt of certificate of satisfactory of ATP and certification of rig from acceptance committee, whichever is later.	**20%**	**T0 + 12 months**
6	**Milestone-6**	Manpower (4 Nos) for Operation, Maintenance and Testing of Aircraft electrical system in the rig after certification of the rig for 1 years	**20%** (with leftover balance if any)	**T0 + 24 months**
Conformance of Acceptability by Bidder		Yes / No	Yes / No	
❖		❖ **T0: Indicates date of the GeM Contract**		

S/N	**Commercial terms of CABS**	**Conformance of Bidder**

2	**Terms of Delivery**: Free Door delivery of the goods/services at CABS, Bengaluru.	
3	**Place of delivery**: CABS, Bangalore	
4	**Delivery Period**: Delivery Period for supply of items/rendering services would be **24 months** from the Effective Date of the Contract. Please note that the Contract can be cancelled unilaterally by the Buyer in case items are not received within the contracted delivery period including installation, Training and acceptance tests etc., as per scope, if any. Extension of contracted delivery period with/ without LD clause will be at the sole discretion of the Buyer.	
5	**Warranty: 24 months** (From the date of final acceptance of stores by Inspection agency/Joint Receipt Inspection or date of installation and commissioning whichever is later , as applicable)	
6	**Product Support after warranty**	Not Applicable
7	**Validity of Quote**: 180 days from the opening of Technical Bids.	
8	**Non-disclosure declaration certificate: (On Bidder Letter Head)** 1. M/s. Bidder hereby declare that I shall not disclose the contract or any provision, specification, plan, design, pattern, sample or information thereof to any third party during and after expiry of Contract. 2. If defying the norms as per the Contract a legal action may be taken against me as per the existing rules	
9	**Bid Security Declaration in lieu of EMD : (On Bidder Letter Head)** Vendor is required to submit **EMD of Rs 15,000,00/-(Rupees Fifteen Lakhs only) or Bid Security Declaration** accepting that if they withdraw or modify their Bids during the period of validity, or if they are awarded the contract and they fail to sign the contract, or to submit a performance security before the deadline defined in the request for bids document, they will be suspended for the period of *upto 2 years* from being eligible to submit Bids for contracts with any procuring entity of DRDO.	
10	**Performance Security Bond & Warranty Bond/e-PBG** **a. Performance Security Bond (PSB):** Successful Bidder shall submit Performance Security Bonds for an amount of 5% of the order value (inclusive of all taxes and duties) within fifteen days from the date of release of Supply Order (SO)/Contract. PSB should be valid up to 60 (sixty) days beyond the Delivery Period/contract. In case the execution of the supply/contract is delayed beyond the delivery period/contracted period and the Buyer grants the extension of delivery period, with or without liquidated damages, the successful Bidder must get the Performance Security Bond revalidated, if not already valid. Performance Security Bond may be submit in the form of Bank Draft, Fixed Deposit Receipt or a Bank Guarantee in favor of Director CABS. b. **Warranty Bond:** Successful Bidder shall submit Warranty Bond (wherever applicable) for an amount of 5% of the order value (inclusive of all taxes and duties) immediately after the delivery/acceptance of stores. Warranty Bond should be valid up to 60 (sixty) days beyond the warranty obligations. Buyer will not release the payment/PCB unless receipt of Warranty Bond. Warranty Bond may be submit in the form of Bank Draft, Fixed Deposit Receipt or a Bank Guarantee in favor of Director CABS. Warranty Bond would be returned to the Seller on successful completion of warranty obligations, under the contract The Performance Security Bond/ Warranty Bond will be forfeited by the Buyer, in case the conditions regarding adherence to delivery schedule and/or other provisions of the contract are not fulfilled by the Bidder. **The Performance Security Bond/ Warranty will be forfeited by the Buyer, in case the conditions regarding adherence to delivery schedule and/or other provisions of the contract are not fulfilled by the Bidder.**	

11	**Free Issue of Material (FIM):**	Not Applicable
12	**Liquidated Damage (LD):** The Buyer may deduct from the Bidder, as agreed, liquidated damages at the rate of 0.5% per week/part thereof, of contract value of the delayed stores which the bidder has failed to deliver within the period agreed for delivery in the contract subject to maximum of 10% of the total order value . In cases where partial delivery does not help in achieving the objective of the contract, LD shall also be levied on the total cost of the ordered quantity delivered by the bidder. This will also include the store(s) supplied within the delivery period. In case of Milestone payment, LD will be levied on milestone wise.	
13	**MSME/Startup confirmation and certification-** **(Mandatory requirement)** a) **MSME:** (If yes, attach latest valid MSME Registration certificate) b) **Startup:** (If yes, attach latest valid Startup Registration certificate)	Yes/No (Certificate attached) Yes/No (Certificate attached)
14	GST No. (Mandatory requirement): Furnish your GST No. without which bill/ invoice cannot be processed for payment	
15	**Name & Mobile No. Contact person. Email**	
16	Whether registered in DRDO / Central Govt / State Govt/ Any other Govt agencies or PSUs. If yes, please quote Reg.No and Validity of Registration.	
17	**Installation & Commissioning : Required**	**Yes**
18	**Training Details : Required** (**For 10 personnel for 15 days in CABS premises**).	**Yes**
19	**Partial Bid Acceptable**	Not Acceptable
20	**Ordering Information** **(Order to be place on)**	Complete details of Bidder with address
21	**Local Content** (Percentage to be indicated): In case of procurement for a value in excess of Rs 10 crores , the 'Class-I local supplier'/ 'Class-II local supplier' shall be required to provide a certificate from the statutory auditor or cost auditor of the company (in the case of companies) or from a practicing cost accountant or practicing chartered accountant (in respect of suppliers other than companies) giving the percentage of local content.	
	a) Class-I Local Supplier(Local content to be minimum 50%.)	Yes/No
	b) Class-II Local Supplier (Local content to be minimum 20%.)	Yes/No
22	**Whether undertaking regarding ban / blacklisting/debarment/ongoing enquiry by CBI/ED/any other Govt. agency has been enclosed (On Bidder Letter Head)**	Yes/No
23	**Certificate regarding Restriction on Public Procurement from certain countries (if applicable)**	Yes/No

Name of the Bidder:

Date: **Place:**

Signature:

Name and Designation of the Authorized Signatory with Seal:

TO BE SUBMITTED ON THE LETTER HEAD OF THE FIRM

Self-declaration by the vendors submitting bids/participating in quoting against Tender Enquiry No. _______________________

TO WHOM SO EVER IT MAY CONCERN

I/we, the undersigned representing M/s ___________________ in the capacity of ________________ in the afore said organisation/firm declare the following truly and honestly in lieu of submission of the Earnest Money Deposit: -

"That we are accepting the clause that at any given point of time if we withdraw or modify the Bids during the period of validity,

or

if we are awarded the contract and then fail to sign/execute the contract at any given time,

or

fail to submit the performance security Bond before the deadline as defined by the purchaser /specified in the RFP document,

we___________

may be suspended for a period **up to 2 years** from being eligible to submit bids for contracts with any of the procuring entities of DRDO/Government Tenders.

(Signature of the Bidder, with Official Seal)

To be printed on Company Letter Head

Certificate on Local Content

1. Reference is made to Tender No. CABS/-----------------------dated ----------.

2. It is certified that the percentage of 'Local Content' being provided by our firm viz, M/s--------------------- in the product ------------------------------- is -----%. The details of 'Local Value Addition' being made by us are as follows:-

Sl No	Description (As per BOQ)	Percentage of 'Local content' (For each line)	Location at which Local Value Addition is made	Percentage of 'Local content' (For total bid value)

3. It is further certified that services such as transportation, insurance, installation, commissioning, training and after sales service support like AMC/CMC etc have not been included in 'Local Content'/ 'Local Value Addition' claimed in the product as per abovementioned details.

Authorised Signatory

Date :